MAMMA ADA CENTRE

In Collaboration with FACM ITALY

UG 1st YEAR

AI FOR REAL-WORLD LEARNING

GAIO

Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

1. **I Demystified** — The Intelligence Behind Machines
2. **Python + AI Starter** — Prompt with Purpose
3. **AI in the Real World** — AI in Your Profession
4. **Navigate the AI Job World** — Build Your AI Skillset
5. **AI Workbench** — AI Coding Companion
6. **Responsible Digital Intelligence** — Data Privacy Matters

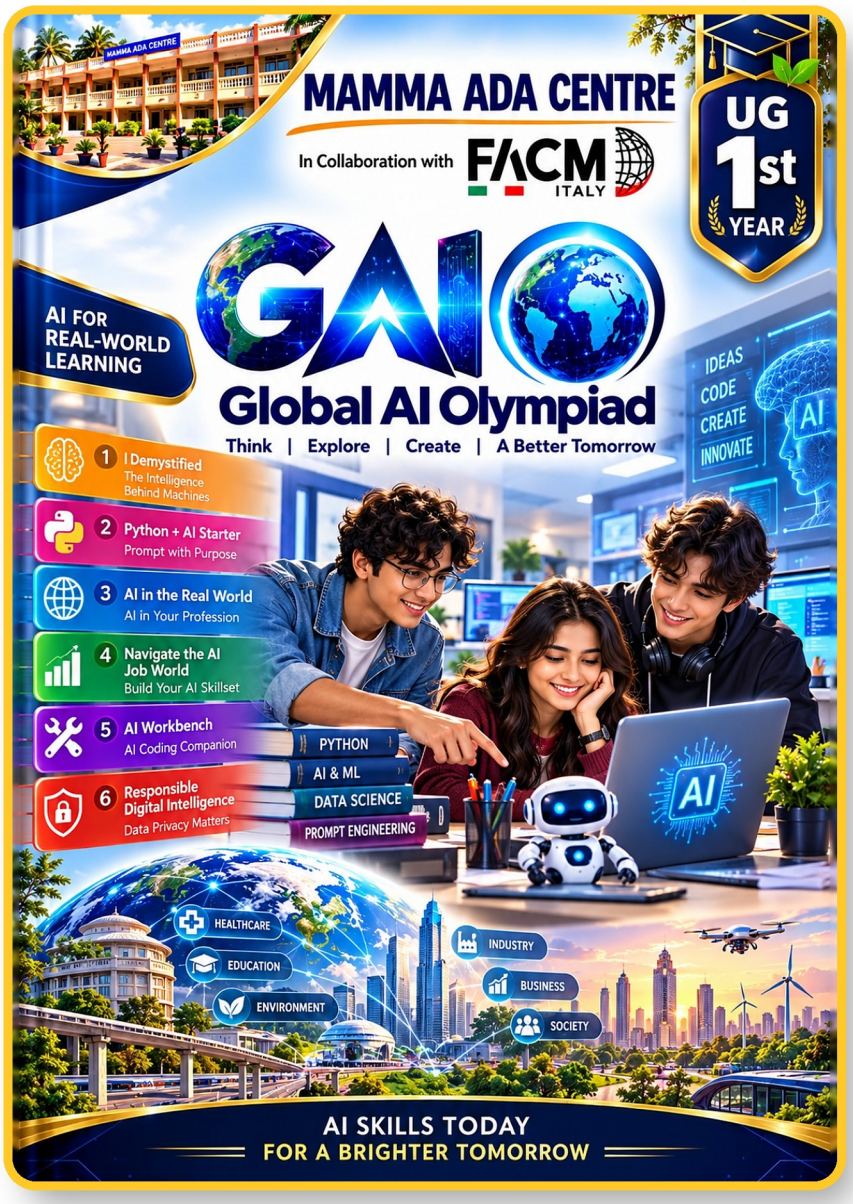
IDEAS
CODE
CREATE
INNOVATE

PYTHON
AI & ML
DATA SCIENCE
PROMPT ENGINEERING

HEALTHCARE
EDUCATION
ENVIRONMENT
INDUSTRY
BUSINESS
SOCIETY

AI SKILLS TODAY
FOR A BRIGHTER TOMORROW

This book belongs to

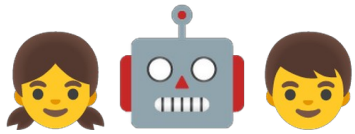


Name: _____

Class & Section: _____

School: _____

📖 PREFACE



AI is transforming every profession, from engineering and medicine to commerce, law and media. As a university student, you must understand how AI works, judge its outputs critically and apply it responsibly in your discipline. This book builds exactly those capabilities.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of UG 1st Year students, with university-level depth.

Every topic opens with an in-depth "Concept Explained" chapter, followed by a Concept Deep Dive, an AI Story, a case study, a lab activity and a Python Code Lab, with worksheets, quizzes, challenge problems and university-style exam preparation.

Responsibility runs through every chapter: you will examine AI's limits, bias, errors, privacy risks and impact on work, and the ethical duties of everyone who builds or uses it.

🤝 A note for teachers and parents

Faculty can use the Concept Explained chapters for lectures, the Code Labs and lab activities for practical sessions, and the exam-prep pages for assessment. Encourage students to verify AI outputs, cite sources and reflect on ethics. An answer key is provided at the end of the book.

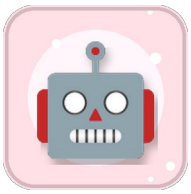
We hope this book helps you think critically, build confidently and lead responsibly in an AI-powered world. Your ideas today can change tomorrow!

With best wishes,

The GAIO Academic Team

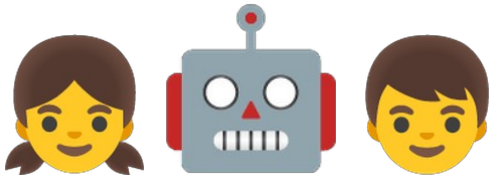
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Innovators!



MEET BOLT!

Welcome to university-level AI! I am Bolt, your AI guide. This year you will write real Python, use AI as a coding partner, study AI in your profession and practise exam-style answers. Let's build your AI edge!



Look for these pictures in your book:

CONCEPT EXPLAINED Concept explained in depth

CONCEPT DEEP DIVE Detailed explanation & code

SCENARIO Real-life scenario

CASE STUDY Analyse a real case

LAB ACTIVITY Hands-on lab

WORKSHEET Practice exercises

PUZZLE & DEFINE Puzzles & definitions

QUIZ TIME MCQ practice

AI STORY An AI story to think about

CHALLENGE PROBLEMS Higher-order thinking

CODE LAB Python code lab

EXAM PREP University-style exam prep

My 6-Month AI Journey

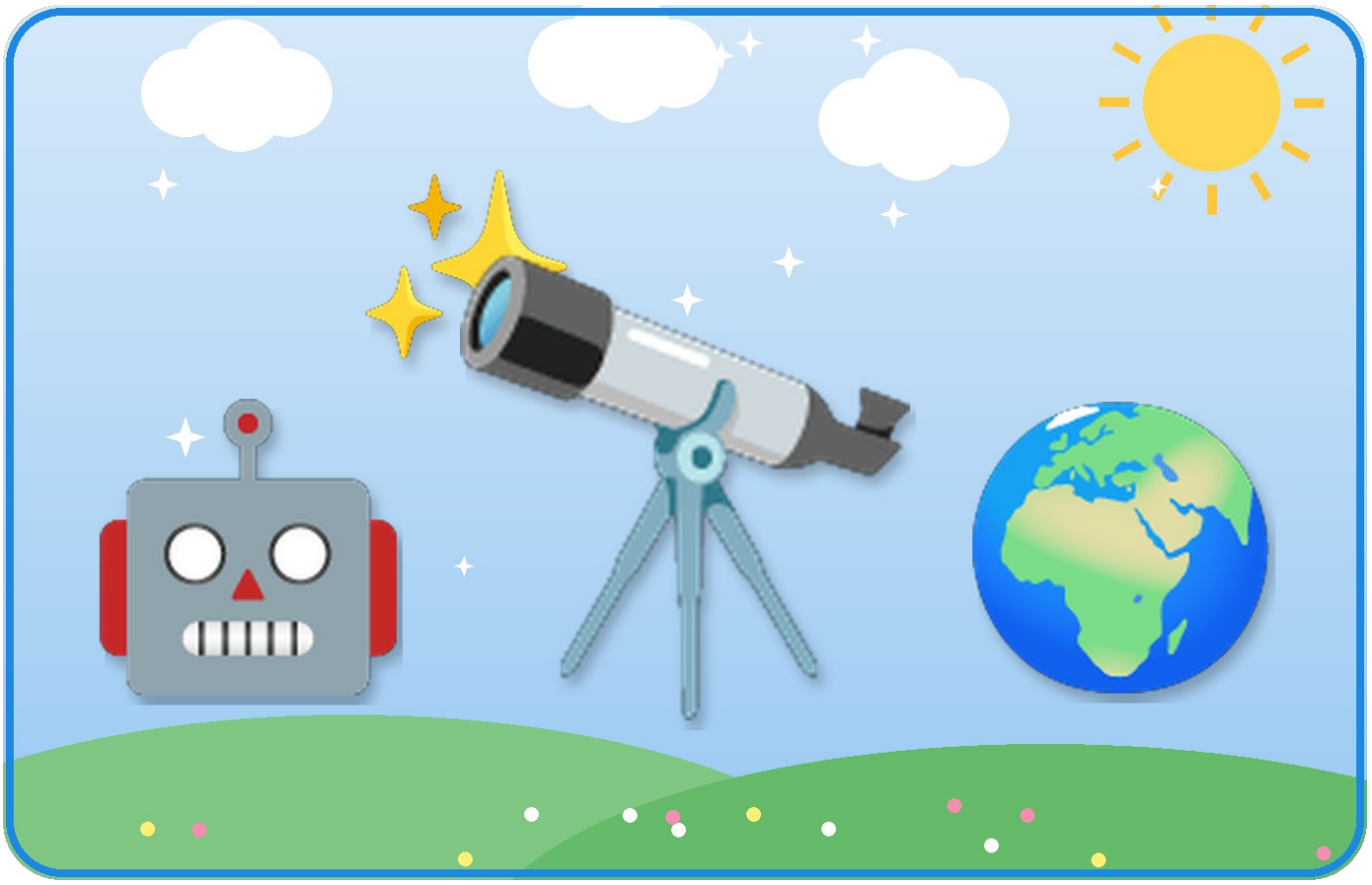
Follow the path. One month at a time!

MONTH 1	**AI DISCOVER** AI Basics	**AI Demystified • The Intelligence Behind Machines**
MONTH 2	**AI CONNECT** Communicate with AI	**Python + AI Starter • Prompt with Purpose**
MONTH 3	**AI SOLVE** AI Applications	**AI in the Real World • AI in Your Profession**
MONTH 4	**AI RISE** AI & Career	**Navigate the AI Job World • Build Your AI Skillset**
MONTH 5	**AI CREATE** AI Tools	**AI Workbench • AI Coding Companion**
MONTH 6	**AI CARE** Responsible AI	**Responsible Digital Intelligence • Data Privacy Matters**

MONTH 1

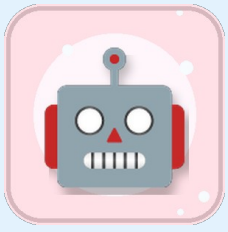
AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:



Topic 1

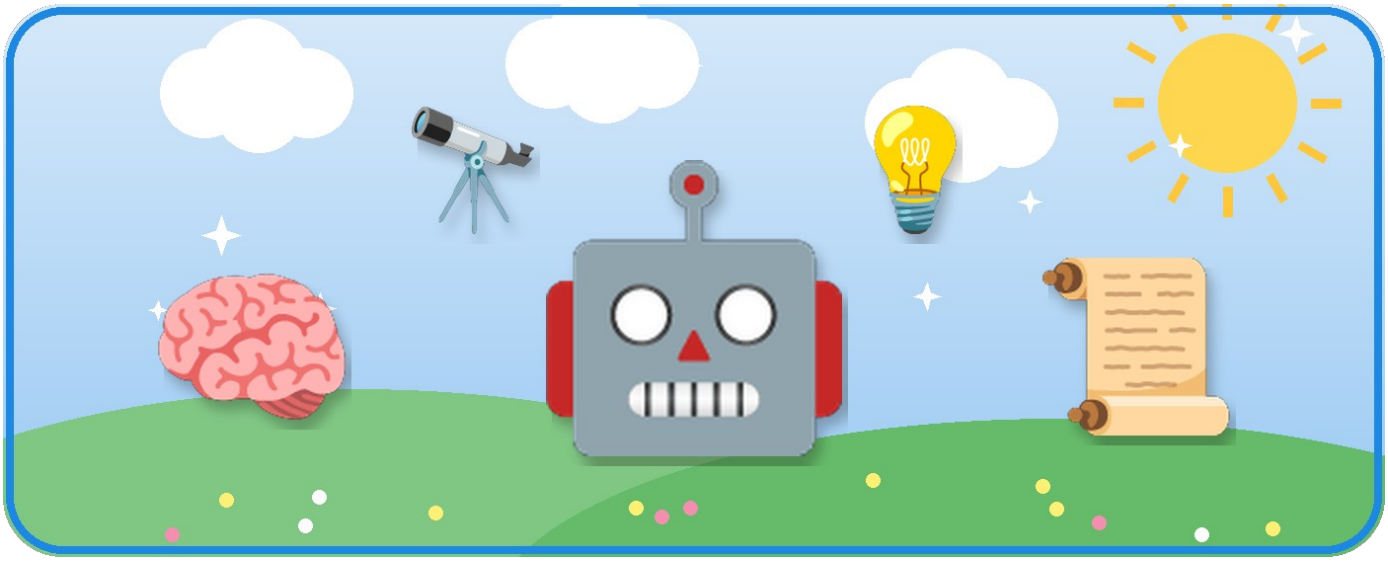
AI Demystified

Topic 2

The Intelligence Behind Machines

TOPIC 1 • MONTH 1 • AI BASICS

AI Demystified



LET'S LEARN

- AI: machines doing tasks that need intelligence.
- AI ⊃ Machine Learning ⊃ Deep Learning.
- Today's AI is narrow: excellent at specific tasks.

Ways machines learn

Supervised

Unsupervised

Reinforcement

Self-supervised

KEY TERMS

Narrow AI Task-specific AI

AGI General human-level AI

Agent Perceives & acts

📖 CONCEPT EXPLAINED: AI Demystified

💡 In this chapter

Artificial Intelligence is one of the most talked-about ideas of our time, and also one of the most misunderstood. Some people imagine thinking robots; others think AI is just clever marketing. In this chapter we will strip away the hype and build a clear, accurate mental model: what AI is, where it came from, how machines actually 'learn', and what today's AI can and cannot do.

🤖 1. What exactly is AI?

Artificial Intelligence (AI) is the field of computer science that builds systems able to perform tasks that normally require human intelligence: recognising images, understanding language, making decisions, planning routes or translating speech. The key word is 'tasks'. AI is not one single machine or one single program; it is a large toolbox of methods for getting computers to behave intelligently in specific situations.

It helps to picture three nested circles. The biggest circle is AI: any technique that makes machines act intelligently, including hand-written rules. Inside it sits Machine Learning (ML): systems that learn patterns from data instead of being told every rule. Inside ML sits Deep Learning (DL): machine learning using large neural networks with many layers. Generative AI tools like chatbots are built mainly on deep learning.

🤔 PAUSE & THINK

Your phone's calculator follows fixed rules. Your phone's face unlock learned from data. Which one is machine learning? Which one is still 'AI' in the broad sense? Explain.

📜 2. A short history: from rules to learning

The term 'Artificial Intelligence' was coined for a 1956 summer workshop at Dartmouth College in the USA. Early researchers were optimistic and built programs that solved logic puzzles and played checkers. In the 1970s and 80s, 'expert systems' captured the knowledge of human specialists as thousands of IF–THEN rules, for example to recommend medicines.

Rules turned out to be brittle: the real world has too many exceptions to write down. From the 1990s, machine learning grew as computers became faster and data became

plentiful. In 2012, a deep neural network called AlexNet dramatically won an image-recognition contest, starting the deep-learning boom. In 2017 the Transformer architecture appeared, and by 2022 large language models had reached millions of everyday users.

✏️ **WORKED EXAMPLE**

Timeline exercise: 1956 Dartmouth → 1997 Deep Blue beats chess champion → 2012 AlexNet → 2016 AlphaGo beats Go champion → 2017 Transformer → 2022 ChatGPT. Notice the pattern: each leap combined better algorithms, more data and more computing power.

🏷️ 3. How machines learn: four styles

In supervised learning, the machine learns from labelled examples: thousands of emails marked 'spam' or 'not spam', or photos labelled 'cat' or 'dog'. It finds the pattern that links inputs to labels, then applies it to new cases. In unsupervised learning there are no labels; the machine discovers structure on its own, such as grouping customers with similar shopping habits.

In reinforcement learning, an agent learns by trial and error, receiving rewards for good actions and penalties for bad ones, much like training a pet. Game-playing AIs and robot controllers often learn this way. Finally, self-supervised learning creates its own labels from raw data, for example by hiding a word in a sentence and learning to predict it. This is how modern language models learn from huge amounts of text.

🖐️ **TRY IT**

Classify each into a learning style: (a) a music app grouping similar songs, (b) a model predicting house prices from past sales, (c) a robot learning to balance by falling and retrying.

🎯 4. Narrow AI, AGI and the hype

Every AI system in use today is narrow AI: it is excellent at specific tasks but cannot transfer that skill to unrelated problems. A chess engine that beats grandmasters cannot drive a car or write a poem. Even powerful chatbots, which seem general, still make basic mistakes, invent facts and lack real-world experience.

Artificial General Intelligence (AGI) refers to a hypothetical system that could learn and perform any intellectual task a human can. Experts strongly disagree about whether and when AGI might arrive. As a student, the healthy stance is curiosity combined with

evidence: ask what a system has actually been shown to do, on which tasks, and how its errors were measured.

🤔 **PAUSE & THINK**

A news headline says 'AI now thinks like a human'. Write two questions you would ask to test whether the claim is true.

✅ **Key takeaways**

- AI is a toolbox of methods for intelligent behaviour; ML and DL are subsets of AI.
- AI moved from hand-written rules (expert systems) to learning from data.
- Learning styles: supervised, unsupervised, reinforcement and self-supervised.
- Today's AI is narrow; AGI remains hypothetical and debated.
- Judge AI claims by evidence: tasks, data and measured errors.

❓ **Quick check: answer in one or two lines.**

1. Draw the AI ⊃ ML ⊃ DL relationship and give one example for each circle.

2. Why did rule-based expert systems struggle in the real world?

3. Give one example each of supervised and reinforcement learning.

🔬 CONCEPT DEEP DIVE: AI Demystified

🔷 AI history: symbolic AI and expert systems (1950s–80s), machine learning (1990s–2000s), deep learning (2012+), generative AI (2020s).

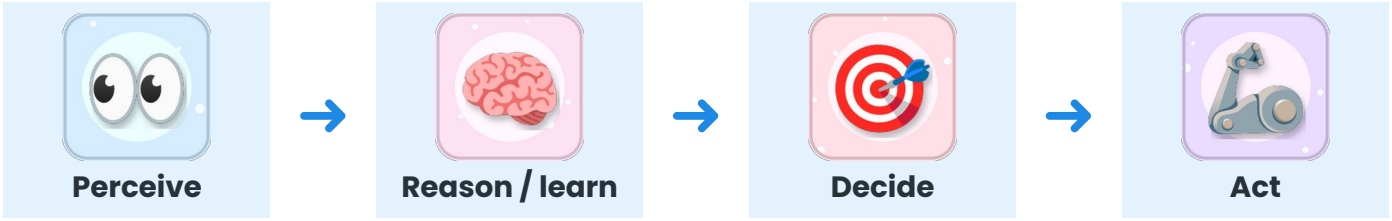
🔷 An intelligent agent perceives its environment through sensors and acts through actuators to achieve goals (PEAS model).

🔷 Supervised learning maps labelled inputs to outputs; unsupervised finds structure; reinforcement learns from rewards.

🔷 Self-supervised learning creates labels from the data itself (e.g., predict the masked word), powering modern LLMs.

🔷 Narrow AI excels at specific tasks; AGI (general intelligence) remains hypothetical and actively debated.

🔄 How it works



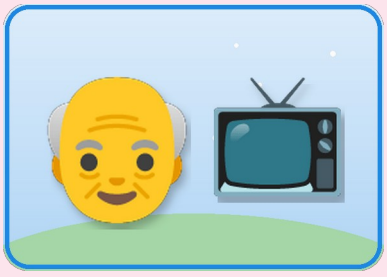
💻 An agent loop in Python

```
while not goal_reached:
    percept = sensors.read()
    action = policy(percept)      # learned or rule-based
    actuators.do(action)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Robot Who Wasn't Coming

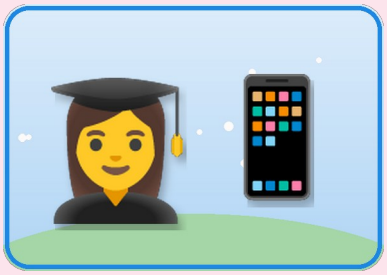
Characters: *Asha (first-year B.Sc student), her grandfather Thatha, Bolt (an AI guide).*



SCENE 1

Thatha switched off the TV news with a sigh. 'They say AI will take every job, Asha. Machines that think like people!' Asha, just three weeks into her first year, frowned. She had heard the same thing in the college canteen, but no one could explain what 'AI' actually was.

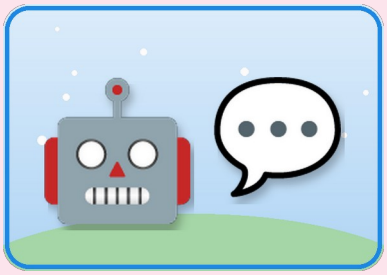
Pause & predict: What do you think Asha will discover about AI?



SCENE 2

That night she opened her phone and counted: face unlock, voice typing, spam filtering, photo search, map routes. 'Thatha, you already use five AI systems every day,' she said. 'And none of them think. Each one learned a single task from data.' Thatha raised an eyebrow. 'So the robots are already here?'

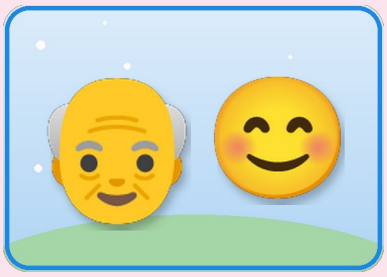
Pause & predict: Is face unlock 'thinking'? Why or why not?



SCENE 3

Asha asked Bolt, her AI study guide, whether it was intelligent. Bolt replied: 'I predict helpful words from patterns in text. I can explain AI history, but I have never seen a sunrise or felt nervous before an exam.' Asha wrote in her notebook: narrow AI, powerful but specific.

Pause & predict: What does Bolt's answer reveal about the limits of today's AI?



SCENE 4

At breakfast Asha drew three circles on a napkin: AI, machine learning, deep learning. 'Jobs will change, Thatha, but people who understand these tools will shape them.' Thatha smiled and folded the napkin into his pocket. 'Then you'd better teach me too.'

⏸ **Pause & predict:** What is one way Asha could explain AI to someone older in her family?

🌟 The big idea

AI is not a single thinking machine; it is many narrow tools learned from data. Understanding them replaces fear with power.

💬 Discuss with your classmates

1. Which AI myth have you heard most often? How would you correct it?

2. Is it fair to call today's AI 'intelligent'? Defend your view.

SCENARIO: AI Demystified

Read the scenario. Think about the choices made at each step.

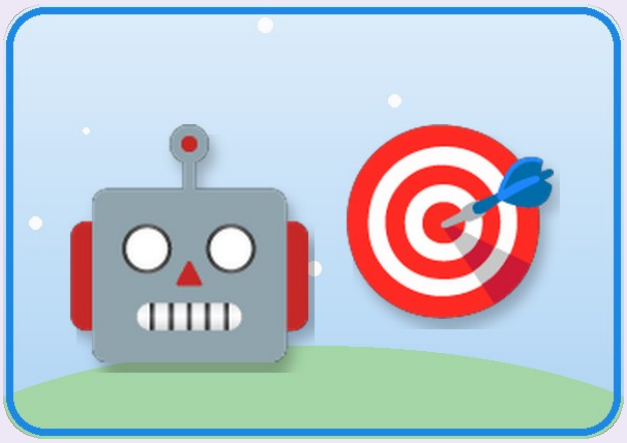
1

Asha, a first-year, hears: 'AI will take all jobs!'

2

She traces AI history: rules, ML, deep learning, GenAI.

3



She sees most AI is narrow: brilliant at one task.

4

Her verdict: AI changes tasks; skills keep you ahead.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI hiding in plain sight

Route planning

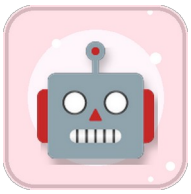
Spam filtering

Recommendations

Fraud detection

Photo search

Voice typing



BOLT SAYS:

AI isn't magic or a mind. It's maths + data + compute, engineered by people!

FUN FACT!

The term 'Artificial Intelligence' was coined for the 1956 Dartmouth workshop!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: ImageNet and the Deep Learning Boom

CASE FILE

In 2012, a deep neural network called AlexNet won the ImageNet image-recognition challenge, cutting the error rate from about 26% to about 15%. It used GPUs and a huge labelled dataset. Within a few years, deep learning dominated vision, speech and language research worldwide.

🔍 Analyse the case

1. Which three ingredients made AlexNet possible?

2. Compute the relative error reduction (26% → 15%).

3. Why did one competition result change an entire field?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Turing Test Showdown

You need:

Chat slips

AI chatbot (teacher)

Teams of 5

Steps:

1		One student and one AI answer the same 5 questions on paper slips.
2		Teacher shuffles and labels slips A and B.
3		Judges guess which answers came from AI and give reasons.
4		Tally accuracy. Discuss: does passing mean 'thinking'?

THINK & TALK

Is imitating intelligence the same as having it?

Record your observations & results:

💻 CODE LAB: A Rule-Based Agent

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  def thermostat(temp):
2      if temp < 18:
3          return 'heat'
4      elif temp > ____:
5          return 'cool'
6      return 'idle'
7
8  for t in [15, 22, 30]:
9      print(t, thermostat(t))
```

COMPLETE **T1. Fill the blank so the agent cools above 26 °C.**

PREDICT **T2. Write the 3 printed lines.**

DEBUG **T3. thermostat('20') crashes. Why? Fix it.**

EXTEND **T4. Add a 'fan' action for 24–26 °C.**

WORKSHEET: AI Demystified (Part A)

Name: _____ Date: _____

1. Match the learning type to its example.

Supervised ●

● **Grouping customers with no labels**

Unsupervised ●

● **Predicting the next word in text**

Reinforcement ●

● **Spam vs not-spam from labelled mail**

Self-supervised ●

● **Robot learns to walk by rewards**

2. Narrow AI or AGI (hypothetical)? Tick ✓.

	Narrow	AGI
Chess engine		
Learns any job like a human		
Speech-to-text app		
Solves any new problem unaided		

WORKSHEET: AI Demystified (Part B)

Name: _____ Date: _____

1. AI eras in order. Write 1, 2, 3, 4.

Generative AI ☐

Rule-based expert systems ☐

Deep learning ☐

Classical machine learning ☐

2. PEAS for a vacuum robot. Match each part.

Performance ● ● Wheels and brushes

Environment ● ● Rooms, furniture, dirt

Actuators ● ● Bump & dirt sensors

Sensors ● ● Cleanliness, battery use

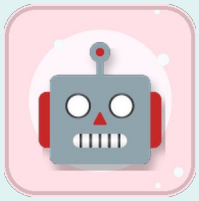
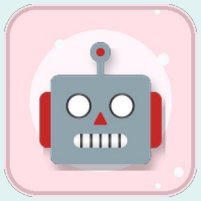
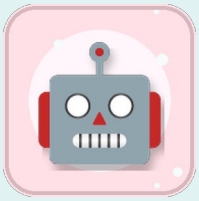
3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: AI Demystified

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
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2. Define it! Write each term in your own words with an example.

Narrow AI

Meaning: _____

Example: _____

AGI

Meaning: _____

Example: _____

Agent

Meaning: _____

Example: _____

? QUIZ TIME: AI Demystified

Circle the right answer!

Q1. Deep learning is a subset of...

a) Machine learning

b) Graphic design

c) Radio

Q2. The term AI was coined in...

a) 2010

b) 1956

c) 1800

Q3. Learning from rewards is...

a) Supervised

b) Clustering

c) Reinforcement

Q4. Today's deployed AI is mostly...

a) Narrow AI

b) Conscious AI

c) Magic

Q5. The Turing Test checks if AI can...

a) Run

b) Imitate human talk

c) Add

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Demystified

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Deep learning is a subset of machine learning.		
	All current AI is general intelligence.		
	Reinforcement learning uses rewards.		
	Expert systems used hand-written rules.		
	AlexNet won ImageNet in 2012.		
	Unsupervised learning needs labels.		

HOME CONNECT

Interview a family member: what do they think AI is? Correct one myth kindly with evidence.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Demystified

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is 'AI ⊃ ML ⊃ DL' a useful mental model?**

COMPARE **Q2. Compare supervised and reinforcement learning with one example each.**

APPLY **Q3. Write the PEAS description for a self-driving taxi.**

EVALUATE **Q4. 'AGI will arrive within 5 years.' Assess this claim.**

CREATE **Q5. Design a timeline poster of 6 AI milestones with one lesson each.**

📝 EXAM PREP: AI Demystified

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Define AI and distinguish it from ML.

2. What is the Turing Test?

3. Give two examples of narrow AI.

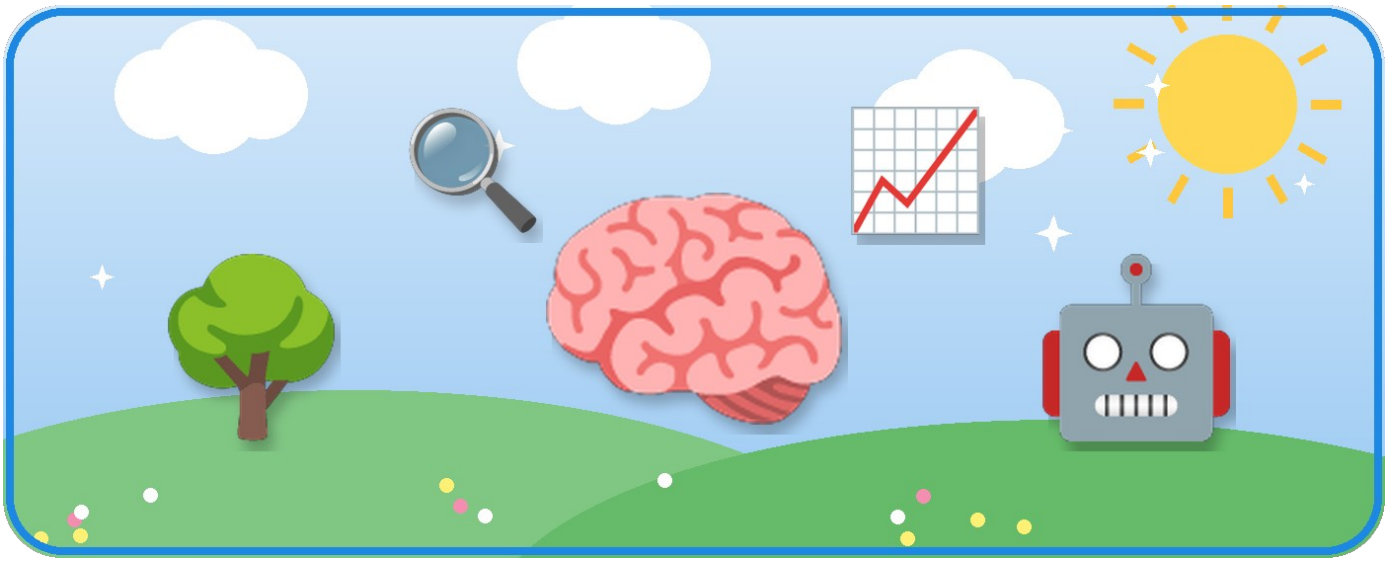
4. What is self-supervised learning?

PART B • Long answer (5 marks)

5. Trace the history of AI from 1956 to today, naming four milestones and explaining why deep learning succeeded after 2012.

TOPIC 2 • MONTH 1 • AI BASICS

The Intelligence Behind Machines



LET'S LEARN

- Classic AI searches and reasons with rules.
- ML learns a function from data: $y \approx f(x)$.
- Models: regression, trees, kNN, neural networks.

Core AI techniques

Search (BFS, A*)

Rules & logic

Regression

Decision trees

KEY TERMS

Heuristic Smart rule of thumb

Regression Predict a number

Decision tree If-then splits

📖 CONCEPT EXPLAINED: The Intelligence Behind Machines

💡 In this chapter

Behind every 'smart' app is an algorithm: a precise, step-by-step method. In this chapter we open the hood and look at the main families of AI techniques: searching for solutions, reasoning with knowledge, and learning from data with models such as regression, decision trees, nearest neighbours and neural networks. You will see that intelligence in machines is built from surprisingly understandable ideas.

🔍 1. Intelligence as search

Many problems can be described as moving from a start state to a goal state: finding a route on a map, solving a puzzle or planning moves in a game. An AI can explore possible steps like branches of a tree. Breadth-first search (BFS) explores all routes one step away, then two steps, and so on: it always finds the shortest path in steps but can be very slow.

Informed search uses a heuristic: a smart estimate of how close a state is to the goal. The famous A* algorithm scores each option with $f(n) = g(n) + h(n)$, where g is the cost so far and h is the estimated remaining cost (for maps, the straight-line distance). By always expanding the most promising option first, A* finds good routes far faster. Navigation apps and video-game characters rely on ideas like this.

✏️ WORKED EXAMPLE

Going from Hostel to Library: path via Canteen has g = 300 m walked, h = 200 m straight-line remaining → f = 500. Path via Gate has g = 150 m, h = 450 m → f = 600. A* explores the Canteen route first because 500 < 600.

📜 2. Reasoning with knowledge

Another classic approach stores knowledge as facts and rules: 'All mammals breathe air', 'A whale is a mammal', therefore 'A whale breathes air'. A program called an inference engine combines rules to reach new conclusions. This kind of logical AI powered the expert systems of the 1980s and still runs in tax software, safety interlocks and eligibility checkers.

Its big advantage is transparency: you can trace exactly why a decision was made. Its weakness is effort and brittleness: experts must write every rule, and rules break when

the world changes or when situations are fuzzy, such as recognising a handwritten digit. That weakness is exactly where machine learning shines.

🤔 **PAUSE & THINK**

Name one decision in your college (e.g., hostel eligibility) that is better handled by clear rules than by machine learning. Why?

📈 3. Learning from data: regression, trees and neighbours

Machine learning models learn a function that maps inputs (x) to outputs (y). Linear regression fits a straight line y = w·x + b through data, choosing w and b so that the average squared error is smallest. It predicts numbers, such as house prices or tomorrow's temperature. Logistic regression adapts this idea to predict probabilities for categories, such as 'pass' or 'fail'.

A decision tree asks a sequence of yes/no questions about features (Is attendance above 75%? Are internal marks above 20?) until it reaches a decision. It is easy to read but can overfit. The k-nearest neighbours (kNN) method simply finds the k most similar past examples and lets them vote. Each method makes different assumptions, so data scientists often try several.

✏️ **WORKED EXAMPLE**

Model y = 2x + 3. For x = 5, prediction = 13. If the true value is 14, the error is 1 and the squared error is 1. Averaging squared errors over all points gives the Mean Squared Error (MSE) that training tries to minimise.

🕸️ 4. Neural networks: learning layered patterns

A neural network is built from simple units called neurons. Each neuron multiplies its inputs by weights, adds a bias and passes the result through an activation function. Stacking many layers lets the network learn layered patterns: in images, early layers detect edges, middle layers detect shapes and later layers detect whole objects such as faces.

Training adjusts millions of weights to reduce error, using an algorithm called backpropagation with gradient descent. Neural networks need lots of data and computing power, but they excel at messy, high-dimensional inputs like images, speech and text, which is why they power today's most impressive AI systems.

TRY IT

A neuron has inputs x = (1, 2), weights w = (0.5, -1) and bias 1. Compute z = w·x + b, then apply ReLU, max(0, z).

Key takeaways

- Search algorithms (BFS, A*) explore states; heuristics make search efficient.
- Knowledge-based AI reasons with facts and rules: transparent but brittle.
- Regression predicts numbers; trees and kNN are intuitive classifiers.
- Neural networks learn layered patterns by adjusting weights.
- No single method wins everywhere; match the method to the problem.

Quick check: answer in one or two lines.

1. Write the A* scoring formula and explain each term.

2. Give one strength and one weakness of decision trees.

3. For y = 3x - 1, what is the prediction at x = 4?

🔬 CONCEPT DEEP DIVE: The Intelligence Behind Machines

🔷 Uninformed search (BFS, DFS) explores blindly; informed search like A* uses f(n) = g(n) + h(n) to prioritise promising paths.

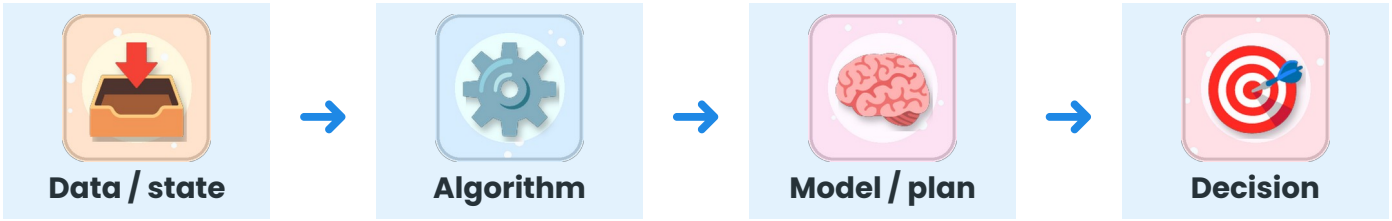
🔷 Knowledge representation stores facts and rules (e.g., IF fever AND rash THEN check measles) for logical inference.

🔷 Linear regression fits y = w·x + b by minimising mean squared error; logistic regression outputs class probabilities.

🔷 Decision trees split data on features that most reduce impurity (entropy / Gini); kNN votes among the k closest points.

🔷 Neural networks stack weighted sums and non-linear activations, learning complex functions from large data.

🔄 How it works



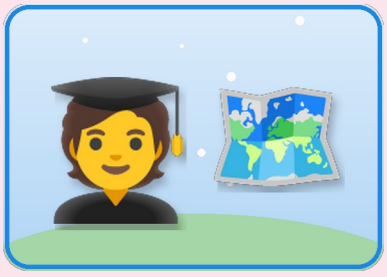
💻 kNN by hand in Python

```
pts = [((1, 1), 'A'), ((2, 1), 'A'), ((8, 9), 'B')]
q = (2, 2)
d = lambda p: (p[0]-q[0])**2 + (p[1]-q[1])**2
print(min(pts, key=lambda t: d(t[0]))[1])   # 'A'
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Lost on Campus

🎭 **Characters:** *Ravi (first-year B.Tech student), Priya (his classmate), a confused fresher named Sam.*



SCENE 1

On orientation day, Ravi found Sam wandering near the hostel, lost. The campus had 40 buildings and no map app. 'I'll write one,' Ravi announced. That evening he coded a program that tried every possible path from the hostel, one step at a time, until it reached the library.

⏸ **Pause & predict:** What kind of search is Ravi's program doing? What problem might appear?

SCENE 2

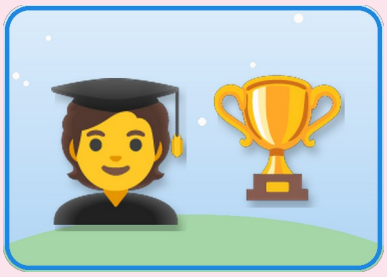
It worked, but slowly. With 40 buildings and hundreds of paths, his laptop fan roared for minutes. Priya peered over his shoulder. 'Why explore paths heading away from the library? Use the straight-line distance as a hint.' Ravi added a heuristic, and the A* version answered in under a second.

⏸ **Pause & predict:** Why did the heuristic make the program so much faster?

SCENE 3

Next, Priya suggested predicting crowds. They collected two weeks of entry counts at each building and trained a simple regression model on time of day and day of week. The canteen at 1 pm was predicted 'very crowded', which matched every student's experience.

⏸ **Pause & predict:** What other features could improve the crowd prediction?



SCENE 4

Their app, 'CampusPath', combined search with learning: A* found routes, and the model avoided crowded paths at peak hours. Sam, now confidently walking to class, became their first tester. The college IT club invited them to present it at the tech fest.

⏸ **Pause & predict:** Which parts of CampusPath use search and which use learning?

🌟 The big idea

Smart systems often combine classic search with machine learning: the right algorithm for each part of the problem.

💬 Discuss with your classmates

1. Where else could search plus prediction help on your campus?

2. What could go wrong if the crowd data were collected only during exam week?

SCENARIO: The Intelligence Behind Machines

Read the scenario. Think about the choices made at each step.

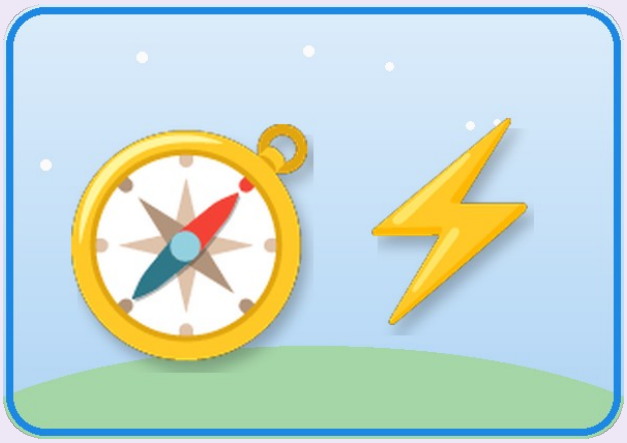
1

Ravi codes a campus navigation app.

2

Breadth-first search finds routes, but slowly.

3



A* with a distance heuristic is far faster!

4

He adds ML to predict crowd levels at each path.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Algorithms at work

A* in maps & games

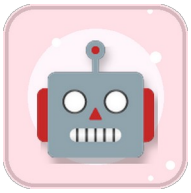
Regression for house prices

Trees for triage

kNN for 'similar movies'

Neural nets for vision

Game-tree search



BOLT SAYS:
Old AI followed rules humans wrote. New AI learns its own rules from examples!

FUN FACT!
IBM's Deep Blue beat world chess champion Garry Kasparov in 1997 using search!

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: Expert Systems: MYCIN's Lesson

CASE FILE

In the 1970s, Stanford's MYCIN expert system used about 600 hand-written rules to recommend antibiotics. In tests, its suggestions matched or beat many doctors. Yet it was never used in hospitals: rules were hard to update, it couldn't explain uncertainty well, and questions of legal responsibility were unresolved.

🔍 Analyse the case

1. Why did a high-performing system still fail to be adopted?

2. How do modern ML systems avoid writing every rule by hand?

3. What adoption barriers from MYCIN still apply today?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Human Decision Tree

You need:

20 animal cards

Chart paper

Teams of 4

Steps:

1		Each team gets 20 animal cards (mammal / bird / fish / reptile).
2		Choose yes/no questions that split the cards best (e.g., 'Has feathers?').
3		Draw the tree; stop when each leaf has one class.
4		Test on 5 new animals. Count correct predictions.

THINK & TALK

Why do the first questions matter most?

Record your observations & results:

💻 CODE LAB: Linear Regression by Hand

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  xs = [1, 2, 3, 4]
2  ys = [3, 5, 7, 9]
3  w, b = 2, 1
4  preds = [w * x + b for x in xs]
5  mse = sum((p - y) ** 2 for p, y in zip(preds, ys)) / ____
6  print(preds, mse)
```

COMPLETE **T1. Fill the blank to compute the mean.**

PREDICT **T2. What are preds and mse?**

MODIFY **T3. Change b to 2. What is the new MSE?**

EXTEND **T4. Write a loop that tries w = 1, 2, 3 and prints each MSE.**

✏️ WORKSHEET: The Intelligence Behind Machines (Part A)

Name: _____ Date: _____

1. Regression or classification? Tick ✓.

	Regress.	Classif.
Predict house price in ₹		
Email: spam or not		
Tomorrow's temperature		
Disease: yes or no		

2. Linear model y = 2x + 3. Use the word box.

WORD BOX: 13 ✿ 3 ✿ 7

x = 2 → y = ____

x = 5 → y = ____

x = 0 → y = ____

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

✏️ WORKSHEET: The Intelligence Behind Machines (Part B)

Name: _____ Date: _____

1. MSE for y = 3x − 2 on points (1,1), (2,5). Use the word box.

WORD BOX: 0.5 ✿ 1 ✿ 4

Prediction at x = 2: ____

Error at x = 2: 5 − 4 = ____

MSE = $(0^2 + 1^2) / 2$ = ____

2. Match the algorithm to its idea.

BFS ● ● **Vote of closest examples**

A* ● ● **Explore level by level**

Decision tree ● ● **Cost + heuristic estimate**

kNN ● ● **Split on best feature**

PUZZLE & DEFINE: The Intelligence Behind Machines

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

2. Define it! Write each term in your own words with an example.

Heuristic

Meaning: _____

Example: _____

Regression

Meaning: _____

Example: _____

Decision tree

Meaning: _____

Example: _____

? QUIZ TIME: The Intelligence Behind Machines

Circle the right answer!

Q1. A* search uses a...

a) Heuristic

b) Dice

c) Melody

Q2. Predicting a number is...

a) Classification

b) Regression

c) Clustering

Q3. kNN decides using...

a) Random guess

b) Longest path

c) Nearest neighbours

Q4. Deep Blue beat Kasparov in...

a) 1997

b) 2016

c) 1956

Q5. ML learns a mapping from...

a) Colours

b) Data examples

c) Radio waves

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: The Intelligence Behind Machines

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A* uses a heuristic estimate.		
	Regression predicts categories only.		
	Decision trees split on features.		
	kNN uses nearby examples to decide.		
	Expert systems learn rules from data automatically.		
	Deep Blue used search.		

HOME CONNECT

Draw a decision tree your family uses for a daily choice (e.g., what to cook). Is it accurate?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: The Intelligence Behind Machines

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does A* find good paths faster than BFS?**

COMPARE **Q2. Compare decision trees and kNN: training cost and explainability.**

APPLY **Q3. For y = 3x - 2, compute MSE for data (1, 1), (2, 5).**

EVALUATE **Q4. Are rule-based systems obsolete?**

CREATE **Q5. Design a 3-level decision tree to decide 'carry umbrella'.**

📝 EXAM PREP: The Intelligence Behind Machines

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. State the A* evaluation function.

2. Differentiate regression and classification.

3. How does kNN classify a point?

4. What is an expert system?

PART B • Long answer (5 marks)

5. Explain how a decision tree is built and used, with an example. Discuss one advantage and one limitation.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. AI ⊃ ML ⊃ ...

a) Deep learning

b) Design

c) Radio

Q2. f(n) = g(n) + h(n) is used by...

a) BFS

b) A*

c) Random

Q3. y = 2x + 3 at x = 5:

a) 10

b) 8

c) 13

Q4. Masked-word prediction is...

a) Self-supervised

b) Reinforcement

c) Clustering

Q5. AlexNet won ImageNet in...

a) 1997

b) 2012

c) 2020

Q6. PEAS: the S stands for...

a) Songs

b) Stars

c) Sensors

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

NARROW AI

AGI

AGENT

HEURISTIC

REGRESSION

SEARCH

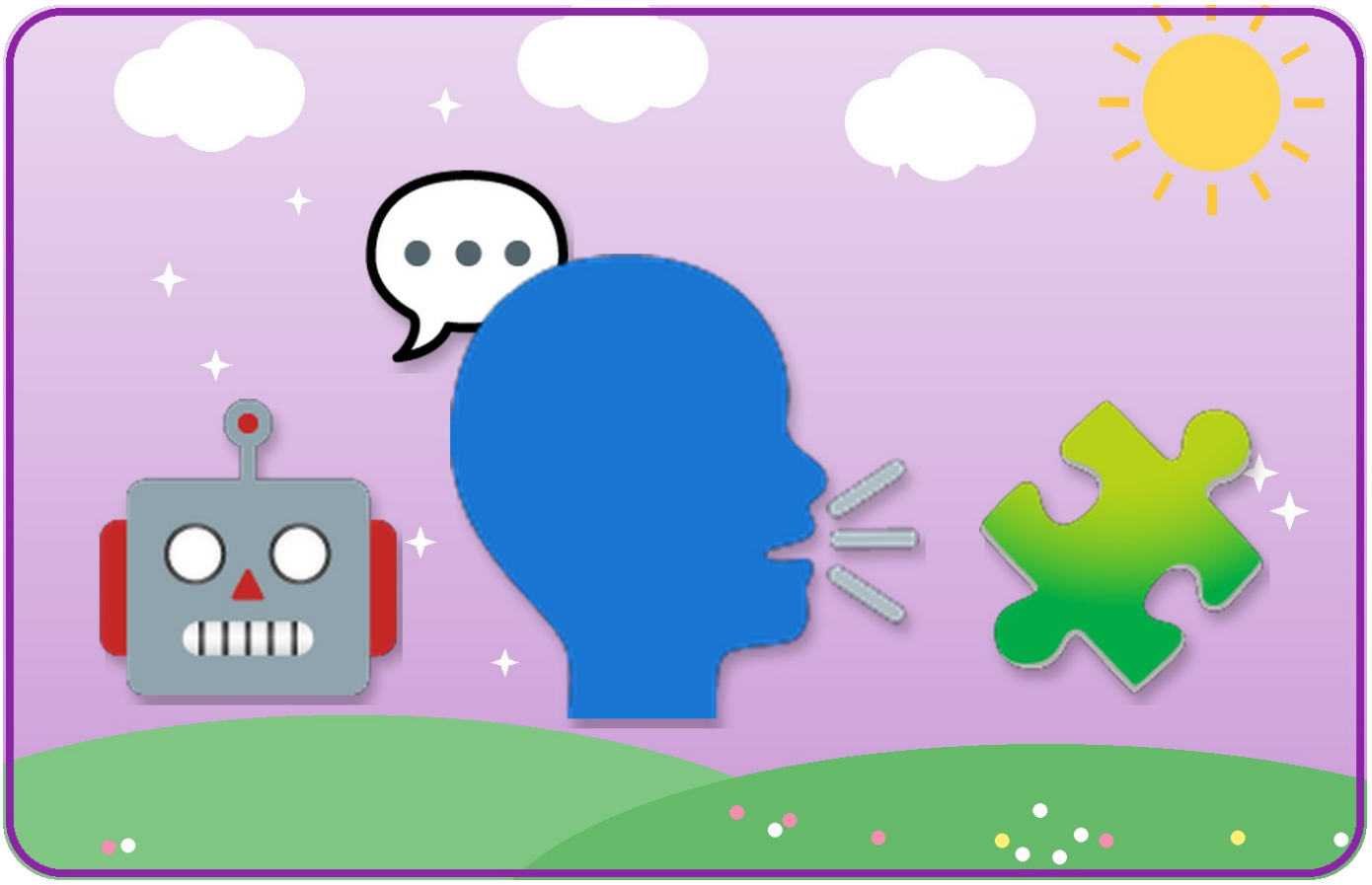
B	K	A	W	U	U	N	A	G	I
R	E	G	R	E	S	S	I	O	N
T	F	E	F	S	H	A	G	T	T
U	H	N	N	S	E	A	R	C	H
R	S	T	M	U	M	P	I	T	W
I	A	N	S	E	T	R	E	E	S
N	N	A	R	R	O	W	A	I	T
G	H	E	U	R	I	S	T	I	C
S	E	N	S	O	R	G	O	B	R
M	U	T	G	S	O	R	M	O	M

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Python + AI Starter

Topic 4

Prompt with Purpose

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Python + AI Starter

LET'S LEARN

- Python is the main language of AI.
- NumPy for arrays, pandas for tables, sklearn for ML.
- ML recipe: load → split → fit → predict → score.

The Python AI stack

NumPy

pandas

Matplotlib

scikit-learn

KEY TERMS

Array Grid of numbers

DataFrame Labelled table

fit() Train a model

📖 CONCEPT EXPLAINED: Python + AI Starter

💡 In this chapter

Python is the everyday language of AI. It is readable, free, and comes with powerful libraries for numbers, tables, charts and machine learning. This chapter takes you from Python basics to training your first model, explaining not just what to type, but why each step matters.

🐍 1. Python essentials for AI

Python programs are built from a few core ideas. Variables store values (marks = 78). Lists hold ordered collections ([78, 85, 90]) and are indexed from 0, so marks[0] is the first item. Dictionaries map keys to values ({'name': 'Asha', 'marks': 78}), which is perfect for records. Loops repeat work, and functions package reusable logic with def.

A favourite Python tool is the list comprehension, which builds a new list in one readable line: squares = [x**2 for x in range(1, 6)] gives [1, 4, 9, 16, 25]. Mastering these basics matters because every AI script, however advanced, is assembled from them.

🖐 TRY IT

Write a function average(nums) that returns the mean of a list. Test it with [70, 80, 90]. What should it return?

🔢 2. NumPy and pandas: the data workhorses

NumPy provides arrays: grids of numbers that support fast, vectorised maths. Instead of looping over a list to double every value, you write arr * 2 and NumPy does it in optimised C code underneath. This speed matters when datasets have millions of values, as in images or sensor readings.

pandas builds on NumPy with the DataFrame, a table with named columns, like a programmable spreadsheet. You can load a CSV with pd.read_csv(), peek with df.head(), summarise with df.describe(), fix missing values with fillna() or dropna(), and aggregate with groupby(). Most real AI work starts here: understanding and cleaning the data.

✏️ **WORKED EXAMPLE**

df.groupby('department')['marks'].mean() returns the average marks for each department in one line, a task that could take many lines of manual loops.

🧠 3. The scikit-learn recipe

scikit-learn is the standard library for classical machine learning. Its design is beautifully consistent: every model is created, then trained with fit(X, y), then used with predict(X) and evaluated with score(X, y). X is the table of input features; y is the column you want to predict.

The full recipe is: load data → split into training and test sets with train_test_split → choose a model → fit on the training set → predict and score on the test set. The test set must stay unseen during training, because it plays the role of the future: it tells you whether the model has truly learned or merely memorised.

🤔 **PAUSE & THINK**

Why would scoring a model on the same data it trained on give a misleadingly high score?

📓 4. Working style: notebooks and good habits

Most learners use Jupyter or Google Colab notebooks, which mix code, output and notes in one document. Run cells top to bottom, name variables clearly, and add short markdown notes explaining what you are doing and why. Set random_state (e.g., 42) whenever splitting or shuffling, so you and others can reproduce the same results.

Errors are normal and useful. Read the last line of an error message first: it usually names the problem (KeyError, ValueError, ModuleNotFoundError). Search the exact message, check spelling and shapes of your data, and change one thing at a time. Professional programmers spend a large part of their time debugging; it is a skill you build deliberately.

🖐️ **TRY IT**

In Colab, load the Iris dataset, split it 80/20 with random_state=42, fit a DecisionTreeClassifier and print the test score.

✅ Key takeaways

- Core Python: variables, lists, dicts, loops, functions, comprehensions.
- NumPy arrays enable fast maths; pandas DataFrames handle tables.
- scikit-learn: fit, predict, score, with a held-out test set.
- Notebooks + random_state + clear notes = reproducible work.
- Reading error messages carefully is a core skill.

❓ Quick check: answer in one or two lines.

1. What does [x*2 for x in [1, 2, 3]] produce?

2. Name the three main scikit-learn methods and what each does.

3. Why must the test set stay unseen during training?

🔬 CONCEPT DEEP DIVE: Python + AI Starter

🔷 Core Python: variables, lists, dictionaries, loops, functions and list comprehensions like [x**2 for x in nums].

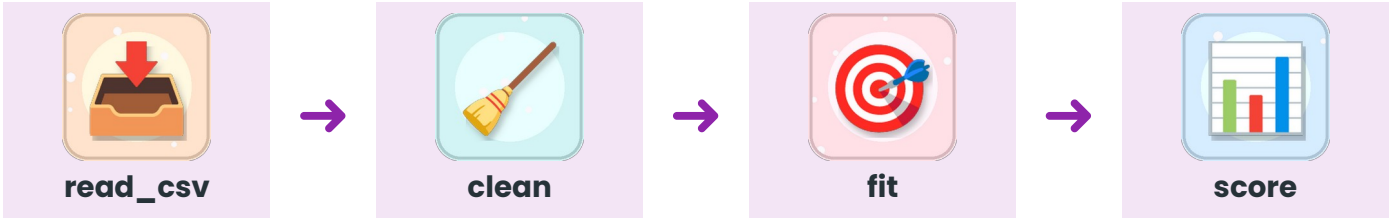
🔷 NumPy arrays support fast vectorised maths: arr * 2 doubles every element without a loop.

🔷 pandas DataFrames: read_csv, head, describe, groupby, dropna and fillna for exploration and cleaning.

🔷 scikit-learn API: every model has fit(X, y), predict(X) and score(X, y); preprocessing via pipelines.

🔷 Always split data before training; set random_state for reproducibility; evaluate on unseen test data.

🔄 How it works



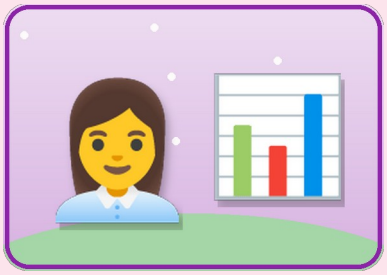
💻 A complete ML script

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
X, y = load_iris(return_X_y=True)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=.2)
print(DecisionTreeClassifier().fit(Xtr, ytr).score(Xte, yte))
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Priya and the Spreadsheet Monster

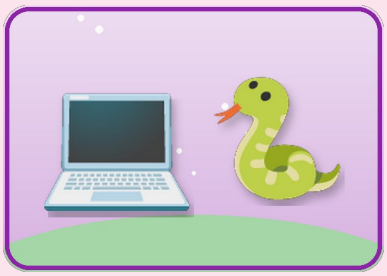
Characters: *Priya (B.Com student), her uncle who runs a textile shop, a Colab notebook.*



SCENE 1

Priya's uncle kept three years of sales in a giant spreadsheet. 'Which month do I sell most sarees? Should I stock more cotton or silk?' he asked. Priya, a commerce student, had always avoided coding. But scrolling 10,000 rows by hand felt impossible.

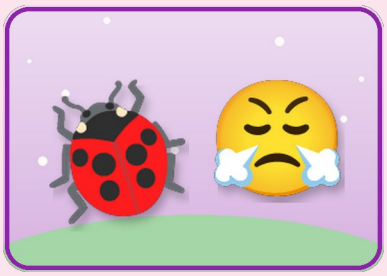
Pause & predict: Why might coding be better than scrolling for this task?



SCENE 2

Her college had just started a Python + AI starter course. In Colab she typed df = pd.read_csv('sales.csv') and then df.head(). Five neat rows appeared. 'That's it?' she laughed. Next, df.groupby('month')['amount'].sum() showed October towering above all other months: festival season.

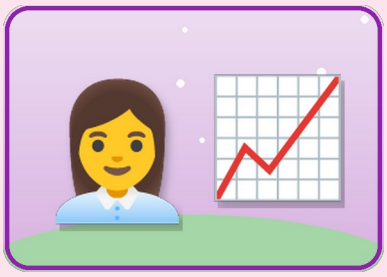
Pause & predict: What other question could Priya answer with groupby?



SCENE 3

Then came an error: KeyError: 'Month'. Frustrated, she nearly quit. Her lab partner pointed at the last line. 'Your column is lowercase month.' One small fix, and the chart appeared. Priya wrote in her notebook: read the last line of the error first.

Pause & predict: What is a good first step when you see a Python error?



SCENE 4

Finally, she trained a simple regression model to forecast next October's sales, checked it on last year's data, and found it reasonably close. Her uncle placed a larger silk order early and avoided the usual festival shortage. 'You're my data scientist now,' he said proudly.

⏸ **Pause & predict:** Why did Priya test her model on last year's data before trusting it?

🌟 The big idea

Python turns overwhelming data into clear answers, and every error message is a clue, not a wall.

💬 Discuss with your classmates

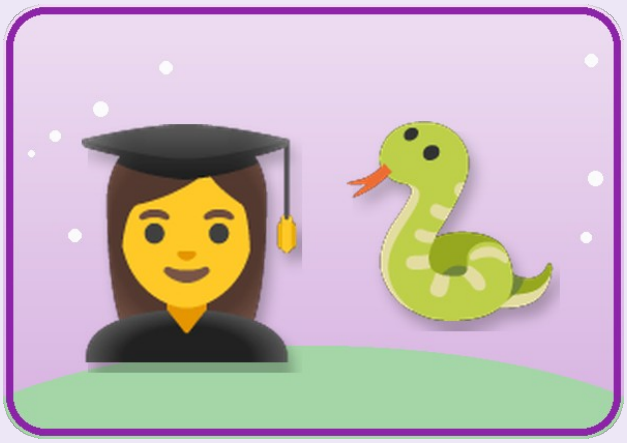
1. What data from your own life or family could you analyse with pandas?

2. Should a shop owner trust a forecast completely? What else should he consider?

SCENARIO: Python + AI Starter

Read the scenario. Think about the choices made at each step.

1

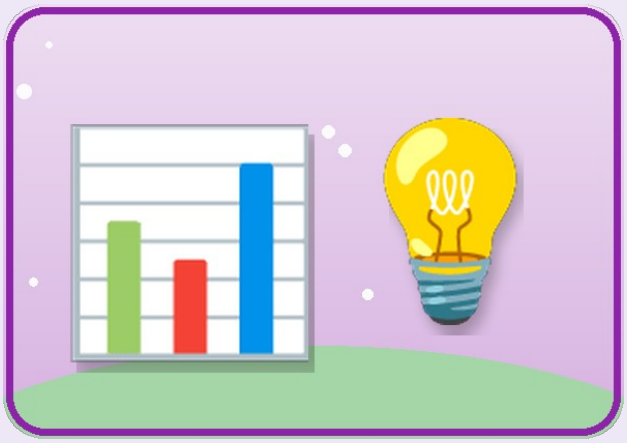


Priya, a commerce student, fears coding.

2

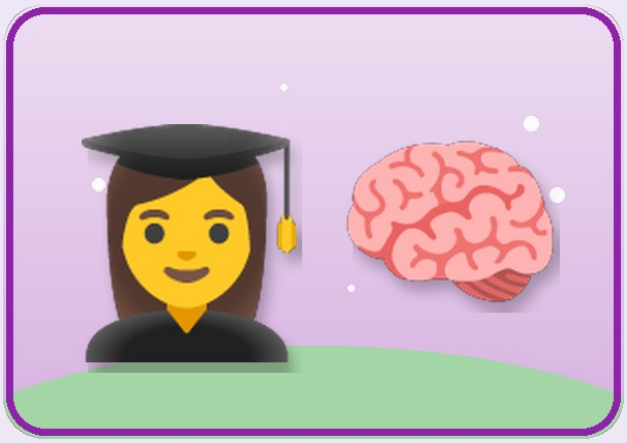
In Colab she loads a sales CSV with pandas.

3



Three lines reveal her best-selling month!

4



Next: a regression to forecast next month's sales.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

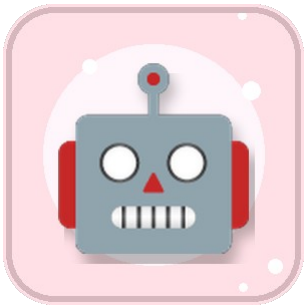
🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Python powering AI

Jupyter / Colab notebooks

PyTorch models

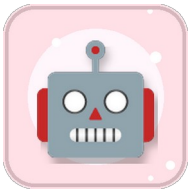
Data dashboards



Chatbot back-ends

Bioinformatics

Satellite analysis



BOLT SAYS:

Five lines of scikit-learn can train a real model. Understanding them takes a semester!

FUN FACT!

Python is named after the comedy group Monty Python, not the snake!

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: Python Spreadsheet Automation at a Firm

CASE FILE

A mid-size accounting firm spent 40 staff-hours a week copying data between spreadsheets. A trainee wrote a 60-line pandas script to merge and validate files automatically. Weekly effort dropped to 4 hours and copy errors almost disappeared. The firm then trained all new joiners in basic Python.

🔍 Analyse the case

1. Calculate the % reduction in weekly effort.

2. Why did errors also fall with automation?

3. What risks come with relying on one trainee's script?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: First Model in Colab

You need:

Laptop / lab PC

Google Colab

Pairs

Steps:

1

Load the Iris dataset: from sklearn.datasets import load_iris.

2

Split 80/20 with train_test_split(random_state=42).

3

Fit a DecisionTreeClassifier; predict on the test set.

4

Print accuracy_score. Try max_depth = 1, 3, 10.

THINK & TALK

Why does max_depth = 10 not always give the best test score?

Record your observations & results:

💻 CODE LAB: Your First Classifier

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  from sklearn.datasets import load_iris
2  from sklearn.model_selection import train_test_split
3  from sklearn.neighbors import KNeighborsClassifier
4  X, y = load_iris(return_X_y=True)
5  Xtr, Xte, ytr, yte = train_test_split(
6      X, y, test_size=0.2, random_state=42)
7  model = KNeighborsClassifier(n_neighbors=____)
8  model.fit(Xtr, ytr)
9  print(model.score(Xte, yte))
```

COMPLETE **T1. Choose k = 5 for the blank.**

PREDICT **T2. How many test samples are there (Iris has 150)?**

DEBUG **T3. model.fit(Xte, ytr) gives an error. Explain.**

EXTEND **T4. Loop k = 1 to 15 and print the best k.**

WORKSHEET: Python + AI Starter (Part A)

Name: _____ Date: _____

1. Match the library to its job.

NumPy ●

pandas ●

Matplotlib ●

scikit-learn ●

● Plot charts

● Fast numeric arrays

● Train ML models

● Read & clean tables

2. The scikit-learn recipe. Write 1, 2, 3, 4.

Score on test data ☐

Load data ☐

model.fit(X_train, y_train) ☐

train_test_split ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

✏️ WORKSHEET: Python + AI Starter (Part B)

Name: _____ Date: _____

1. Predict the output. Use the word box.

WORD BOX: [2, 4, 6] ✿ 12 ✿ 6

sum([3, 4, 5]) → ____

[x*2 for x in [1, 2, 3]] → ____

len('Python') → ____

2. Match the pandas method to its job.

df.head() ●

df.describe() ●

df.dropna() ●

df.groupby() ●

● **Summary statistics**

● **Remove missing rows**

● **First 5 rows**

● **Aggregate by category**

PUZZLE & DEFINE: Python + AI Starter

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Array

Meaning: _____

Example: _____

DataFrame

Meaning: _____

Example: _____

fit()

Meaning: _____

Example: _____

? QUIZ TIME: Python + AI Starter

Circle the right answer!

Q1. Tables in Python are handled by...

a) pandas

b) Paint

c) Radio

Q2. len([3, 6, 9]) returns...

a) 9

b) 3

c) 18

Q3. model.fit() is used to...

a) Delete

b) Plot

c) Train

Q4. Python lists start at index...

a) 0

b) 1

c) -1

Q5. A test set measures...

a) Speed

b) Generalisation

c) Volume

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Python + AI Starter

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Python lists are zero-indexed.		
	pandas is used for tabular data.		
	model.predict() trains the model.		
	train_test_split creates train and test sets.		
	NumPy arrays are slower than lists for maths.		
	random_state helps reproducibility.		

HOME CONNECT

Track your daily expenses for a week in a CSV. Use pandas to find your top spending category.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Python + AI Starter

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do we split data into train and test sets?**

COMPARE **Q2. Compare a Python list and a NumPy array.**

APPLY **Q3. Write a list comprehension for squares of even numbers 1–10.**

EVALUATE **Q4. Is 100% training accuracy good news?**

CREATE **Q5. Plan a pandas analysis of your class marks: 4 steps.**

📝 EXAM PREP: Python + AI Starter

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four Python libraries used in AI and their roles.

2. What does train_test_split do?

3. Write a list comprehension for cubes of 1–5.

4. Why set random_state?

PART B • Long answer (5 marks)

5. Describe the full workflow to train and evaluate a classifier in scikit-learn, with code for each step.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Prompt with Purpose

LET'S LEARN

- Purposeful prompts state goal, audience & limits.
- Patterns: few-shot, chain-of-thought, persona, critique.
- Ask for structured output (tables, JSON) to reuse.

Prompt patterns

Few-shot

Step-by-step

Persona

Self-critique

KEY TERMS

Structured output JSON / table format

Chain-of-thought Step-wise reasoning

Constraint Limit on output

📖 CONCEPT EXPLAINED: Prompt with Purpose

💡 In this chapter

Large language models respond to whatever we type, but the quality of their answers depends heavily on how we ask. Prompting with purpose means treating a prompt like a precise brief to a capable assistant. In this chapter you will learn the anatomy of a strong prompt, proven prompt patterns, how to get structured output, and how to test prompts systematically.

🎯 1. Anatomy of a purposeful prompt

A vague prompt ('Explain thermodynamics') forces the model to guess your level, length and goal, so it gives a generic answer. A purposeful prompt states the goal (revise for an exam), the audience (first-year B.Sc students), the context (we covered the first law last week), constraints (150 words, one everyday analogy) and the format (three bullet points plus two MCQs).

Think of it as briefing a new teammate. The more clearly you describe what 'done' looks like, the less the model has to invent. Adding success criteria, such as 'the analogy must use something from a kitchen', makes outputs more consistent and easier to judge.

✏️ WORKED EXAMPLE

Weak: 'Tell me about overfitting.' Strong: 'Explain overfitting to first-year B.Sc students in under 120 words, using one cricket analogy, then give 2 MCQs with answers.'

🧩 2. Patterns that work

Few-shot prompting includes one or more solved examples so the model copies the format and style. Chain-of-thought prompting asks the model to reason step by step, which helps on maths and logic problems and lets you check its working. Persona prompting ('Act as a strict examiner') shapes tone and focus.

Two more patterns are powerful for students. Self-critique asks the model to review its own answer for errors and improve it. Clarify-first asks the model to list questions it needs answered before starting. Both reduce wrong assumptions, especially on open-ended tasks like project planning.

🤔 PAUSE & THINK

For a physics numerical, which pattern would you use: persona, few-shot or chain-of-thought? Justify in one line.

3. Structured output and using AI from code

When an answer will be used by a program, ask for structured output, usually JSON: 'Return JSON with keys term, definition, example.' JSON is easy for code to parse with json.loads(). But models sometimes add extra text or break the format, so always validate the output and handle errors gracefully.

Settings matter too. Temperature controls randomness: low values (0–0.3) give focused, repeatable answers for facts and code; higher values give more varied, creative text. Max tokens limits length. Choosing these deliberately is part of prompting with purpose.

TRY IT

Write a prompt that returns three Python data types as a JSON array of objects with keys name and example.

4. Testing prompts like a scientist

Because model outputs vary, one good answer proves little. Treat prompts like experiments: fix a small set of test inputs, define a rubric (accuracy, clarity, level, format), run each prompt version several times and record scores. Change one element at a time so you know what caused any improvement.

Finally, remember responsibility. Verify facts against textbooks or trusted sources, never paste confidential data into public tools, and follow your college's rules on disclosing AI assistance. Purposeful prompting is about getting reliable help, not outsourcing your thinking.

PAUSE & THINK

You compared two prompts once and prompt B looked better. What two things should you do before concluding B is better?

✅ Key takeaways

- Purposeful prompts state goal, audience, context, constraints and format.
- Patterns: few-shot, chain-of-thought, persona, self-critique, clarify-first.
- Ask for JSON when code will use the output, and validate it.
- Low temperature for facts and code; higher for creative tasks.
- Test prompts with fixed inputs, rubrics and repeated runs.

❓ Quick check: answer in one or two lines.

1. List the five parts of a purposeful prompt.

2. When would you use chain-of-thought prompting?

3. Why must JSON output from an LLM be validated?

🔬 CONCEPT DEEP DIVE: Prompt with Purpose

🔷 Purposeful prompts specify: goal, audience, context, constraints (length, tone), output format and success criteria.

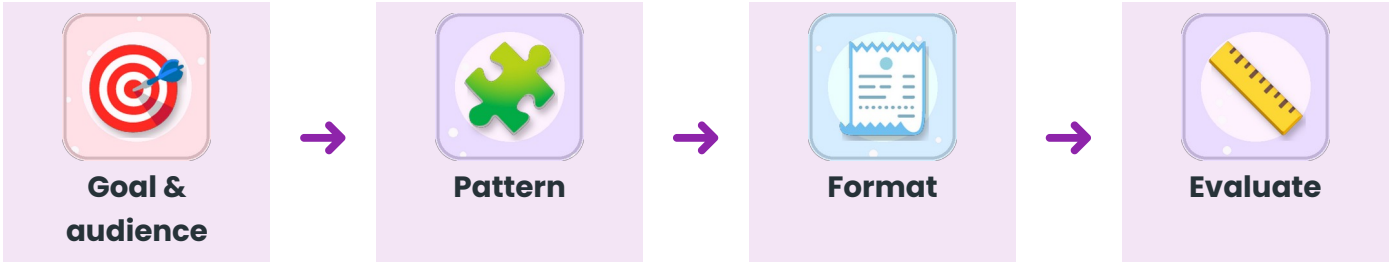
🔷 Few-shot examples teach format and style; chain-of-thought improves multi-step reasoning on maths and logic.

🔷 Structured output: ask for JSON with named keys so code can parse it reliably; validate before using.

🔷 Self-critique and 'ask me clarifying questions first' patterns reduce wrong assumptions.

🔷 Evaluate prompts systematically: fixed test inputs, a rubric, multiple runs, and change one variable at a time.

🔄 How it works



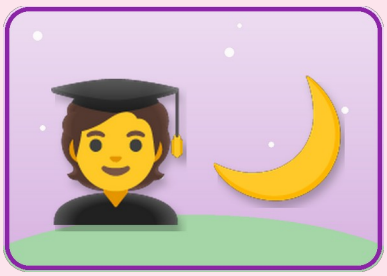
💻 Calling an LLM with structured output

```
prompt = '''Return JSON with keys term, definition, example.
Term: overfitting. Audience: first-year B.Sc.'''
reply = llm(prompt, temperature=0.2)
data = json.loads(reply)   # validate before use!
```

✍️ In your own words: explain this topic to a Class 10 student.

📚 AI STORY: The Night Before the Exam

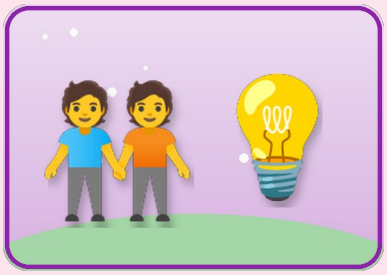
🎭 **Characters:** *Karthik (B.Sc Physics student), his roommate Imran, an AI chatbot.*



SCENE 1

It was 11 pm, and Karthik's thermodynamics exam was the next morning. He typed into a chatbot: 'Explain thermodynamics.' Two pages of dense text appeared, full of terms he had never heard. He felt even more confused.

⏸ **Pause & predict:** What was wrong with Karthik's prompt?



SCENE 2

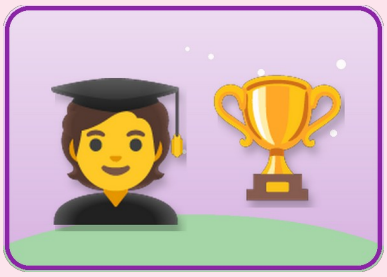
Imran, a computer science student, laughed kindly. 'You asked a question the size of a textbook. Tell it who you are, what you need and how long.' Karthik tried: 'I'm a first-year B.Sc student. Explain the first law of thermodynamics in 150 words with one kitchen analogy, then give 3 MCQs with answers.'

⏸ **Pause & predict:** Identify the goal, audience, constraint and format in the new prompt.

SCENE 3

The answer compared a pressure cooker to energy conservation, clearly and briefly. The MCQs were perfect for revision. But one answer claimed a formula he didn't recognise. Karthik checked his textbook: the chatbot had made a sign error. He corrected it and wrote a note: always verify.

⏸ **Pause & predict:** Why is it risky to trust the chatbot's answers without checking?



SCENE 4

By midnight he had a one-page summary, ten practice questions and confidence. The next day, a question on the first law appeared, and his kitchen analogy helped him explain it. Later he shared his 'purposeful prompt' template with his whole class.

⏸ **Pause & predict:** How could Karthik use the same method for other subjects?

🌟 The big idea

Clear prompts produce clear answers, but your own judgement must check them.

💬 Discuss with your classmates

1. Write a purposeful prompt for a topic you are studying this week.

2. Where should students draw the line between AI help and doing their own work?

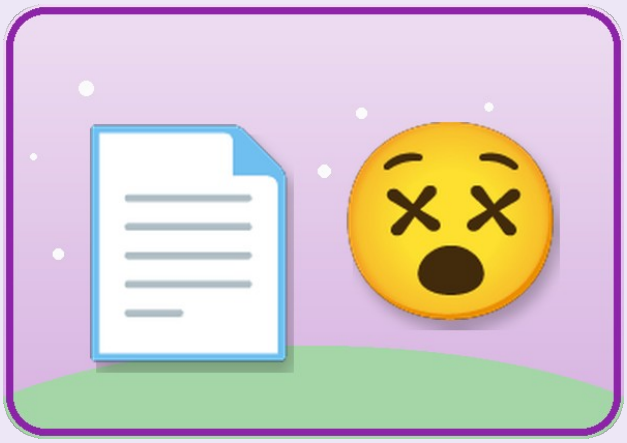
SCENARIO: Prompt with Purpose

Read the scenario. Think about the choices made at each step.

1

Karthik asks AI: 'Explain thermodynamics.'

2



He gets 2 pages of dense, generic text.

3

New prompt: goal, level, 150 words, 1 analogy, 3 MCQs.

4

Perfect revision notes, and he checks facts in his textbook.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Purposeful prompts on campus

Concept explainers

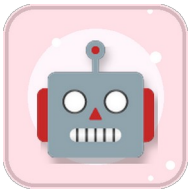
Practice question banks

Data cleaning help

Code explanations

Formal emails

Literature summaries



BOLT SAYS:

Vague prompt, vague answer. Precise prompt with constraints? Precise answer!

FUN FACT!

Asking an LLM to answer in JSON makes its output easy for programs to use!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Prompting a University Help-Desk

CASE FILE

A university trialled an AI help-desk for admission queries. Initial prompts were vague, and the bot gave wrong fee figures in 18% of test cases. After adding the official fee table as context, a strict 'answer only from context' rule, and JSON output with a 'source' field, errors fell to 3%.

🔍 Analyse the case

1. Which prompt changes most likely reduced errors? Why?

2. Why include a 'source' field in the output?

3. Is 3% error acceptable for fee information? Justify.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Prompt Lab Experiment

You need:

AI tool

Results table

Pairs

Steps:

1.
   Task: explain 'overfitting' to first-year students.
2.
   Run 4 prompts: plain, + audience, + few-shot example, + 'check your answer'.
3.
   Score each output 1–5 on accuracy, clarity, length.
4.
   Plot scores. Which addition helped most?

THINK & TALK

Is a single run enough to conclude which prompt is best?

Record your observations & results:

💻 CODE LAB: Parsing Structured LLM Output

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import json
2  reply = '{"term": "overfitting", "level": "UG1", "mcqs": 3}'
3  data = json.loads(reply)
4  print(data['term'].upper())
5  print(data['mcqs'] * ____)
6  bad = "Sure! {'term': 'AI'}"
7  json.loads(bad)
```

COMPLETE **T1. Fill the blank so it prints 6.**

PREDICT **T2. What does line 4 print?**

DEBUG **T3. Why does line 7 fail? Give two reasons.**

EXTEND **T4. Wrap line 7 in try/except and print 'Invalid JSON'.**

WORKSHEET: Prompt with Purpose (Part A)

Name: _____ Date: _____

1. Match the pattern to its prompt.

Few-shot ●

Chain-of-thought ●

Persona ●

Self-critique ●

● **Now review your answer for errors**

● **Act as an examiner**

● **Here are 2 solved examples...**

● **Solve step by step**

2. Vague or purposeful prompt? Tick ✓.

	Vague	Purposeful
Tell me about AI		
Explain CNNs in 150 words for B.Sc students with 1 example		
Write something on climate		
List 5 Python data types as a JSON array		

WORKSHEET: Prompt with Purpose (Part B)

Name: _____ Date: _____

1. Systematic prompt testing. Write 1, 2, 3, 4.

Compare scores across versions ☐

Fix a set of test inputs ☐

Score outputs with a rubric ☐

Run each prompt version several times ☐

2. Will this output parse as JSON? Tick ✓.

	Valid	Invalid
{"term": "AI", "year": 1956}		
{term: AI, year: 1956}		
["NumPy", "pandas"]		
Sure! Here is the JSON: {...}		

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Prompt with Purpose

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Structured output

Meaning: _____

Example: _____

Chain-of-thought

Meaning: _____

Example: _____

Constraint

Meaning: _____

Example: _____

? QUIZ TIME: Prompt with Purpose

Circle the right answer!

Q1. Giving solved examples is...

a) Few-shot

b) Zero-shot

c) No-shot

Q2. 'Answer in under 100 words' is a...

a) Persona

b) Constraint

c) Rhyme

Q3. Machine-readable output format:

a) Painting

b) Speech

c) JSON

Q4. Step-wise reasoning prompt:

a) Chain-of-thought

b) Link chain

c) Iron chain

Q5. Comparing prompts fairly needs...

a) One run

b) Several runs

c) No checks

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Prompt with Purpose

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
🎯 Specifying audience improves relevance.		
🧾 LLM JSON output never needs validation.		
🪜 Chain-of-thought helps multi-step problems.		
🔁 One run is enough to compare prompts.		
🚫 Constraints can control length and tone.		
❓ Asking for clarifying questions can reduce wrong assumptions.		

HOME CONNECT

Write two prompts for a family recipe: one vague, one purposeful. Compare outputs together.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Prompt with Purpose

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does few-shot prompting improve format consistency?**

COMPARE **Q2. Compare chain-of-thought and direct answering for a physics numerical.**

APPLY **Q3. Write a prompt that returns 3 MCQs on Ohm's law as JSON.**

EVALUATE **Q4. Should students use AI to generate their own practice questions?**

CREATE **Q5. Design a 5-point rubric to evaluate AI explanations.**

📝 EXAM PREP: Prompt with Purpose

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. List four elements of a purposeful prompt.

2. What is few-shot prompting?

3. Why request JSON output?

4. What is chain-of-thought prompting?

PART B • Long answer (5 marks)

5. Design and justify an experiment to compare three prompt versions for generating revision notes.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. [x*2 for x in [1,2,3]] gives...

a) [2, 4, 6] | b) [1, 2, 3] | c) 6

Q2. df.dropna() removes...

a) Numbers | b) Missing values | c) Columns

Q3. Every sklearn model has...

a) paint() | b) play() | c) fit()

Q4. Prompt with solved examples:

a) Few-shot | b) Zero-shot | c) Frozen

Q5. Before using LLM JSON, you should...

a) Trust it | b) Validate it | c) Delete it

Q6. len('Python') is...

a) 5 | b) 7 | c) 6

🧩 WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

🔢 ARRAY	🐼 DATAFRAME	🎯 FIT()	🚫 CONSTRAINT	⭐ PYTHON	⭐ NUMPY

U	C	J	S	O	N	D	H	D	S
E	O	I	P	H	G	A	B	O	B
N	N	B	R	M	M	T	F	H	A
U	S	A	O	C	D	A	F	C	R
M	T	P	M	A	P	F	I	B	R
P	R	A	P	A	Y	R	T	M	A
Y	A	N	T	I	T	A	(	E	Y
F	I	D	F	S	H	M	)	A	N
U	N	A	B	H	O	E	E	B	A
M	T	S	W	D	N	K	L	R	A

🏅 **I finished AI CONNECT! Teacher's sign: _____**

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI in the Real World

Topic 6

AI in Your Profession

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI in the Real World

LET'S LEARN

- AI creates value by predicting, detecting, generating.
- Real deployments face data, cost & trust limits.
- Judge AI by outcomes, not demos.

What AI does

Predict

Detect

Generate

Optimise

KEY TERMS

Prediction
Estimate unknowns

Data drift Data changes over time

Pilot Small real trial

📖 CONCEPT EXPLAINED: AI in the Real World

💡 In this chapter

AI demos are impressive, but the real test is whether AI improves outcomes in messy, everyday settings: hospitals, farms, cities and offices. In this chapter we look at what AI actually does in the world, how value is created from data to impact, and why many promising systems struggle after deployment.

🔮 1. Four things AI does

Most real-world AI applications perform one of four jobs. Predict: estimate something unknown, such as next week's electricity demand or a patient's risk of readmission. Detect: spot patterns or anomalies, such as cracks in bridge photos or fraudulent card payments. Generate: create content, such as summaries, translations or images. Optimise: choose the best plan, such as delivery routes or exam timetables.

Naming the job clarifies what data you need and how to measure success. Prediction needs historical outcomes; detection needs examples of the rare event; generation needs quality checks; optimisation needs clear objectives and constraints.

🖐 TRY IT

Classify: (a) a weather app's rain forecast, (b) an app suggesting the fastest bus route, (c) a tool spotting fake currency notes.

🔗 2. The AI value chain

Value does not come from a model alone. It flows through a chain: collect relevant data → build and validate a model → integrate it into a real workflow → people take action → outcomes improve and are measured. A crop-disease detector creates value only if farmers receive the alert in time, understand it, can afford the treatment and act on it.

Weak links anywhere break the chain. Many projects fail not because the model is inaccurate but because it is not connected to decisions, the users do not trust it, or no one owns the results. That is why successful AI teams include domain experts, designers and operations staff, not just data scientists.

✏️ **WORKED EXAMPLE**

Cyclone early warning: satellite & weather data → AI-assisted forecast → alerts in local languages via SMS and loudspeakers → evacuation by local officials → lives saved. Each step is essential.

📉 3. Why deployments struggle

Data drift happens when the world changes after training: shopping habits shift during a festival, or sensors are replaced with new models. A model trained on last year's data slowly becomes wrong. Teams must monitor performance and retrain regularly.

Other challenges include poor data quality, high infrastructure costs, integration with old software, low connectivity in rural areas, multiple languages and literacy levels, and legal or ethical constraints. A pilot, meaning a small real-world trial, helps uncover these problems before a full rollout.

🤔 **PAUSE & THINK**

An attendance-prediction model worked well in 2024 but poorly in 2026 after timetables changed. What is this called, and what should the team do?

📊 4. Measuring real impact

Model metrics like accuracy matter, but organisations care about outcomes: lives saved, money saved, hours freed, errors avoided, satisfaction improved. A fraud model with 99% accuracy is useless if it blocks so many genuine customers that they leave the bank.

Good evaluation compares the AI-assisted process with a baseline, ideally through a controlled trial, and checks who benefits and who might be left out. Public-interest AI in particular must work fairly across regions, languages and communities.

🤔 **PAUSE & THINK**

Suggest one outcome metric (not accuracy) for an AI tool that helps doctors read X-rays.

✅ Key takeaways

- AI's four jobs: predict, detect, generate, optimise.
- Value flows through data → model → integration → action → outcome.
- Data drift, costs, trust and integration often block real impact.
- Pilots reveal real-world problems early.
- Measure outcomes against a baseline, not just accuracy.

❓ Quick check: answer in one or two lines.

1. Name AI's four jobs with one example each.

2. What is data drift?

3. Why can an accurate model still fail to create value?

🔬 CONCEPT DEEP DIVE: AI in the Real World

🔷 AI creates value through four verbs: predict (demand), detect (fraud), generate (content), optimise (routes, energy).

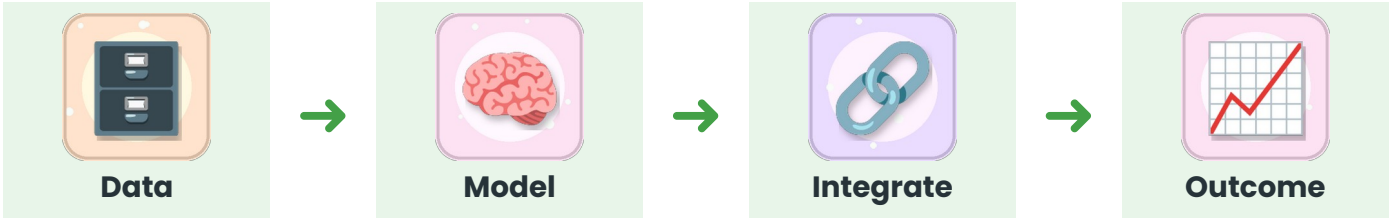
🔷 The AI value chain: data collection → model → integration into workflows → human action → measurable outcome.

🔷 Deployment challenges: data quality, drift, infrastructure costs, user trust, and integration with legacy systems.

🔷 Evaluate real deployments with outcome metrics (lives saved, costs cut), not just model accuracy.

🔷 Public-interest AI (health, agriculture, climate) must work across languages, connectivity levels and literacy.

🔄 How it works



💻 Detecting data drift (idea)

```
old_mean = train['rainfall'].mean()
new_mean = live['rainfall'].mean()
if abs(new_mean - old_mean) / old_mean > 0.2:
    print('Drift alert: retrain the model!')
```

✍️ In your own words: explain this topic to a Class 10 student.

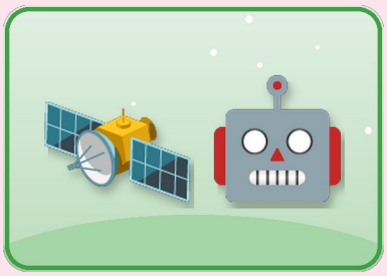
AI STORY: Three Days Before the Storm

Characters: *Lakshmi (a fisherwoman in Odisha), her son Arjun (a college student), a district officer.*

SCENE 1

Lakshmi had fished the Bay of Bengal for twenty years. The sky looked calm, but her son Arjun's phone buzzed with an alert in Odia: 'Severe cyclone likely to reach the coast in 72 hours. Do not go to sea.' Lakshmi hesitated; a day without fishing meant a day without income.

Pause & predict: Why might Lakshmi doubt the alert?



SCENE 2

Arjun, who studied AI in college, explained. 'Satellites and weather stations send huge amounts of data. Forecasters now use AI models alongside physics models, so the warnings come earlier.' He showed her the cyclone's predicted path on a map, curving straight toward their village.

Pause & predict: Why do forecasters use AI together with physics models instead of AI alone?

SCENE 3

The district officer arrived with loudspeakers and buses. Shelters in schools were ready. 'The forecast is only useful if people act on it,' he said, helping elders board. Lakshmi secured her boat and joined her neighbours at the shelter.

Pause & predict: Which links of the AI value chain does the district officer represent?

SCENE 4

The cyclone struck with fierce winds two nights later. Homes were damaged, but everyone in the village was safe. Lakshmi looked at Arjun's phone and said quietly, 'That message saved us. But so did the people who carried us here.'

⏸ **Pause & predict:** What does Lakshmi's last line teach about technology and people?

🌟 The big idea

AI creates real value only when predictions reach people in time and people are ready to act.

💬 Discuss with your classmates

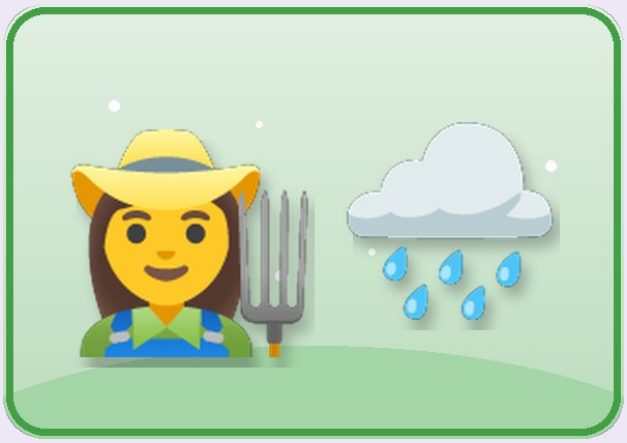
1. Identify each link of the value chain in this story.

2. What could break the chain in a village with poor mobile coverage?

SCENARIO: AI in the Real World

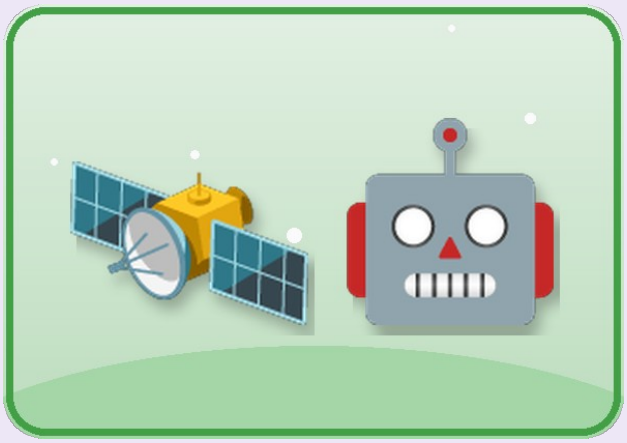
Read the scenario. Think about the choices made at each step.

1



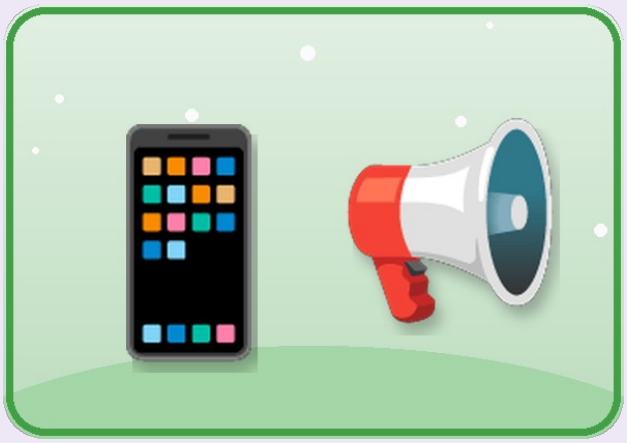
Odisha villages face a fast-approaching cyclone.

2



AI-assisted forecasts predict landfall days ahead.

3



Alerts reach phones in local languages.

4

Timely evacuation saves lives. Data + people + plans!

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: AI across society

Health screening

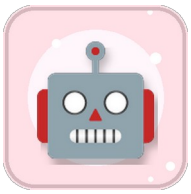
Crop advisories

Smart traffic

Power-grid balancing

Digital public services

Cyclone forecasting



BOLT SAYS:

A demo works on 10 examples. The real world throws a million messy ones at me!

FUN FACT!

AI weather models can now produce global forecasts in minutes instead of hours!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: AI Weather Forecasting Leap

CASE FILE

AI weather models such as Google DeepMind's GraphCast, trained on decades of historical data, produced 10-day global forecasts in under a minute and matched or beat traditional systems on many measures. Meteorological agencies began testing them alongside physics-based models rather than replacing those models outright.

🔍 Analyse the case

1. Why might agencies run AI and physics models side by side?

2. What risks arise if extreme events are rare in training data?

3. How could faster forecasts help farmers in India?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: AI Opportunity Audit

You need:

Campus map

Audit sheet

Teams of 4

Steps:

1.
   Walk your campus; list 6 repetitive or data-rich problems.
2.
   Tag each: predict, detect, generate or optimise.
3.
   Score value (1–5), data availability (1–5), risk (1–5).
4.
   Pick the best use case: (value × data) - risk. Present it.

THINK & TALK

Which high-value idea scored low because of data or risk?

Record your observations & results:

💻 CODE LAB: Detecting Data Drift

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import statistics as st
2  train = [100, 110, 95, 105, 90]
3  live  = [130, 140, 120, 135, 125]
4  m1, m2 = st.mean(train), st.mean(live)
5  change = abs(m2 - m1) / ____
6  print(m1, m2, round(change, 2))
7  if change > 0.2: print('Retrain!')
```

COMPLETE **T1. Fill the blank for relative change.**

PREDICT **T2. Write m1, m2 and change.**

PREDICT **T3. Will 'Retrain!' print?**

EXTEND **T4. Also compare standard deviations using st.stdev.**

WORKSHEET: AI in the Real World (Part A)

Name: _____ Date: _____

1. Match the AI capability to the use case.

Predict ●

Detect ●

Generate ●

Optimise ●

● Bus routes with least fuel

● Next week's electricity demand

● Draft a public notice

● Cracks in bridge photos

2. Score = (Value × Data) − Risk. Use the word box.

WORD BOX: 21 ✿ 11 ✿ 8

Traffic: (5 × 5) − 4 = ____

Canteen: (3 × 4) − 1 = ____

Health: (5 × 2) − 2 = ____

WORKSHEET: AI in the Real World (Part B)

Name: _____ Date: _____

1. Drift check: |new - old| ÷ old. Use the word box.

WORD BOX: 30% ✿ Yes ✿ No

Old 100 mm, new 130 mm → change = ____

Above 20% threshold? ____

Old 100, new 110 → above 20%? ____

2. AI value chain. Write 1, 2, 3, 4.

Measurable outcome ☐

Collect data ☐

Integrate into workflow ☐

Build model ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: AI in the Real World

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Prediction

Meaning: _____

Example: _____

Data drift

Meaning: _____

Example: _____

Pilot

Meaning: _____

Example: _____

? QUIZ TIME: AI in the Real World

Circle the right answer!

Q1. Spotting cracks in photos is...

a) Detection

b) Generation

c) Composition

Q2. Real data slowly changing is...

a) Data party

b) Data drift

c) Data freeze

Q3. AI value should be judged by...

a) Demos

b) Hype

c) Real outcomes

Q4. A small real-world trial:

a) Pilot

b) Captain

c) Flight

Q5. Least-fuel bus routes is...

a) Generation

b) Optimisation

c) Detection

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: AI in the Real World

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Data drift can reduce model performance.		
A good demo guarantees real-world success.		
Route planning is an optimisation task.		
AI weather models can forecast very quickly.		
Public AI only needs to work in English.		
Outcomes matter more than accuracy alone.		

HOME CONNECT

Find one AI service used by a local government body. Who benefits and who might be left out?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI in the Real World

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can a 95%-accurate model still fail in deployment?**

COMPARE **Q2. Compare AI for prediction and for optimisation with examples.**

APPLY **Q3. Old mean rainfall 100 mm, new 130 mm. Is drift > 20%? Show working.**

EVALUATE **Q4. Should AI forecasts alone trigger evacuations?**

CREATE **Q5. Map the AI value chain for a crop-disease alert service.**

📝 EXAM PREP: AI in the Real World

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name the four value verbs of AI.

2. What is data drift?

3. Why is a pilot useful?

4. Give one public-interest AI use in India.

PART B • Long answer (5 marks)

5. Using a real example, explain the AI value chain and three challenges that can block real-world impact.

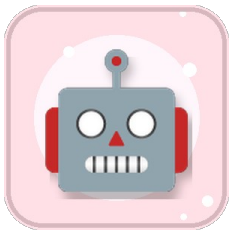
TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI in Your Profession

LET'S LEARN

- Every profession is being reshaped by AI.
- AI automates tasks, not whole jobs (mostly).
- Domain expertise + AI skills = high value.

Task impact types



Automate

Augment

Create new tasks

Unchanged

KEY TERMS

Augmentation
AI assists humans

Automation AI does the task

Task analysis
Break job into tasks

📖 CONCEPT EXPLAINED: AI in Your Profession

💡 In this chapter

Whatever your discipline (engineering, commerce, medicine, law, arts or agriculture), AI is already changing the work you will do. This chapter shows how to analyse a profession task by task, distinguish automation from augmentation, and identify the 'AI + X' skills that will keep you valuable.

🧩 1. Jobs are bundles of tasks

Headlines often ask whether AI will 'replace' a job. A more useful question is which tasks within a job AI can do or assist. A doctor's job includes listening to patients, examining them, reading scans, writing notes, deciding treatments and comforting families. AI can draft notes and flag patterns in scans, but examining, deciding and comforting remain deeply human.

Task analysis means listing the tasks of a role, estimating how much time each takes, and judging how exposed each is to AI. This produces a far more accurate picture than guessing about whole job titles, and it shows exactly which skills will grow in importance.

🖐 TRY IT

List six tasks of a teacher. Mark each as automate, augment or unchanged.

🤝 2. Automation, augmentation and new tasks

Automation means AI performs a task with little human involvement, like extracting invoice data into a spreadsheet. Augmentation means AI makes a human faster or better, like a lawyer using AI to search thousands of past judgements before building an argument. In practice, most professional work moves towards augmentation, with humans responsible for final judgement.

AI also creates new tasks: writing prompts, checking AI outputs, curating training data, explaining AI decisions to customers and monitoring systems for errors. People who can do these well alongside their domain work become especially valuable.

WORKED EXAMPLE

Accounting firm: data entry (automated) → anomaly review (augmented: AI flags, accountant investigates) → client advisory (unchanged, but enriched with AI-generated insights).

3. AI across disciplines

Engineering uses AI for generative design, simulation and predictive maintenance. Medicine uses it for imaging, triage and clinical note-writing. Law uses it for contract review and legal research. Commerce uses it for forecasting, fraud detection and customer analytics. Agriculture uses satellite images and sensors for crop health and yield prediction. Arts and media use it for concept generation, editing and translation.

In each field, the professionals who benefit most understand both the domain deeply and AI well enough to judge its outputs. They know where AI is reliable, where it fails and when to say no.

PAUSE & THINK

Pick your own discipline. Name one task where AI could help and one task where you would not trust AI. Explain why.

4. Becoming an 'AI + X' professional

'X' is your field; 'AI' is the literacy to use and question AI tools. Core AI + X skills include data literacy (reading charts, spotting bad data), prompting and tool use, critical evaluation of outputs, ethics and privacy awareness, and communication of results to non-experts.

Start now: use AI tools in your coursework where allowed, keep a log of where they helped or failed, learn basic Python and data analysis, and discuss AI's impact with professionals in your field. Your degree remains valuable; AI makes your judgement and expertise more important, not less.

TRY IT

Write a 3-line 'AI + X' skill plan for this semester for your own discipline.

✅ Key takeaways

- Analyse tasks, not job titles, to understand AI's impact.
- Automation replaces tasks; augmentation strengthens humans.
- AI creates new tasks: prompting, checking, curating, explaining.
- Every discipline is adopting AI in its own ways.
- AI + X: domain expertise plus AI literacy and judgement.

❓ Quick check: answer in one or two lines.

1. Differentiate automation and augmentation with an example each.

2. Why is task analysis better than asking 'will AI replace this job?'

3. List three 'AI + X' skills.

🔬 CONCEPT DEEP DIVE: AI in Your Profession

🔷 Jobs are bundles of tasks; AI exposure varies by task, so analyse tasks rather than whole job titles.

🔷 Automation replaces a task; augmentation boosts human performance; AI also creates new tasks (prompting, oversight).

🔷 Discipline examples: engineering (generative design), medicine (imaging), law (e-discovery), commerce (forecasting).

🔷 Domain experts are needed to define problems, judge outputs and take responsibility for decisions.

🔷 Graduates need 'AI + X' skills: their discipline (X) plus AI literacy, data skills and critical evaluation.

🔄 How it works

💻 Task exposure score in Python

```
tasks = {'notes': 0.8, 'diagnosis': 0.3, 'empathy': 0.05}
hours = {'notes': 15, 'diagnosis': 20, 'empathy': 5}
saved = sum(tasks[t] * hours[t] for t in tasks)
print(saved)   # 18.25 hours/week potentially assisted
```

✍️ In your own words: explain this topic to a Class 10 student.

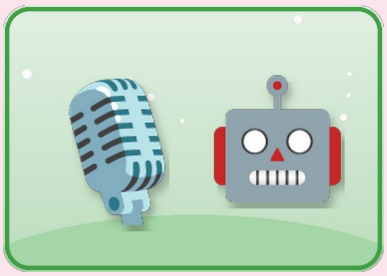
AI STORY: Doctor Imran's Extra Hour

Characters: *Dr. Imran (a district hospital physician), Nurse Kavitha, an AI scribe tool.*

SCENE 1

Dr. Imran saw sixty patients a day and spent nearly three hours every evening typing notes. 'I became a doctor to care for people, not to type,' he told Nurse Kavitha. His patients often waited, and he went home exhausted.

Pause & predict: Which of Dr. Imran's tasks seems most suitable for AI support?



SCENE 2

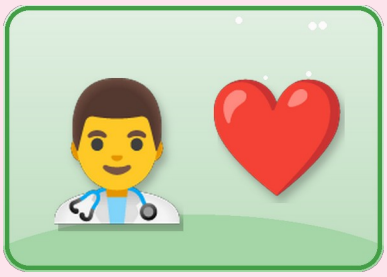
The hospital piloted an AI scribe. With patient consent, it listened to consultations and drafted structured notes: symptoms, history, examination, plan. On the first day, Dr. Imran noticed it wrote 'no allergies' for a patient who had mentioned a penicillin allergy.

Pause & predict: Why is this error dangerous? What should happen next?

SCENE 3

He made a rule: he would read and correct every draft before signing. Kavitha added a checklist for allergies and medicines. Errors became rare, and when they occurred, they were caught. The AI drafted; the humans decided and took responsibility.

Pause & predict: Is this an example of automation or augmentation? Explain.



SCENE 4

Within a month, Dr. Imran saved an hour every day. He used it to explain diagnoses more patiently and to train junior doctors. 'AI didn't replace me,' he said. 'It gave me back the parts of my job that need a human.'

⏸ **Pause & predict:** What new task did AI create for Dr. Imran?

🌟 The big idea

In most professions, AI handles parts of tasks, while humans keep the judgement, empathy and responsibility.

💬 Discuss with your classmates

1. Which task in your future profession would you happily hand to AI?

2. Which task would you never hand over? Why?

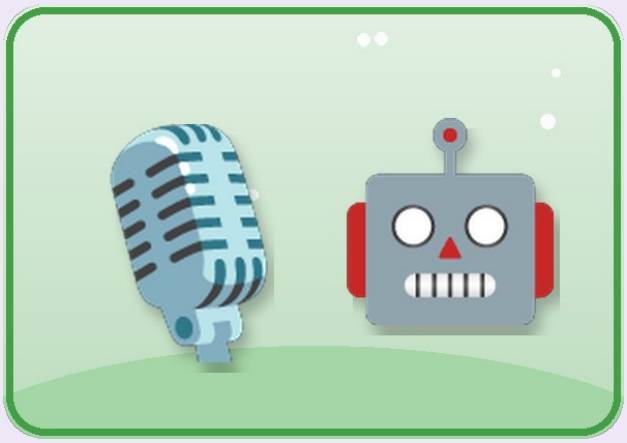
🎞️ SCENARIO: AI in Your Profession

Read the scenario. Think about the choices made at each step.

1

Dr. Imran spends 3 hours daily typing notes.

2

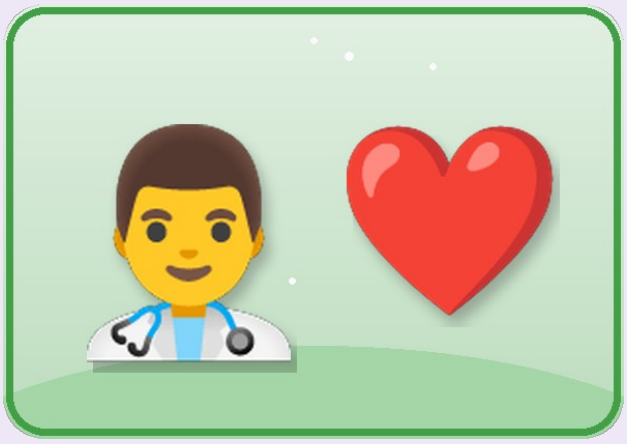


An AI scribe drafts notes from the consultation.

3

He reviews and corrects each note before signing.

4



He gains an hour a day for patients.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI by discipline

Engineering: design & simulation

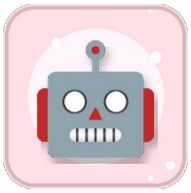
Medicine: imaging & notes

Law: contract review

Commerce: forecasting

Arts: concept generation

Agriculture: yield & pests



BOLT SAYS:

I won't replace a good doctor. But a doctor using AI may replace one who doesn't!

FUN FACT!

AI scribes can draft clinical notes, giving doctors more time with patients!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: AI Contract Review at a Law Firm

CASE FILE

A law firm used AI to review 5,000 supplier contracts for risky clauses. The AI flagged 600 contracts in two days, a job that previously took junior lawyers six weeks. Senior lawyers reviewed every flagged clause. Junior lawyers moved to client negotiation and strategy tasks.

🔍 Analyse the case

1. Which tasks were automated and which augmented?

2. Why must senior lawyers still review flags?

3. How should law students prepare for this change?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: My Profession Task Map

You need:

Task grid

Coloured pens

Dept. groups

Steps:

1

List 10 tasks of a professional in your field.

2

Colour-code: automate, augment, same.

3

Estimate weekly hours AI could save.

4

Present the new skills graduates need.

THINK & TALK

Which tasks will stay human for a long time? Why?

Record your observations & results:

💻 CODE LAB: Task Exposure Calculator

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
tasks = {
  'reports':   (0.8, 10),   # (exposure, hours/week)
  'meetings':  (0.2, 8),
  'site_visit':(0.05, 12)}
assisted = sum(e * h for e, h in tasks.____())
total = sum(h for _, h in tasks.values())
print(round(assisted, 1), round(assisted / total * 100))
```

COMPLETE **T1. Fill the blank (dictionary method).**

PREDICT **T2. What two numbers are printed?**

MODIFY **T3. Add 'emails': (0.6, 5). New assisted hours?**

EXTEND **T4. Print the task with the highest exposure.**

WORKSHEET: AI in Your Profession (Part A)

Name: _____ Date: _____

1. Automate or augment? Tick ✓.

	Automate	Augment
Copying invoice data into forms		
Doctor's final diagnosis with AI scan		
Resetting a password		
Lawyer drafting strategy with AI research		

2. Match the profession to its AI tool.

Civil engineer	●	●	Contract clause review
Accountant	●	●	Structural simulation
Lawyer	●	●	Satellite crop monitoring
Agronomist	●	●	Anomaly detection in ledgers

WORKSHEET: AI in Your Profession (Part B)

Name: _____ Date: _____

1. Assisted hours = Σ exposure × hours. Use the word box.

WORD BOX: 12 ✿ 6 ✿ 2.5

0.6 × 10 h = ____

6 + (0.2 × 30) = ____

0.05 × 50 h = ____

2. Circle tasks likely to stay mainly human.

Comforting a patient

Formatting a report

Negotiating a deal

Summing a spreadsheet

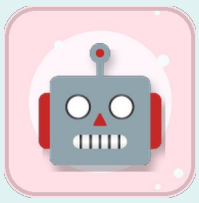
Final legal judgement

Mentoring a junior

PUZZLE & DEFINE: AI in Your Profession

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Augmentation

Meaning: _____

Example: _____

Automation

Meaning: _____

Example: _____

Task analysis

Meaning: _____

Example: _____

? QUIZ TIME: AI in Your Profession

Circle the right answer!

Q1. AI mostly changes jobs by...

a) Changing tasks

b) Ending all jobs

c) Nothing

Q2. AI assisting a human is...

a) Automation

b) Augmentation

c) Stopping

Q3. Most valuable combo:

a) Luck + hype

b) Rest + AI

c) Domain + AI skills

Q4. Contract review AI helps...

a) Lawyers

b) Painters

c) Chefs

Q5. Breaking a job into tasks is...

a) Task art

b) Task analysis

c) Task music

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI in Your Profession

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Jobs are bundles of tasks.		
AI will automate every task in every job.		
Augmentation means AI helps humans.		
'AI + X' combines your field with AI skills.		
AI legal reviews need no human checking.		
AI creates new tasks too.		

HOME CONNECT

Ask a professional relative which part of their work AI already helps with. Note their worries.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI in Your Profession

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why analyse tasks instead of job titles for AI impact?**

COMPARE **Q2. Compare AI's impact on a radiologist and a nurse.**

APPLY **Q3. Exposure 0.6 on 10 h/week task, 0.2 on 30 h. Assisted hours?**

EVALUATE **Q4. 'AI will make degrees useless.' Do you agree?**

CREATE **Q5. Design an 'AI + X' elective for your department.**

📝 EXAM PREP: AI in Your Profession

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Differentiate automation and augmentation.

2. Why analyse tasks, not jobs?

3. What are 'AI + X' skills?

4. Give one AI tool used in your discipline.

PART B • Long answer (5 marks)

5. Analyse how AI will change a profession of your choice over the next 10 years. Classify tasks and recommend skills.

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Least-fuel delivery routes:

a) Optimise | b) Generate | c) Detect

Q2. Mean rainfall 100 → 130 is...

a) 3% change | b) 30% change | c) 130% change

Q3. AI assisting a doctor's notes:

a) Unchanged | b) Full automation | c) Augmentation

Q4. 0.6 × 10 + 0.2 × 30 hours =

a) 12 | b) 40 | c) 8

Q5. AI impact is best analysed by...

a) Job titles | b) Tasks | c) Luck

Q6. GraphCast is an AI for...

a) Music | b) Games | c) Weather

WORD SEARCH: AI SOLVE

🔍 **Find these words. They go across → or down ↓.**

PREDICTION

DATA DRIFT

PILOT

AUTOMATION

DRIFT

AUGMENT

I	K	P	I	L	O	T	D	C	R
R	C	M	C	O	E	A	K	O	T
O	A	U	G	M	E	N	T	D	A
B	W	W	B	N	U	L	T	I	S
S	H	B	K	A	C	D	W	T	K
B	G	O	K	W	V	A	L	U	E
I	E	B	L	L	D	R	I	F	T
M	D	A	T	A	D	R	I	F	T
P	R	E	D	I	C	T	I	O	N
A	U	T	O	M	A	T	I	O	N

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

Navigate the AI Job World

Topic 8

Build Your AI Skillset

TOPIC 7 • MONTH 4 • AI & CAREER

Navigate the AI Job World

LET'S LEARN

- AI jobs span data, engineering, product & policy.
- Read job descriptions as skill checklists.
- Internships and projects bridge degree and job.

AI job families

Data analyst

ML engineer

AI product manager

AI governance

KEY TERMS

JD Job description

Skill gap Missing skills

Freelancing Project-based work

📖 CONCEPT EXPLAINED: Navigate the AI Job World

💡 In this chapter

The AI job market can look like a maze: new job titles, long lists of required skills, and endless online postings. This chapter gives you a map. You will learn the main AI job families, how to read a job description like a skills checklist, where real opportunities come from, and how to avoid fake offers.

🧭 1. The main AI job families

AI careers cluster into a few families. Data roles (data analyst, data scientist) turn data into insights and predictions. Engineering roles (ML engineer, MLOps engineer) build, deploy and maintain models in real systems. Product and design roles (AI product manager, UX designer) decide what to build and how people will use it. Research roles invent new methods. Governance roles (AI policy, risk and audit) ensure AI is safe, fair and legal.

There are also 'AI-enabled' roles in every industry: a marketing analyst using AI tools, a lawyer specialising in technology, an agronomist using satellite data. You do not need to be a computer science graduate to work with AI; you need the right mix of domain knowledge and AI skills.

🤔 PAUSE & THINK

Which job family fits your current interests best? Name one skill you already have that matches it.

📋 2. Reading job descriptions like a pro

A job description (JD) is a skills checklist written by the employer. It usually lists responsibilities, must-have skills, nice-to-have skills and qualifications. Read several JDs for the same role and count how often each skill appears. Skills that appear in most postings, such as Python and SQL for data roles, are the core you should build first.

Do not be discouraged if you do not match every line. Many employers consider candidates who meet most must-have skills and show strong projects and willingness to learn. Use the gaps you find as your learning plan.

✏️ **WORKED EXAMPLE**

Skill tally from 4 data-analyst JDs: Python 4, SQL 4, Excel 3, Power BI 2, statistics 2, Spark 1. Priority order: Python & SQL → Excel → visualisation & statistics → Spark later.

🚪 3. Where opportunities come from

Openings appear on job portals, company career pages and professional networks, but many first opportunities come through other routes: internships, campus placement drives, hackathons, open-source contributions, research projects with faculty, and referrals from people who have seen your work.

Freelancing and short projects can also build experience, but choose reputable platforms and clear contracts. Every route rewards the same things: visible evidence of skills, clear communication and reliability.

🖐️ **TRY IT**

List three companies or organisations in your region that use AI. Find one internship or project route for each.

🚩 4. Staying safe: fake offers and scams

Unfortunately, students are common targets for fake job and internship offers. Red flags include requests to pay a fee for 'training', 'registration' or a 'kit'; email addresses from free or look-alike domains; offers made without any interview; urgent deadlines ('accept within an hour'); and spelling or formatting errors.

Protect yourself by verifying the company's official website and careers page, checking with your placement cell, never sharing OTPs or bank details, and remembering a simple rule: genuine employers pay you; they do not ask you to pay them.

🤔 **PAUSE & THINK**

An email offers you an 'AI internship' with a ₹4,999 registration fee from hr.techcorp@gmail.com. List every red flag you can find.

✅ Key takeaways

- AI job families: data, engineering, product/design, research, governance.
- AI-enabled roles exist in every industry.
- Read multiple JDs and tally skills to set learning priorities.
- Internships, hackathons, open source and referrals open doors.
- Never pay for a job; verify every offer.

❓ Quick check: answer in one or two lines.

1. Name four AI job families.

2. How can tallying skills across JDs help you plan?

3. List three red flags of a fake offer.

🔬 CONCEPT DEEP DIVE: Navigate the AI Job World

🔷 AI job families: data (analyst, scientist), engineering (ML, MLOps), product & design, research, governance & policy.

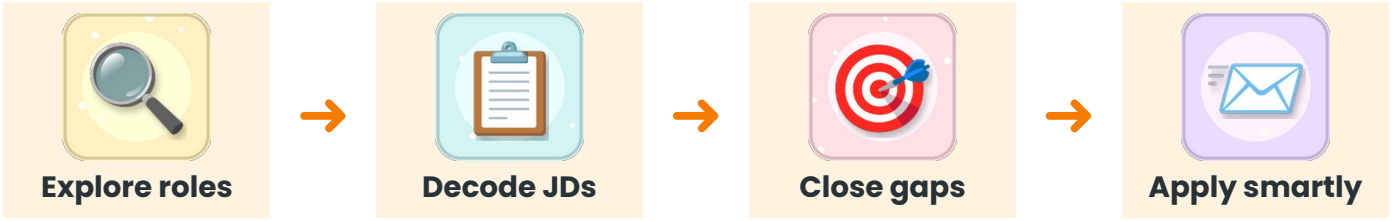
🔷 Job descriptions list must-have and preferred skills; aim to meet most must-haves before applying.

🔷 Entry routes: internships, apprenticeships, campus placements, hackathons, open-source contributions, freelancing.

🔷 Hiring processes often include online tests, technical interviews, case studies and portfolio reviews.

🔷 Beware fake job offers: never pay to get a job; verify company domains and offer letters.

🔄 How it works



💻 Tallying JD skills in Python

```
from collections import Counter
jds = [['python','sql'], ['sql','excel'],
       ['python','sql','stats']]
c = Counter(s for jd in jds for s in jd)
print(c.most_common(2))   # [('sql', 3), ('python', 2)]
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Offer That Was Too Good

Characters: *Neha (B.Sc Computer Science student), her placement officer Mr. Joseph, an unknown 'recruiter'.*

SCENE 1

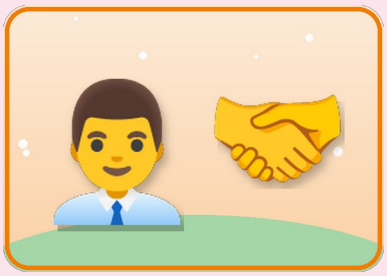
Neha had applied to forty internships and heard nothing. Then an email arrived: 'Congratulations! You are selected for an AI Data Internship at TechNova Global. Stipend ₹25,000. Pay ₹4,999 registration within 24 hours to confirm.' Her heart raced; finally, success!

Pause & predict: What should Neha check before replying?

SCENE 2

Something felt odd. There had been no interview. The email came from technova.hr.global@gmail.com. She searched the official TechNova website: its careers page listed no such internship and warned, 'We never charge candidates any fee.'

Pause & predict: List the red flags Neha noticed.



SCENE 3

Neha showed the email to Mr. Joseph at the placement cell. He sighed. 'Three students received the same message this week.' He issued a campus alert and then sat with Neha to review her applications. 'Let's look at what real job descriptions are asking for.'

Pause & predict: How did reporting the scam help other students?

SCENE 4

Together they tallied skills from ten genuine data-analyst postings: Python and SQL in all ten, Excel in eight, dashboards in six. Neha built a sales dashboard project, joined a hackathon, and two months later earned a real, paid internship through a referral from a hackathon mentor.

⏸ **Pause & predict:** Which strategies led to Neha's genuine internship?

🌟 The big idea

Genuine employers pay you; they never ask you to pay them. Real opportunities come from real skills and real connections.

💬 Discuss with your classmates

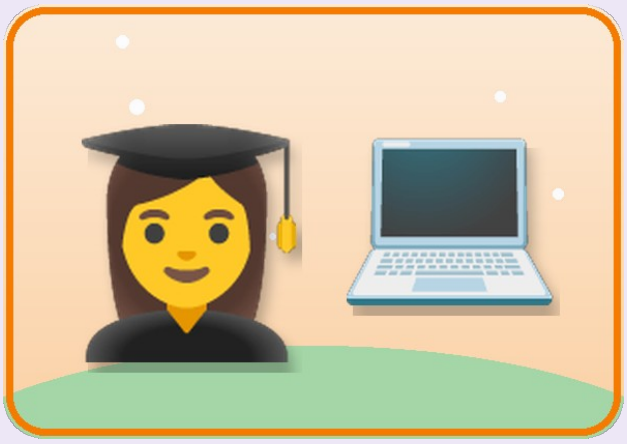
1. Have you or someone you know received a suspicious offer? What happened?

2. Which job family from this chapter interests you most? Why?

SCENARIO: Navigate the AI Job World

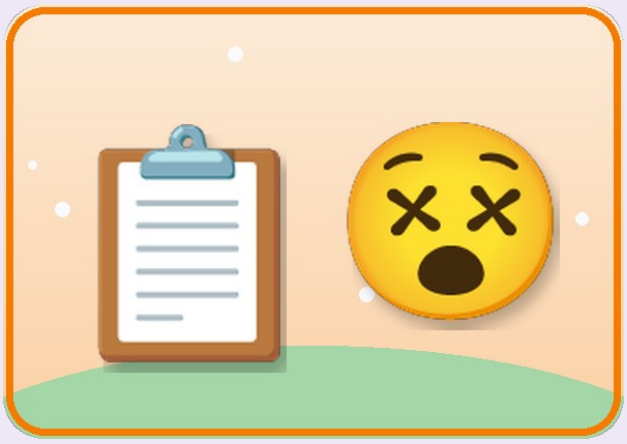
Read the scenario. Think about the choices made at each step.

1



Neha searches 'AI jobs for freshers'. 2,000 results!

2

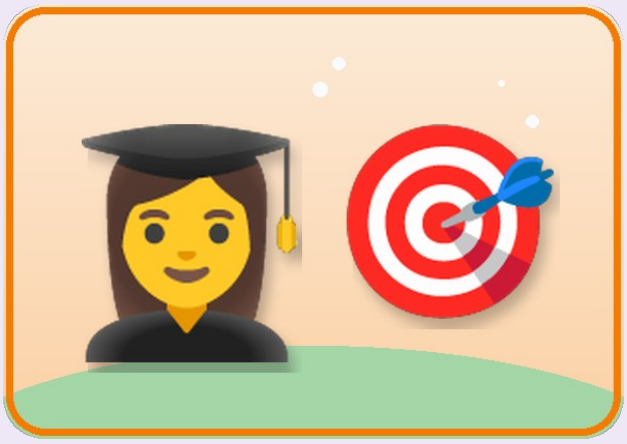


Every post lists different skills. She feels lost.

3

She tallies skills across 20 posts: SQL, Python, stats top.

4



She focuses on those 3 and lands an internship!

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Where to find AI opportunities

Job portals

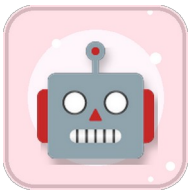
Professional networks

Campus placements

Hackathons

Open-source

Freelance platforms



BOLT SAYS:

A job description is a treasure map. Every skill listed is a clue to your next step!

FUN FACT!

Many AI job postings now ask for 'AI literacy' even in non-tech roles!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The Fake Internship Offer

CASE FILE

Several engineering students received 'AI internship' offers by email from a company with a look-alike domain. Each was asked to pay ₹4,999 for 'training kit and registration'. One student checked the official company website and careers page, found no such programme, and alerted the placement cell, which warned all students.

🔍 Analyse the case

1. List three red flags in this offer.

2. Which verification steps worked?

3. Draft a 3-point safety checklist for job seekers.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Decode the Job Description

You need:

3 real AI job posts

Highlighters

Pairs

Steps:

1.
   Read 3 AI job posts (analyst, ML engineer, product).
2.
   Highlight: must-have skills (yellow), nice-to-have (green).
3.
   Tally the most common skills across posts.
4.
   List your top 3 skill gaps and one action for each.

THINK & TALK

Which skill appeared in all three posts? Why?

Record your observations & results:

💻 CODE LAB: Mining Job Descriptions

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  from collections import Counter
2  jds = ['python sql excel', 'sql tableau python',
3         'python statistics sql', 'excel powerbi']
4  words = [w for jd in jds for w in jd.____()]
5  top = Counter(words).most_common(3)
6  print(top)
```

COMPLETE **T1. Fill the blank to split each JD into words.**

PREDICT **T2. Write the top-3 list.**

MODIFY **T3. How would you ignore case ('Python' vs 'python')?**

EXTEND **T4. Print skills that appear in only one JD.**

WORKSHEET: Navigate the AI Job World (Part A)

Name: _____ Date: _____

1. Match the role to its daily work.

Data analyst ●

● Deploys and monitors models

ML engineer ●

● Decides what AI feature to build

AI product manager ●

● Dashboards and SQL insights

AI governance lead ●

● Risk policies and audits

2. Must-have or nice-to-have for a data analyst? Tick ✓.

	Must	Nice
SQL		
Excel / spreadsheets		
Video editing		
Deep learning research		

WORKSHEET: Navigate the AI Job World (Part B)

Name: _____ Date: _____

1. Circle red flags in a job offer.

Asks you to pay a fee

Official company email domain

'Accept in 1 hour or lose it'

Clear role and stipend details

Spelling errors in the offer

No interview at all

2. Skill tally from 4 JDs. Use the word box.

WORD BOX: Python ✿ SQL ✿ Spark

Appears in 4 of 4 JDs: ____

Appears in 3 of 4 JDs: ____

Appears in only 1 JD: ____

PUZZLE & DEFINE: Navigate the AI Job World

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

JD

Meaning: _____

Example: _____

Skill gap

Meaning: _____

Example: _____

Freelancing

Meaning: _____

Example: _____

? QUIZ TIME: Navigate the AI Job World

Circle the right answer!

Q1. JD stands for job...

a) Description

b) Demo

c) Doughnut

Q2. Who deploys models to production?

a) Designer

b) ML engineer

c) Marketer

Q3. Missing skills for a role:

a) Skill gift

b) Skill party

c) Skill gap

Q4. Core tool for data analysts:

a) SQL

b) Guitar

c) Knitting

Q5. Bridge between degree and job:

a) Sofa

b) Internships

c) TV

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Navigate the AI Job World

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	JDs list must-have and preferred skills.		
	Paying a fee for an internship is normal.		
	Hackathons can lead to job opportunities.		
	SQL is common in data analyst roles.		
	AI jobs exist only in software firms.		
	Verify a company's domain before replying.		

HOME CONNECT

Discuss with your family how to verify a job offer. Share the 3-point safety checklist.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Navigate the AI Job World

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why should you decode several JDs, not just one?**

COMPARE **Q2. Compare a data analyst and a data scientist role.**

APPLY **Q3. Skills across 4 JDs: Python 4, SQL 3, Excel 2, Spark 1. Which 2 first?**

EVALUATE **Q4. Should you apply if you meet 60% of must-haves?**

CREATE **Q5. Design a weekly job-search routine for final year.**

📝 EXAM PREP: Navigate the AI Job World

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four AI job families.

2. What is a skill gap?

3. List two red flags of fake job offers.

4. Name two entry routes into AI jobs.

PART B • Long answer (5 marks)

5. Explain how to decode job descriptions and build a 6-month plan to become employable as a data analyst.

TOPIC 8 • MONTH 4 • AI & CAREER

Build Your AI Skillset

LET'S LEARN

- Be T-shaped: broad basics + one deep skill.
- Learn by building: projects beat passive videos.
- Track progress with a 90-day learning plan.

The AI skill stack

Programming

Maths & stats

ML concepts

Communication

KEY TERMS

T-shaped Broad + deep skills

Deliberate practice Focused improvement

Roadmap Learning plan

📖 CONCEPT EXPLAINED: Build Your AI Skillset

💡 In this chapter

Knowing which skills to build is only half the challenge; the other half is building them effectively. This chapter explains the T-shaped skill model, why projects beat passive learning, how deliberate practice works, and how to design a realistic 90-day plan that turns intention into ability.

🔠 1. The T-shaped profile

Imagine the letter T. The horizontal bar represents broad foundations useful across all AI work: Python programming, mathematics and statistics, data handling, basic machine learning concepts, and communication. The vertical bar represents depth in one area, such as natural language processing, computer vision, data analytics or AI in your own discipline.

Breadth lets you collaborate and adapt; depth makes you distinctive and valuable. In your first year, focus mostly on the horizontal bar, while exploring a few areas to decide where to go deep later.

🤔 PAUSE & THINK

Sketch your own T. What is already on your horizontal bar? What might your vertical bar be?

🛠️ 2. Why projects beat passive learning

Watching tutorials feels productive, but recognition is not the same as ability. When you build a project, you must frame a problem, find and clean data, choose methods, debug errors and explain results. These are exactly the skills employers and examiners test.

Start small: analyse your class's attendance data, predict cricket scores, build a quiz generator. Finish what you start, write a short README and share it. A few completed projects teach more, and prove more, than dozens of half-watched courses.

✏️ WORKED EXAMPLE

Project ladder: Month 1, analyse a CSV with pandas and charts. Month 2, train a simple classifier with scikit-learn. Month 3, build a small web demo with Streamlit and publish it on GitHub.

🏋️ 3. Deliberate practice

Deliberate practice, a concept from research on expert performance, means practising with specific goals, focusing on your weaknesses, getting quick feedback and reflecting on mistakes. Playing the same easy songs on a guitar does not make you better; practising the difficult passage slowly, again and again, does.

For AI skills, this means solving problems slightly beyond your comfort zone, comparing your code with better solutions, asking mentors or AI tools for feedback, and keeping a learning log of what you got wrong and why.

🖐️ TRY IT

Identify one weak skill (e.g., SQL joins). Write a specific practice goal for this week.

🗓️ 4. Designing a 90-day plan

Break 90 days into three 30-day sprints. In each sprint: learn the core concepts (week 1–2), build something with them (week 2–3), share it and get feedback (week 4), then reflect and adjust. Set measurable goals, such as '1 project + 1 blog post', not vague ones like 'learn ML'.

Protect regular time (e.g., 6 hours a week), track your hours and outputs, and find an accountability partner. Free resources such as NPTEL courses, Kaggle Learn, official library documentation and university open courseware are more than enough to start.

🤔 PAUSE & THINK

What could stop you from following a 90-day plan? Write one strategy to overcome it.

✅ Key takeaways

- Be T-shaped: broad foundations plus one area of depth.
- Projects build and prove skills better than passive watching.
- Deliberate practice targets weaknesses with feedback.
- Plan in 30-day sprints: learn, build, share, reflect.
- Free, high-quality resources are widely available.

? Quick check: answer in one or two lines.

1. Explain the T-shaped skill model.

2. Give two features of deliberate practice.

3. Write one measurable goal for your next 30 days.

🔬 CONCEPT DEEP DIVE: Build Your AI Skillset

🔷 T-shaped profile: broad foundations (Python, maths, data, communication) plus depth in one area (NLP, vision, analytics).

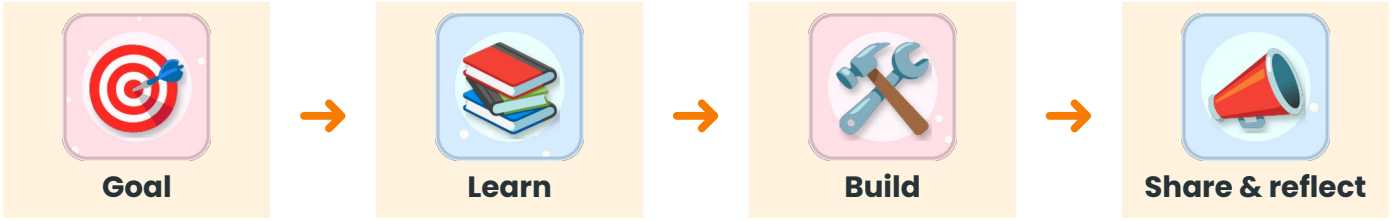
🔷 Deliberate practice: specific goals, focused effort on weaknesses, immediate feedback and reflection.

🔷 Project-based learning builds transferable skills: problem framing, debugging, documentation and teamwork.

🔷 Credentials help (NPTEL, university certificates), but portfolios with real projects signal ability more strongly.

🔷 Plan in sprints (e.g., 30 days): learn → build → share → reflect; track hours and outputs.

🔄 How it works



💻 Tracking your learning in Python

```
log = {'week1': 6, 'week2': 8, 'week3': 4, 'week4': 9}   # hours
total = sum(log.values())                                 # 27
avg = total / len(log)                                    # 6.75
print('On track!' if avg >= 6 else 'Adjust plan')
```

✍️ In your own words: explain this topic to a Class 10 student.

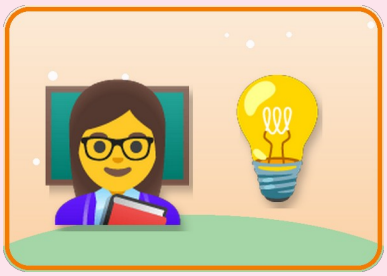
AI STORY: Eighty Hours of Videos

Characters: *Arjun (B.Tech student), Meera (his senior and mentor), a stubborn bug.*

SCENE 1

By the end of his first semester, Arjun had watched eighty hours of AI tutorials. He could name every algorithm. But when a friend asked him to build a simple spam classifier, he froze. He didn't know where to start.

Pause & predict: Why could Arjun recognise concepts but not build anything?



SCENE 2

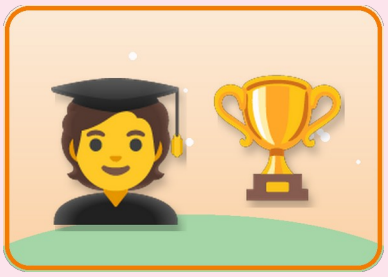
Meera, a final-year student, listened and smiled. 'Watching is like watching cricket. You only get better by batting.' She gave him a challenge: build one small, complete project each month, and write down every mistake he made.

Pause & predict: What is Meera's advice an example of?

SCENE 3

The first project, a fake-news classifier, broke thirty times: missing values, wrong shapes, a model that predicted 'real' for everything. Each fix went into his learning log. By week four, it reached 82% accuracy, and he understood why each step mattered.

Pause & predict: How did the learning log help Arjun improve?



SCENE 4

Three months and three projects later, Arjun entered the college hackathon. His team built a crop-price predictor in 24 hours and won second place. A judge asked to see his GitHub. 'Your READMEs explain everything,' she said. 'That's rare.'

⏸ **Pause & predict:** Which parts of Arjun's T-shaped profile grew during these months?

🌟 The big idea

Skills grow by building, failing and fixing, not just by watching.

💬 Discuss with your classmates

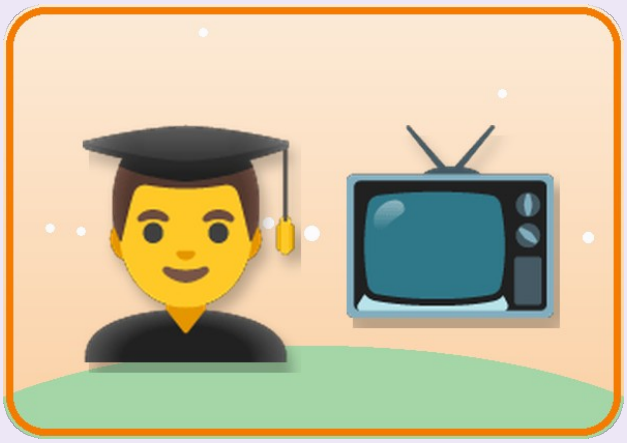
1. What small project could you complete this month?

2. How will you track your mistakes and learning?

SCENARIO: Build Your AI Skillset

Read the scenario. Think about the choices made at each step.

1



Arjun watched 80 hours of AI videos. Still stuck.

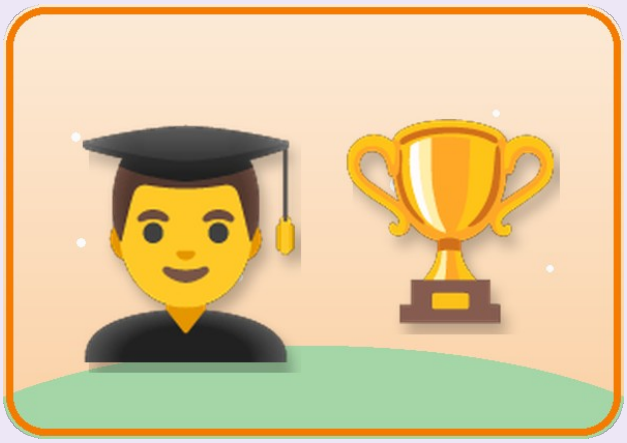
2

His mentor says: 'Build one small project per month.'

3

He builds a fake-news classifier and fixes 30 bugs.

4



3 projects later, he wins a college hackathon!

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Places to build skills

Online courses (NPTEL, etc.)

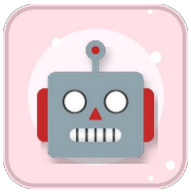
Kaggle competitions

GitHub projects

Student tech clubs

Research internships

Documentation reading



BOLT SAYS:
Watching 100 tutorials teaches less than building 3 projects and fixing every bug!

FUN FACT!
NPTEL offers free courses from IITs and IISc, many with certification exams!

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: From Kaggle Beginner to Data Scientist

CASE FILE

A B.Com student started Kaggle's beginner competitions in her first year. She published notebooks explaining every step, which earned upvotes and comments. Over two years she reached 'Expert' tier. A fintech start-up hired her as a junior data scientist after reviewing her public notebooks, ahead of candidates with only certificates.

Analyse the case

1. Why did public notebooks matter more than certificates?

2. How did explaining her work improve her skills?

3. Design a similar 2-year plan for your discipline.

My verdict: Was AI used well here? Why?

LAB ACTIVITY: 90-Day Skill Sprint Plan

You need:

Planner sheet

Course catalogues

Accountability pairs

Steps:

1.
   Pick one target role and its top 4 skills.
2.
   Plan 3 sprints of 30 days: learn → build → share.
3.
   Set measurable goals (e.g., 1 project, 1 certificate).
4.
   Swap plans with your partner; agree on weekly check-ins.

THINK & TALK

What will you stop doing to make time for this plan?

Record your observations & results:

💻 CODE LAB: Learning Tracker

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  hours = [6, 8, 4, 9, 7, 3]
2  target = 6
3  weeks_met = sum(1 for h in hours if h >= ____)
4  avg = sum(hours) / len(hours)
5  print(weeks_met, round(avg, 2))
6  streak = 0
7  for h in hours: streak = streak + 1 if h >= target else 0
8  print(streak)
```

COMPLETE **T1. Fill the blank using the variable.**

PREDICT **T2. Write weeks_met and avg.**

PREDICT **T3. What is the final streak value?**

EXTEND **T4. Compute the longest streak, not the final one.**

WORKSHEET: Build Your AI Skillset (Part A)

Name: _____ Date: _____

1. Effective skill cycle. Write 1, 2, 3, 4.

Share & get feedback ☐

Set a specific goal ☐

Build a project ☐

Learn the basics ☐

2. Active or passive learning? Tick ✓.

	Active	Passive
Binge-watching tutorials		
Building a model from scratch		
Debugging your own code		
Only reading slides		

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

✏️ WORKSHEET: Build Your AI Skillset (Part B)

Name: _____ Date: _____

1. Learning log: 6, 8, 4, 9 hours. Use the word box.

WORD BOX: 27 ✿ 6.75 ✿ Yes

Total hours = ____

Average per week = ____

Meets 6 h/week goal? ____

2. Draw your T-shaped skill profile: broad bar on top, deep bar down.

My deep specialism: _____

PUZZLE & DEFINE: Build Your AI Skillset

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

T-shaped

Meaning: _____

Example: _____

Deliberate practice

Meaning: _____

Example: _____

Roadmap

Meaning: _____

Example: _____

? QUIZ TIME: Build Your AI Skillset

Circle the right answer!

Q1. Broad basics + one deep skill:

a) T-shaped

b) O-shaped

c) Flat

Q2. Best way to learn AI skills:

a) Sleep

b) Build projects

c) Only watch

Q3. Free IIT/IISc courses platform:

a) Steam

b) Netflix

c) NPTEL

Q4. Kaggle is known for...

a) Data competitions

b) Pizza

c) Cars

Q5. Focused practice on weaknesses:

a) Random practice

b) Deliberate practice

c) No practice

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Build Your AI Skillset

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	T-shaped means broad + deep skills.		
	Watching videos alone builds strong skills.		
	Deliberate practice targets weaknesses.		
	Public projects can impress employers.		
	Certificates matter more than any project.		
	Sprint plans help track progress.		

HOME CONNECT

Share your 90-day skill plan with a family member and ask them to check in every two weeks.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Build Your AI Skillset

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is project-based learning more effective than passive watching?**

COMPARE **Q2. Compare a certificate and a portfolio project as proof of skill.**

APPLY **Q3. Weekly hours: 6, 8, 4, 9. Average? On track for a 6 h target?**

EVALUATE **Q4. Is it better to be a generalist or a specialist in AI?**

CREATE **Q5. Write a 30-day sprint with 4 weekly goals for learning pandas.**

📝 EXAM PREP: Build Your AI Skillset

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is a T-shaped skill profile?

2. Define deliberate practice.

3. Why are portfolios valuable?

4. Name two free learning platforms.

PART B • Long answer (5 marks)

5. Design a 90-day AI learning plan with sprints, measurable goals and evaluation methods. Justify your choices.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. Asked to pay for an internship?

a) Red flag | b) Normal | c) Bonus

Q2. Data analysts commonly need...

a) Guitar | b) SQL | c) Sculpture

Q3. Hours 6, 8, 4, 9: average?

a) 6 | b) 27 | c) 6.75

Q4. Broad + deep skills:

a) T-shaped | b) Round | c) Flat

Q5. Strongest proof of applied skill:

a) Attendance | b) Portfolio projects | c) Selfies

Q6. NPTEL courses come from...

a) Film studios | b) Restaurants | c) IITs & IISc

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

JD

SKILL GAP

T-SHAPED

ROADMAP

SKILL

KAGGLE

I	F	R	E	S	U	M	E	M	O
U	L	S	K	I	L	L	G	A	P
T	S	H	A	P	E	D	T	G	L
S	K	I	L	L	D	B	J	D	H
I	N	T	E	R	N	I	U	W	H
D	L	K	A	G	G	L	E	F	K
P	A	R	O	A	D	M	A	P	B
M	S	P	R	I	N	T	C	K	L
A	L	K	L	E	O	W	C	K	W
G	P	K	E	I	N	W	F	L	U

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

AI Workbench

Topic 10

AI Coding Companion

TOPIC 9 • MONTH 5 • AI TOOLS

AI Workbench

LET'S LEARN

- A workbench: notebook + data + libraries + Git.
- Colab gives free GPUs in the browser.
- Reproducible work: versions, seeds & READMEs.

Your AI workbench

Jupyter / Colab

Datasets

Git & GitHub

No-code tools

KEY TERMS

Notebook Code + notes + output

Version control Track code changes

Reproducible Same steps, same result

📖 CONCEPT EXPLAINED: AI Workbench

💡 In this chapter

Every craftsperson needs a workbench. For AI work, yours includes notebooks to experiment, datasets to learn from, version control to track changes, and habits that make your results reproducible. This chapter sets up that workbench and explains why each tool matters.

📓 1. Notebooks: your lab journal

Jupyter notebooks and Google Colab let you write code in cells, run them one at a time and see results immediately, including tables and charts. You can add text cells to explain your reasoning. This makes notebooks ideal for exploring data, learning and teaching.

Colab runs in the browser, needs no installation and offers free (limited) GPU access, which is useful for deep learning. Good notebook habits include running cells top to bottom, restarting and re-running before sharing, and keeping notebooks focused on one task.

🖐 TRY IT

Open Colab, create a notebook titled 'My Workbench', and add one text cell describing a dataset you would like to explore.

🗄 2. Finding and judging datasets

Datasets are the raw material of AI. Good sources include Kaggle, the UCI Machine Learning Repository, India's Open Government Data platform (data.gov.in) and model hubs that host datasets alongside pre-trained models. Many are free for learning and research.

Before using any dataset, check its licence (can you use it, share it, use it commercially?), how it was collected, what time period it covers, and whether some groups are missing or under-represented. A dataset is never neutral: it reflects the choices of whoever created it.

🤔 **PAUSE & THINK**

A dataset of 'Indian student performance' was collected only from city schools. What could go wrong if you used it to predict results for rural students?

🌿 3. Version control with Git and GitHub

Git records snapshots of your project over time. You stage changes with git add, save a snapshot with git commit -m 'message', and upload to an online repository such as GitHub with git push. If something breaks, you can go back to an earlier working version.

GitHub also lets teams collaborate, review each other's code and showcase projects publicly. A clean GitHub profile with well-documented projects is one of the best portfolios an AI student can have.

✏️ **WORKED EXAMPLE**

Workflow: git init → git add analysis.ipynb → git commit -m 'Add EDA of attendance data' → git push. Clear commit messages describe what changed and why.

🔁 4. Reproducibility: making results trustworthy

A result is reproducible if someone else can follow your steps and get the same answer. This requires fixing random seeds, recording library versions in a requirements.txt file, keeping data files accessible (or describing how to obtain them), and writing a README that explains how to run the project.

Reproducibility is not bureaucracy; it is the foundation of trust in science and engineering. Major AI conferences now encourage authors to share code and data for exactly this reason.

🖐 **TRY IT**

Write the six sections you would include in a README for a student project.

✅ Key takeaways

- Notebooks mix code, results and explanations.
- Choose datasets carefully: licence, collection method, coverage.
- Git tracks versions; GitHub shares and showcases work.
- Seeds, versions, data access and READMEs make results reproducible.
- Your workbench habits are part of your professional reputation.

❓ Quick check: answer in one or two lines.

1. Name three dataset sources.

2. What do git add, git commit and git push do?

3. List three practices that improve reproducibility.

🔬 CONCEPT DEEP DIVE: AI Workbench

🔷 Notebooks (Jupyter, Colab) mix code, outputs and notes, ideal for exploration and teaching.

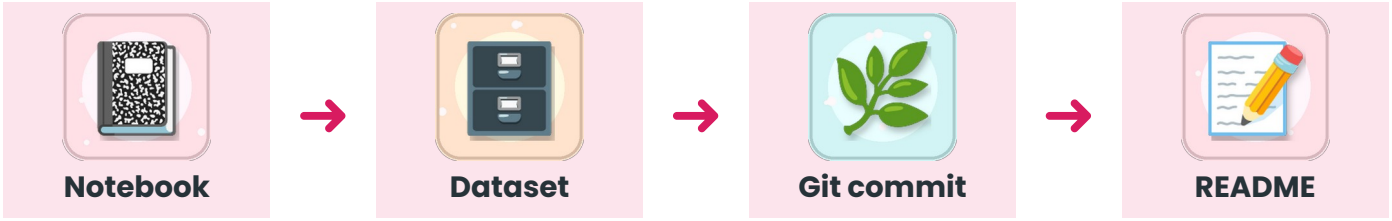
🔷 Datasets: Kaggle, UCI, data.gov.in; always check licence, collection method and known biases.

🔷 Git tracks versions: add → commit → push; branches allow safe experiments; GitHub hosts and shares code.

🔷 Reproducibility: fix random seeds, record library versions (requirements.txt), document steps in a README.

🔷 No-code and low-code tools let non-programmers prototype ML, but understanding the data is still essential.

🔄 How it works



💻 Making results reproducible

```
import random, numpy as np
random.seed(42); np.random.seed(42)
# requirements.txt: pandas==2.2.2  scikit-learn==1.5.0
# README: data source, steps, how to run, results
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: It Works on My Laptop

Characters: *Divya and Rahul (project partners), Professor Rao.*

SCENE 1

Divya proudly showed her model: 91% accuracy predicting student dropouts. 'Send it to me,' said her partner Rahul. He opened the notebook on his laptop and pressed Run All. Red errors filled the screen.

Pause & predict: Guess three reasons the notebook might fail on Rahul's laptop.

SCENE 2

The dataset file was missing. The pandas version was different. When they finally made it run, the accuracy was 86%, not 91%. 'Did you change something?' Rahul asked. Divya had not; the random split was different every time.

Pause & predict: Why did the accuracy change between runs?

SCENE 3

Professor Rao explained: 'If no one else can reproduce your result, it's not yet a result.' He showed them how to set random seeds, write a requirements.txt file, store the data path clearly, use Git for versions and write a README with steps to run the project.

Pause & predict: Which of the professor's practices would you adopt first?

SCENE 4

They rebuilt the project properly and pushed it to GitHub. A classmate cloned it on a lab computer and got exactly 88.4%, the same as theirs. At the project review, the examiner ran it live without a single error. Rahul grinned: 'Now it works on every laptop.'

⏸ **Pause & predict:** Why is reproducibility important for teamwork and trust?

🌟 The big idea

A result you can't reproduce is a rumour. Good workbench habits turn experiments into trustworthy work.

💬 Discuss with your classmates

1. Have you ever had code that worked only on one computer? Why?

2. Write the first three lines of a README for your latest project.

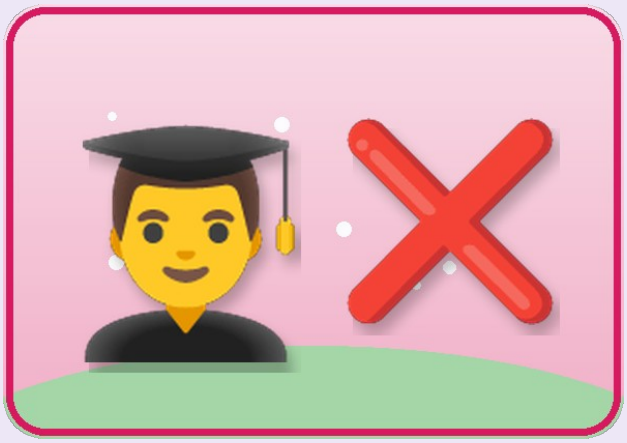
SCENARIO: AI Workbench

Read the scenario. Think about the choices made at each step.

1

Divya's model works on her laptop only.

2

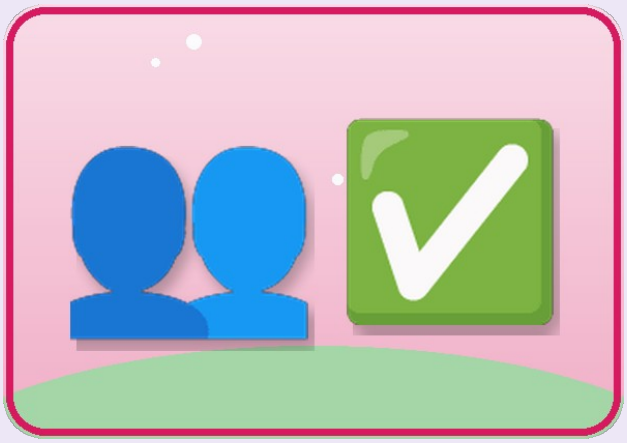


Her teammate gets errors: missing files, wrong versions.

3

They add Git, requirements.txt, a seed and a README.

4



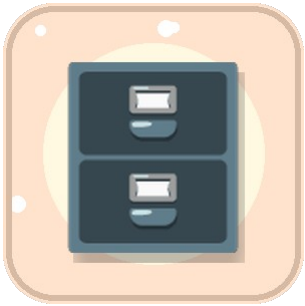
Now anyone can reproduce the results in Colab!

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Workbench in action



data.gov.in datasets

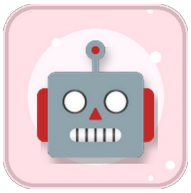
Kaggle notebooks

Pre-trained model hubs

Dashboards with Streamlit

Team code on GitHub

No-code ML builders



BOLT SAYS:
If your result can't be reproduced, it's a rumour, not a result!

FUN FACT!
India's Open Government Data platform (data.gov.in) hosts thousands of public datasets!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The Reproducibility Crisis in ML

CASE FILE

Researchers who tried to reproduce published machine-learning results found that many could not be fully reproduced: code was missing, data was unavailable or hyperparameters were unreported. Major AI conferences responded by introducing reproducibility checklists and encouraging authors to share code and data.

🔍 Analyse the case

1. Name three causes of irreproducible ML results.

2. How do checklists improve research quality?

3. Apply a checklist to your last project: what's missing?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Reproducibility Relay

You need:

Colab

Small CSV dataset

Teams of 3

Steps:

1. Student A builds a notebook that trains a model; saves it to GitHub.
2. Student B re-runs it from scratch on a new machine.
3. Note every failure: missing files, versions, random seeds.
4. Student C writes a README that fixes all issues.

THINK & TALK

Why did results differ when the random seed was not set?

Record your observations & results:

💻 CODE LAB: Reproducible Experiment

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import random
2  random.seed(____)
3  data = list(range(10))
4  random.shuffle(data)
5  train, test = data[:8], data[8:]
6  print(len(train), len(test))
7  print(test)
```

COMPLETE **T1. Use seed 42 in the blank.**

PREDICT **T2. What does line 6 print?**

EXPLAIN **T3. Will line 7 print the same list on every run? Why?**

EXTEND **T4. Write the requirements.txt line for this code.**

WORKSHEET: AI Workbench (Part A)

Name: _____ Date: _____

1. Match the tool to its use.

Colab ●

● Share & version code

GitHub ●

● Find ready datasets

Kaggle Datasets ●

● Run notebooks with GPU

Streamlit ●

● Build a quick web app

2. Git basics. Write 1, 2, 3, 4.

git push ☐

git init / clone ☐

git commit -m 'msg' ☐

git add file.py ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

WORKSHEET: AI Workbench (Part B)

Name: _____ Date: _____

1. Circle items that improve reproducibility.

Fixed random seed

requirements.txt

Unnamed data files

README with steps

Random untracked edits

Git commit history

2. Match the dataset source to its strength.

data.gov.in ●

Kaggle ●

UCI repository ●

Model hubs ●

● Pre-trained models & datasets

● Classic academic datasets

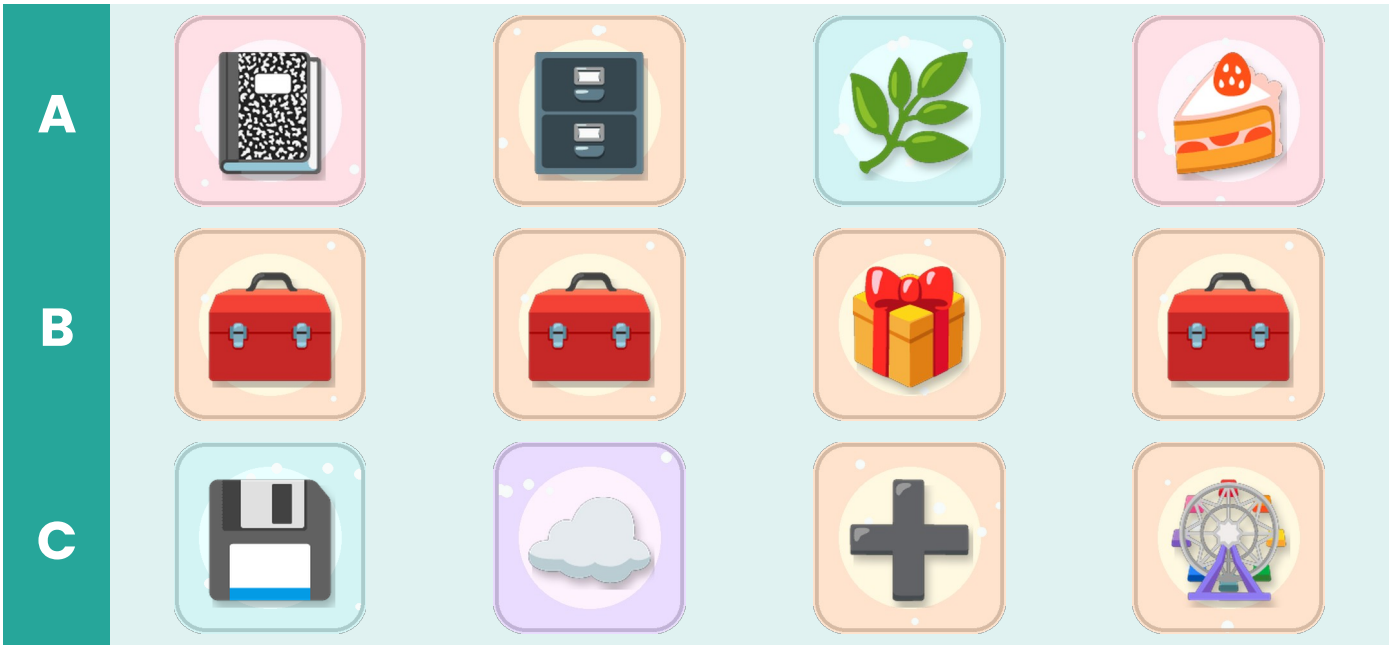
● Indian government data

● Competitions & notebooks

PUZZLE & DEFINE: AI Workbench

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Notebook

Meaning: _____

Example: _____

Version control

Meaning: _____

Example: _____

Reproducible

Meaning: _____

Example: _____

? QUIZ TIME: AI Workbench

Circle the right answer!

Q1. Free browser notebooks with GPUs:

a) Google Colab

b) Notepad

c) Paint

Q2. git commit saves...

a) An email

b) A snapshot

c) A song

Q3. Same steps, same result:

a) Random

b) Chaotic

c) Reproducible

Q4. Indian public datasets portal:

a) data.gov.in

b) game.in

c) food.in

Q5. A README explains...

a) Recipes

b) How to use it

c) Music

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Workbench

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Random seeds help reproduce results.		
Git tracks changes to code.		
Dataset licences can be ignored.		
requirements.txt records library versions.		
Colab requires installing Python locally.		
A README helps others run your project.		

HOME CONNECT

Create a free GitHub account (with guidance) and upload one small project with a README.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Workbench

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does setting a random seed matter?**

COMPARE **Q2. Compare notebooks and .py scripts for AI projects.**

APPLY **Q3. Write the 4 Git commands to publish a new project.**

EVALUATE **Q4. Are no-code ML tools good for beginners?**

CREATE **Q5. Design a README template with 6 sections.**

📝 EXAM PREP: AI Workbench

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is a Jupyter/Colab notebook?

2. List the basic Git workflow.

3. Why record library versions?

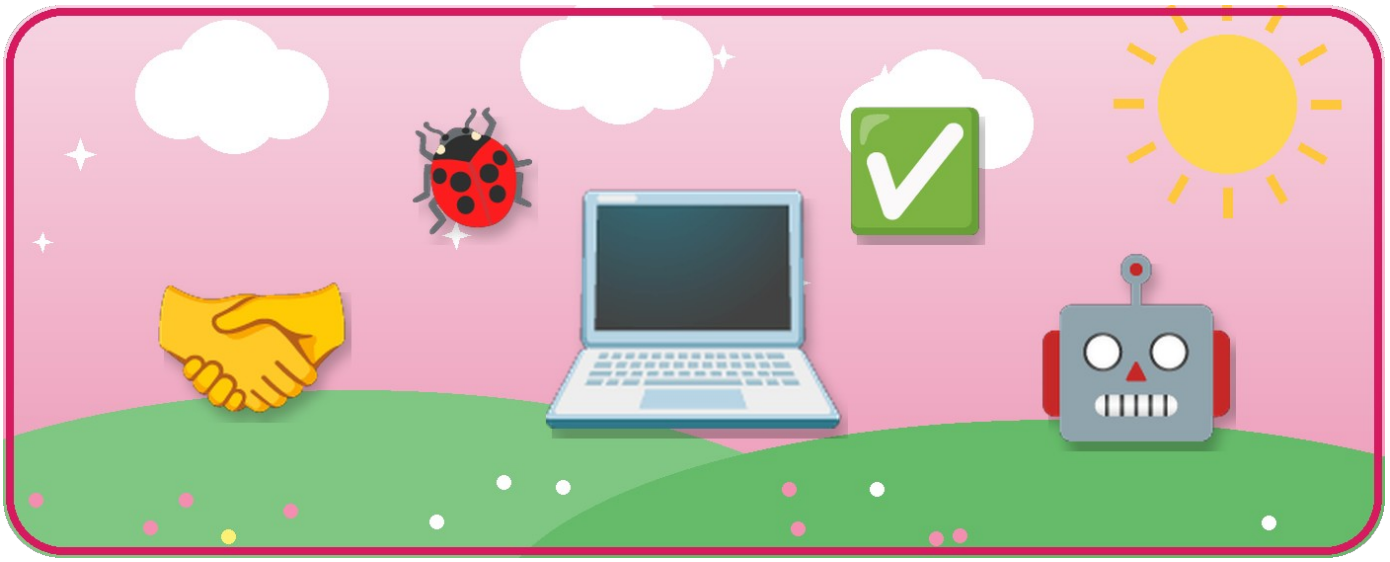
4. Name two dataset sources.

PART B • Long answer (5 marks)

5. Explain reproducibility in AI projects and list six practices that ensure others can reproduce your results.

TOPIC 10 • MONTH 5 • AI TOOLS

AI Coding Companion



LET'S LEARN

- AI assistants generate, explain & debug code.
- You stay the engineer: review every line.
- Tests prove code works; AI output is a draft.

Ways to use an AI coder

Generate

Explain

Debug

Write tests

KEY TERMS

Pair programming Two minds, one code

Unit test Checks one function

Hallucinated API Made-up function

📖 CONCEPT EXPLAINED: AI Coding Companion

💡 **In this chapter**

AI coding assistants can write functions, explain unfamiliar code and suggest fixes in seconds. Used well, they make you faster and help you learn. Used carelessly, they introduce hidden bugs and security risks. This chapter shows how to work with an AI coding companion like a responsible engineer.

✨ 1. What AI coding assistants do

AI coding assistants are language models trained on large amounts of code and text. They predict likely code from your instructions and context. Common uses include generating boilerplate code, completing lines as you type, explaining what a piece of code does, translating between languages, writing documentation and suggesting unit tests.

For learners, the 'explain' mode is especially valuable: paste a confusing snippet and ask for a line-by-line explanation, then try to rewrite it yourself. This turns the assistant into a patient tutor rather than a shortcut.

🖐 **TRY IT**

Ask an AI assistant to explain the list comprehension [x for x in nums if x % 2 == 0]. Then explain it in your own words.

👻 2. How AI code goes wrong

AI-generated code can look perfect and still be wrong. Typical problems include logic errors on edge cases (empty lists, negative numbers), outdated library functions, 'hallucinated' functions or packages that do not exist, and insecure patterns such as building database queries from raw user input.

Because the code sounds confident and runs on simple examples, these errors are easy to miss. The responsibility for any code you submit or deploy remains yours, not the tool's.

✏️ **WORKED EXAMPLE**

An AI wrote def avg(nums): return sum(nums) / len(nums). It works for [2, 4, 6] but crashes with ZeroDivisionError for an empty list. A test case for [] would catch this immediately.

🧪 3. Testing: your safety net

A unit test checks that a small piece of code behaves correctly. With pytest, you write functions such as def test_avg(): assert avg([2, 4]) == 3. Good tests cover normal cases, edge cases and error cases. If tests pass, you gain confidence; if they fail, you know exactly where to look.

A powerful workflow is to write or review the tests first, then ask the AI to produce code that passes them. The tests become a precise specification, and the AI becomes a fast implementer under your control.

🤔 **PAUSE & THINK**

What three test cases would you write for a function that returns the largest number in a list?

🛡️ 4. Security, licences and integrity

Never paste passwords, API keys or confidential code into public AI tools. Store secrets in environment variables. Before installing any package an AI suggests, check that it exists on the official registry and is well maintained, because attackers sometimes register fake packages with names AI tools commonly invent.

Respect licences: generated code may resemble existing licensed code. In coursework, follow your institution's policy and disclose AI assistance where required. The goal is to become a better programmer, not to hide behind a tool.

🤔 **PAUSE & THINK**

Write one personal rule you will follow when using AI for coding assignments.

✅ Key takeaways

- AI assistants generate, complete, explain and translate code.
- AI code can hide edge-case bugs, fake packages and security flaws.
- Unit tests are your safety net; consider writing them first.
- Protect secrets and verify every dependency.
- You remain responsible for any code you submit.

❓ Quick check: answer in one or two lines.

1. Give two common failure modes of AI-generated code.

2. What is a unit test?

3. Why should you verify packages suggested by AI?

🔬 CONCEPT DEEP DIVE: AI Coding Companion

🔷 AI coding assistants predict code from context: they can generate functions, explain code, refactor and write tests.

🔷 Common failure modes: hallucinated libraries, outdated APIs, subtle logic bugs, insecure patterns (e.g., SQL injection).

🔷 Test-driven use: write tests first or immediately; treat AI code as a draft that must pass tests and review.

🔷 Security & privacy: never paste secrets, keys or confidential code into tools without approval.

🔷 Licensing & integrity: check if generated code matches licensed code; disclose AI use in coursework as required.

🔄 How it works

Specify

Draft

Test

Review & commit

💻 A unit test with pytest

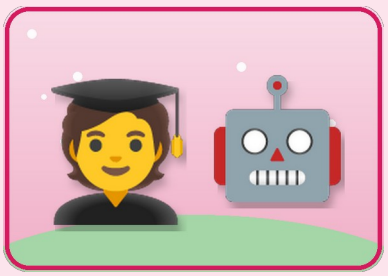
```
def clean(num): return num.replace(' ', '')[-10:]

def test_clean():
    assert clean('+91 98765 43210') == '9876543210'
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Phone Number Bug

Characters: *Sameer (B.Sc student), his senior Fatima, an AI coding assistant.*



SCENE 1

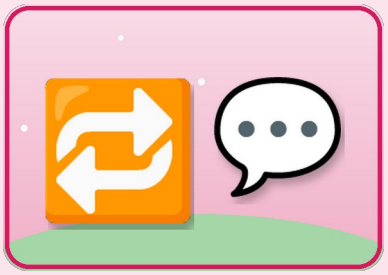
Sameer needed a function to clean phone numbers for his college club's contact list. He asked an AI assistant, which instantly produced neat code. He tested it on three numbers. Perfect! He was about to submit it to the club's app.

Pause & predict: Is testing on three examples enough? Why?

SCENE 2

Fatima, who maintained the app, asked, 'Where are your tests?' She wrote quick pytest cases: a number with +91, one with dashes, one with spaces and an empty string. Two tests failed. The AI's code had chopped off digits when numbers began with +91.

Pause & predict: What kinds of cases did Fatima's tests cover?



SCENE 3

Sameer pasted the failing tests back into the assistant and asked for a fix. The new code passed. Then the AI suggested installing a package called 'phonecleanr'. Fatima checked the official package registry: it did not exist. 'Never install what you haven't verified,' she warned.

Pause & predict: Why could installing an AI-suggested package be dangerous?

SCENE 4

With all tests green, Sameer committed the code and wrote in the README: 'Function drafted with AI assistance, tested with 6 cases.' The club's contact list worked flawlessly. Sameer now writes tests first, every time.

⏸ **Pause & predict:** Why did Sameer disclose AI assistance?

🌟 The big idea

AI can write code fast, but tests, verification and honesty make it trustworthy.

💬 Discuss with your classmates

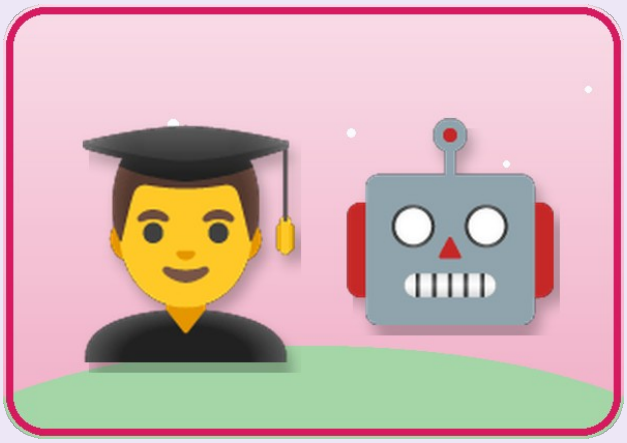
1. What rules should your class follow when using AI coding tools?

2. Write two test cases for a function that calculates a grade from marks.

🎞️ SCENARIO: AI Coding Companion

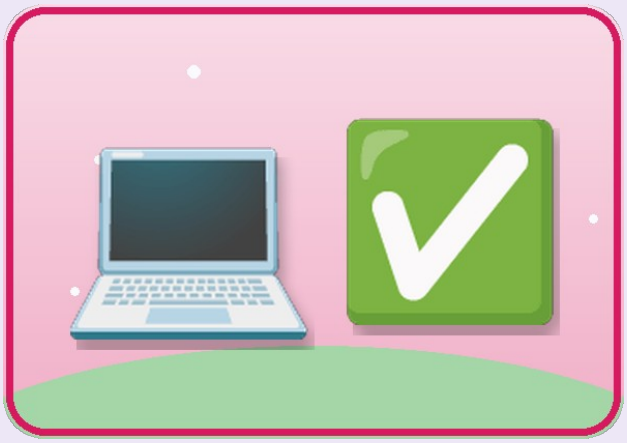
Read the scenario. Think about the choices made at each step.

1



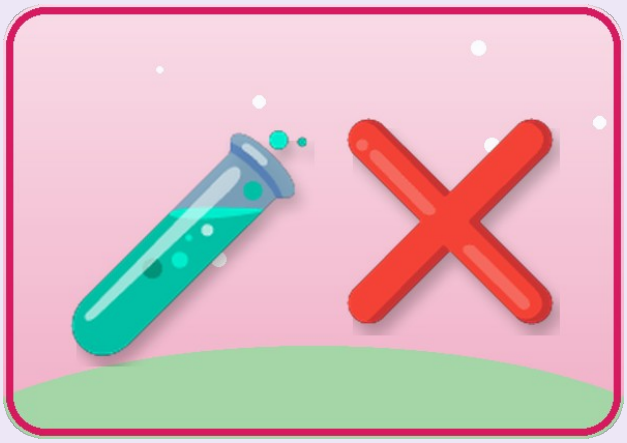
Sameer asks AI to write a function to clean phone numbers.

2



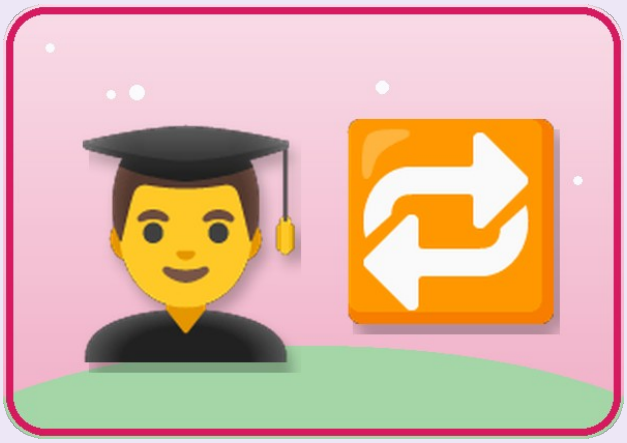
The code looks perfect and runs on his 3 examples.

3



A unit test with '+91' numbers fails!

4



He asks AI to fix it, adds more tests, then commits.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI pair programming

Autocomplete in IDEs

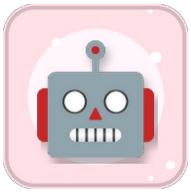
Explaining legacy code

Converting languages

Generating unit tests

Writing docstrings

Spotting security bugs



BOLT SAYS:
I can write code fast. But I can also invent functions that don't exist. Run and test me!

FUN FACT!
Studies have found AI-suggested code can contain security flaws if not reviewed!

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: Slopsquatting: Fake Packages from AI

CASE FILE

Security researchers noticed AI coding tools sometimes recommend Python or JavaScript packages that don't exist. Attackers began registering those invented names with malicious code, hoping developers would install them. Teams responded by verifying every dependency against official registries and using lock files.

Analyse the case

1. Explain how a hallucinated package becomes a security risk.

2. Which checks would stop this attack?

3. Write a team rule for adding new dependencies.

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Human vs AI Debug Duel

You need:

Laptops

5 buggy snippets

Timer

Steps:

1.
   Each pair gets 5 short buggy Python snippets.
2.
   Round 1 (10 min): fix bugs without AI. Record time.
3.
   Round 2: fix 5 new bugs with an AI assistant.
4.
   Write a test for each fix. Did any AI fix fail the test?

THINK & TALK

When did the AI's fix look right but fail the test?

Record your observations & results:

💻 CODE LAB: Test the AI's Code

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  def clean(num):
2      digits = ''.join(c for c in num if c.isdigit())
3      return digits[-10:]
4
5  def test_clean():
6      assert clean('98765 43210') == '9876543210'
7      assert clean('+91-98765-43210') == ____
8      assert clean('12') == '12'
```

COMPLETE **T1. Fill the expected value in line 7.**

PREDICT **T2. Do all three asserts pass?**

DEBUG **T3. Should clean('12') be accepted? Suggest a fix.**

EXTEND **T4. Write a test for an empty string input.**

WORKSHEET: AI Coding Companion (Part A)

Name: _____ Date: _____

1. Good or risky AI-coding practice? Tick ✓.

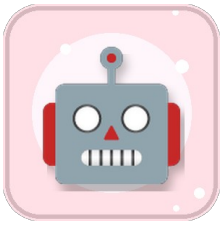
	Good	Risky
Run tests on AI-written code		
Paste API keys into a public AI chat		
Ask AI to explain unfamiliar code		
Submit AI code without reading it		

2. AI-assisted coding loop. Write 1, 2, 3, 4.

Commit when tests pass ☐

Describe the task clearly ☐

Review & test the code ☐



Get AI draft code ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

WORKSHEET: AI Coding Companion (Part B)

Name: _____ Date: _____

1. Predict the output of clean(num). Use the word box.

WORD BOX: **9876543210** ✿ **1234567890** ✿ **Fail**

clean('98765 43210') → ____

clean('+91 12345 67890') → ____

assert clean('123') == '0000000123' → ____

2. Match the AI-code risk to its defence.

Hallucinated package	**Unit tests & review**
Leaked API key	**Check registry & lock files**
Logic bug	**Licence scanning**
Licence issue	**Never paste secrets; use env vars**

PUZZLE & DEFINE: AI Coding Companion

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Pair programming

Meaning: _____

Example: _____

Unit test

Meaning: _____

Example: _____

Hallucinated API

Meaning: _____

Example: _____

? QUIZ TIME: AI Coding Companion

Circle the right answer!

Q1. Checks one function works:

a) Unit test

b) Mic test

c) Road test

Q2. A made-up library function is a...

a) Bonus

b) Hallucinated API

c) Password

Q3. Who is responsible for AI-written code?

a) The AI

b) Nobody

c) The developer

Q4. Never paste into public AI chats:

a) Secret keys

b) Python syntax

c) Comments

Q5. AI code output should be treated as...

a) Final

b) A draft

c) Garbage

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Coding Companion

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	AI code should be tested before use.		
	Pasting API keys into public chats is safe.		
	AI can suggest packages that don't exist.		
	The developer is responsible for committed code.		
	Code that runs once is always correct.		
	Disclosing AI use may be required in coursework.		

HOME CONNECT

Ask an AI tool to explain a short program, then explain it back to a friend without notes.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Coding Companion

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why must AI-generated code be tested even if it runs?**

COMPARE **Q2. Compare asking AI to 'write code' vs 'explain this code'.**

APPLY **Q3. Write a test for add(a, b) covering negatives and zero.**

EVALUATE **Q4. Should first-year exams allow AI coding assistants?**

CREATE **Q5. Write 4 team rules for responsible AI-assisted coding.**

📝 EXAM PREP: AI Coding Companion

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. List three uses of AI coding assistants.

2. What is a hallucinated API?

3. Why write unit tests?

4. Why never paste secrets into AI tools?

PART B • Long answer (5 marks)

5. Describe a responsible AI-assisted coding workflow and explain how it prevents three common failures.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. Fixing seeds improves...

a) Reproducibility | b) Colours | c) Volume

Q2. git add → git commit → ...

a) git delete | b) git push | c) git play

Q3. Library versions are listed in...

a) diary.txt | b) song.mp3 | c) requirements.txt

Q4. AI invented a package. Risk:

a) Malicious install | b) Free bonus | c) Speed-up

Q5. AI-written code must be...

a) Trusted | b) Tested | c) Ignored

Q6. Free GPU notebooks:

a) Notepad | b) Paint | c) Colab

WORD SEARCH: AI CREATE

Find these words. They go across → or down ↓.

NOTEBOOK

UNIT TEST

COLAB

GITHUB

SEED

TEST

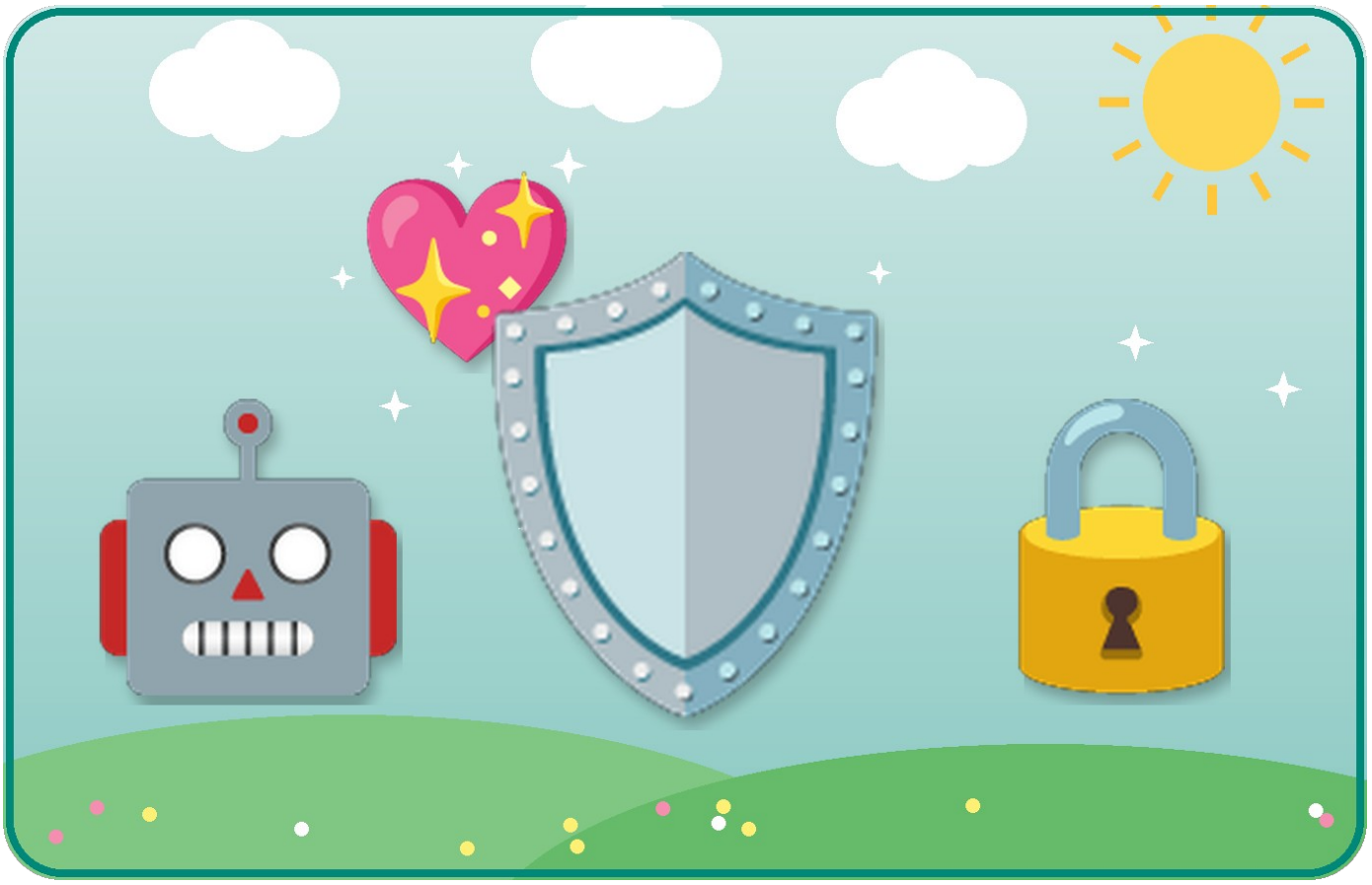
R	F	C	B	I	A	M	N	A	S
T	G	O	O	M	N	U	T	A	E
P	I	L	B	F	N	W	E	U	E
G	T	A	D	H	O	P	S	N	D
M	H	B	S	M	T	S	T	I	I
P	U	D	D	U	E	M	K	T	B
O	B	C	E	G	B	L	S	T	W
M	E	L	B	I	O	T	C	E	K
L	K	F	U	C	O	E	K	S	R
F	B	C	G	W	K	T	N	T	B

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

Responsible Digital Intelligence

Topic 12

Data Privacy Matters

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Responsible Digital Intelligence

LET'S LEARN

- Responsible AI: fair, transparent, safe, accountable.
- Bias enters through data, design and deployment.
- Use AI with integrity in study and work.

Responsible AI principles

Fairness

Transparency

Safety

Accountability

KEY TERMS

Bias Systematic unfairness

Transparency Openness about AI

Integrity Honest academic work

📖 CONCEPT EXPLAINED: Responsible Digital Intelligence

💡 In this chapter

As AI shapes decisions about jobs, loans, news and education, responsibility becomes as important as capability. This chapter introduces the core principles of responsible AI, explains how bias enters systems, and discusses everyday responsibilities: academic integrity, misinformation and digital wellbeing.

⚖️ 1. Principles of responsible AI

Across international and Indian frameworks (such as those from the OECD, UNESCO and NITI Aayog), similar principles appear. Fairness: AI should not discriminate unjustly. Transparency: people should know when AI is used and how it works at a suitable level. Accountability: named people and organisations must answer for outcomes. Safety and reliability: systems should be tested and monitored. Privacy: personal data must be protected.

Principles become meaningful only when translated into practices: testing outcomes across groups, documenting models, providing human review and appeal, and monitoring systems after launch.

🤔 PAUSE & THINK

Which principle matters most for an AI that shortlists scholarship applicants? Justify.

📊 2. How bias enters AI

Bias can enter at every stage. Historical bias: data reflects past unfairness, such as hiring records dominated by one group. Sampling bias: data does not represent everyone, such as surveys conducted only in cities. Label bias: human annotators bring their own prejudices. Proxy bias: an innocent-looking feature, like pin code or college name, may stand in for caste, religion or income.

Feedback loops can amplify bias after deployment: if a system sends more inspections to one area, more problems are recorded there, which makes the system send even more inspections. Detecting bias requires measuring outcomes for different groups, not just overall accuracy.

✏️ **WORKED EXAMPLE**

Four-fifths rule check: Group A selection rate 50%, Group B 30%. Ratio = 30 / 50 = 0.6. Since 0.6 < 0.8, the system shows possible adverse impact and needs review.

📰 3. Misinformation and synthetic media

Generative AI can produce convincing fake text, images, voices and videos in seconds. This creates risks of misinformation, fraud and harassment. Responsible users pause before sharing, check sources, look for credible confirmation, and label AI-generated content they create themselves.

Platforms and governments are introducing labels, watermarks and rules, but no technical tool is perfect. Critical thinking remains the strongest defence.

🖐️ **TRY IT**

Find a viral claim online. Check it with two independent, reliable sources. Record what you found.

🎓 4. Integrity and wellbeing in the AI age

Academic integrity means the work you submit reflects your own understanding. Using AI to brainstorm, explain concepts or check grammar may be allowed; submitting AI-written work as your own usually is not. Always follow your institution's policy, disclose AI use when required, and be ready to explain your work in your own words.

Digital wellbeing also matters: recommendation systems are designed to capture attention. Set limits, notice how apps make you feel and use AI as a tool, not a substitute for thinking, friendship or rest.

🤔 **PAUSE & THINK**

Write your own one-sentence 'AI integrity pledge' for this semester.

✅ Key takeaways

- Core principles: fairness, transparency, accountability, safety, privacy.
- Bias enters via history, sampling, labels, proxies and feedback loops.
- Measure outcomes by group; use checks like the four-fifths rule.
- Verify before sharing; label your own AI-generated content.
- Follow integrity policies and protect your digital wellbeing.

❓ Quick check: answer in one or two lines.

1. List four responsible-AI principles.

2. What is a proxy feature? Give an example.

3. Group A 60%, Group B 42% selected. Does the four-fifths rule flag it?

🔬 CONCEPT DEEP DIVE: Responsible Digital Intelligence

🔷 Responsible AI principles (common across OECD, UNESCO, NITI Aayog): fairness, transparency, accountability, safety, privacy.

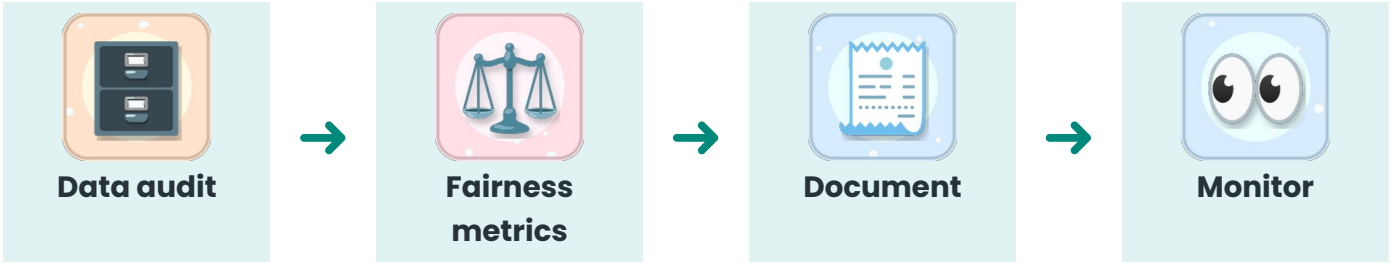
🔷 Bias sources: historical data, sampling gaps, label bias, proxy features and feedback loops after deployment.

🔷 Mitigation: diverse data, fairness metrics across groups, human review, documentation (model cards) and audits.

🔷 Misinformation risks rise with GenAI; responsible users verify, label AI content and avoid spreading unverified claims.

🔷 Academic integrity: follow institutional AI policies, disclose use, and ensure the reasoning submitted is your own.

🔄 How it works



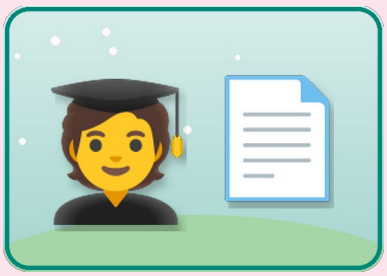
💻 Selection-rate ratio (four-fifths rule)

```
rate_a = 50 / 100    # group A selected
rate_b = 30 / 100    # group B selected
ratio = rate_b / rate_a          # 0.6
print('Possible bias' if ratio < 0.8 else 'OK')
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Viva

Characters: *Lakshmi and Vikram (classmates), Professor Anand.*



SCENE 1

The night before the deadline, Vikram pasted the assignment question into a chatbot and submitted the polished essay it produced. Lakshmi spent three evenings researching, using AI only to brainstorm ideas and check her grammar, and wrote a note at the end disclosing that use.

Pause & predict: Which approach do you think Professor Anand will value? Why?

SCENE 2

Professor Anand announced short vivas for every submission. He asked Vikram to explain one argument from his essay. Vikram stumbled; he couldn't explain terms he had 'written'. The essay also cited two journal articles that did not exist.

Pause & predict: What problems did the viva reveal?

SCENE 3

The professor explained the college's AI policy: AI may support learning, but submitted work must reflect your own understanding, and AI use must be disclosed. Vikram's essay received zero, with a chance to redo it. Lakshmi explained her ideas confidently and received an A.

Pause & predict: Do you think the professor's response to Vikram was fair? Explain.

SCENE 4

Later, Vikram asked Lakshmi how she used AI. 'As a study partner, not a ghost-writer,' she said. Together they drafted an 'AI integrity pledge' for their class. Professor Anand liked it so much that he shared it with the whole department.

⏸ **Pause & predict:** What would you include in your class's AI integrity pledge?

🌟 The big idea

Responsible AI use means honesty about how you used it and ownership of what you submit.

💬 Discuss with your classmates

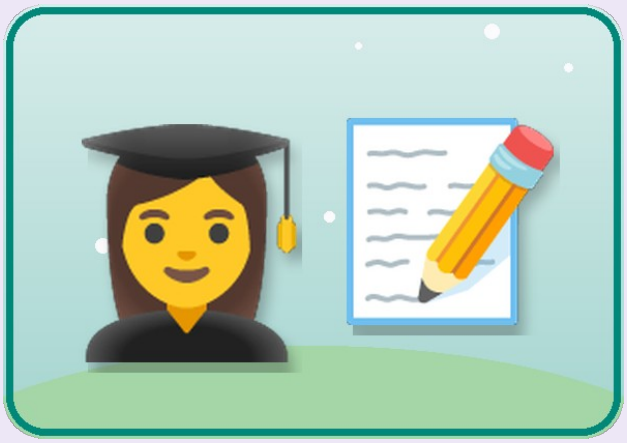
1. Where is the line between using AI to learn and using it to cheat?

2. How should colleges design assessments in the age of AI?

SCENARIO: Responsible Digital Intelligence

Read the scenario. Think about the choices made at each step.

1



Lakshmi's friend submits an AI-written assignment.

2

The professor asks him to explain it in a viva. He can't.

3

The college policy: disclose AI use; your thinking must be yours.

4

Lakshmi uses AI to brainstorm, writes herself, and discloses.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Responsibility in practice

Academic integrity policies

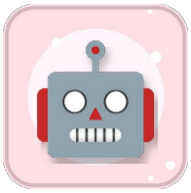
Fair lending audits

Misinformation labels

Bias-tested hiring tools

Energy-aware AI

Digital wellbeing



BOLT SAYS:
Being smart with AI isn't enough. Being responsible with AI is what earns trust!

FUN FACT!
Training very large AI models can use as much electricity as many homes use in a year!

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: Biased Hiring Algorithm Scrapped

CASE FILE

A large tech company built an AI tool to rank job applicants using 10 years of past resumes, mostly from men. The model learnt to downgrade resumes containing words like 'women's' (e.g., 'women's chess club'). Engineers could not guarantee fairness, and the company abandoned the tool.

🔍 Analyse the case

1. Identify the bias source in this case.

2. Why didn't simply deleting the word 'women's' fix it?

3. Suggest two safeguards before deploying hiring AI.

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Ethics Case Clinic

You need:

4 dilemma cards

Principles chart

Teams of 4

Steps:

1.
   Each team gets a dilemma (e.g., AI-written assignment, biased hiring tool).
2.
   Map which principles are at stake.
3.
   Argue 2 options; list stakeholders affected.
4.
   Recommend a decision with a safeguard.

THINK & TALK

Did any principles conflict with each other? How did you decide?

Record your observations & results:

💻 CODE LAB: Fairness Check in Code

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  selected = {'A': 50, 'B': 30}
2  applied  = {'A': 100, 'B': 100}
3  rate = {g: selected[g] / applied[g] for g in selected}
4  ratio = min(rate.values()) / ____(rate.values())
5  print(rate, round(ratio, 2))
6  print('Review!' if ratio < 0.8 else 'OK')
```

COMPLETE **T1. Fill the blank (built-in function).**

PREDICT **T2. Write rate and ratio.**

PREDICT **T3. What is printed on the last line?**

EXTEND **T4. Add group C: 40 of 80 selected. New ratio?**

WORKSHEET: Responsible Digital Intelligence (Part A)

Name: _____ Date: _____

1. Match the principle to its practice.

Fairness ●

● **Named owner answers for harms**

Transparency ●

● **Tell users AI is being used**

Safety ●

● **Test outcomes across groups**

Accountability ●

● **Red-team before release**

2. Integrity: OK or misconduct? Tick ✓.

	OK	Misconduct
Using AI to brainstorm, then writing yourself (disclosed)		
Submitting AI essay as your own		
Asking AI to explain a concept		
Citing AI-invented references		

WORKSHEET: Responsible Digital Intelligence (Part B)

Name: _____ Date: _____

1. Four-fifths rule. Use the word box.

WORD BOX: **0.6** ✿ **Possible bias** ✿ **0.36**

Group B rate: 36/100 = ____

Ratio: 0.36 ÷ 0.60 = ____

Ratio < 0.8 means ____

2. Match the bias source to its example.

Historical bias ●

Sampling bias ●

Label bias ●

Feedback loop ●

● **Annotators' prejudice in tags**

● **Past hiring favoured one group**

● **Policing data sends more police to same areas**

● **Survey only in cities**

PUZZLE & DEFINE: Responsible Digital Intelligence

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Bias

Meaning: _____

Example: _____

Transparency

Meaning: _____

Example: _____

Integrity

Meaning: _____

Example: _____

? QUIZ TIME: Responsible Digital Intelligence

Circle the right answer!

Q1. Systematic unfair AI outcomes:

a) Algorithmic bias

b) Art

c) Melody

Q2. Telling users AI is used is...

a) Secrecy

b) Transparency

c) Chance

Q3. Submitting AI essays as yours is...

a) Fine

b) Clever

c) Misconduct

Q4. Bias can enter through...

a) Data & design

b) Moonlight

c) Balloons

Q5. Large AI training uses lots of...

a) Bread

b) Energy

c) Salt

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Responsible Digital Intelligence

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Proxy features can cause bias.		
	Removing a sensitive column guarantees fairness.		
	Model cards document a model's limits.		
	Disclosing AI use supports academic integrity.		
	Feedback loops can amplify bias.		
	AI has no environmental impact.		

HOME CONNECT

Discuss with family: where could an AI system be unfair to someone you know? What safeguard would help?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Responsible Digital Intelligence

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. How can a model be biased even if 'gender' is not a feature?**

COMPARE **Q2. Compare transparency and explainability.**

APPLY **Q3. Group A 60/100 selected, B 36/100. Apply the four-fifths rule.**

EVALUATE **Q4. Should colleges ban AI tools entirely?**

CREATE **Q5. Draft a 5-line AI-use declaration for assignments.**

📝 EXAM PREP: Responsible Digital Intelligence

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four responsible-AI principles.

2. What is a proxy feature?

3. State the four-fifths rule.

4. What is a model card?

PART B • Long answer (5 marks)

5. Using a real case, explain how bias enters AI systems and propose a lifecycle of safeguards.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Data Privacy Matters

LET'S LEARN

- Personal data identifies a person directly or indirectly.
- India's DPDP Act 2023 requires consent & purpose.
- Protect data: minimise, secure, anonymise.

Privacy-by-design steps

Minimise

Consent

Encrypt

Anonymise

KEY TERMS

Data principal
Person the data is about

Data fiduciary
Org deciding data use

Pseudonymise
Replace identity with code

📖 CONCEPT EXPLAINED: Data Privacy Matters

💡 In this chapter

Every app, form and device collects data about us, and AI systems are hungry for it. Privacy is about having control over information that identifies us. This chapter explains what personal data is, the key ideas of India's Digital Personal Data Protection (DPDP) Act 2023, why 'anonymous' data often isn't, and practical techniques to protect privacy.

👤 1. What counts as personal data?

Personal data is any information that identifies a person directly or indirectly. Direct identifiers include name, phone number and email. Indirect identifiers, sometimes called quasi-identifiers, include age, pin code, gender, college and job title. Individually they seem harmless, but combined they can single out one person.

Some data is especially sensitive: health, financial, biometric and location data, and data about children. Misuse can cause discrimination, fraud or harm, so it deserves stronger protection.

🤔 PAUSE & THINK

Your age, pin code and college are listed together in a public spreadsheet. Could someone identify you? How?

⚖️ 2. India's DPDP Act 2023: key ideas

The Digital Personal Data Protection Act 2023 governs how organisations (data fiduciaries) handle the personal data of individuals (data principals). Key principles include processing data only for a lawful purpose, obtaining consent that is free, specific, informed and unambiguous, collecting only what is necessary, keeping data accurate, protecting it with security safeguards, and deleting it when no longer needed.

Data principals have rights, including the right to information about how their data is processed, the right to correction and erasure, and the right to grievance redressal. Special care is required for children's data, including verifiable parental consent. Serious breaches can attract very large penalties.

✏️ **WORKED EXAMPLE**

A college attendance app should state: what it collects (roll number, timestamps), why (attendance records), how long it keeps data (until graduation), who to contact for corrections, and how to withdraw consent where applicable.

🎭 3. Anonymisation is harder than it looks

Pseudonymisation replaces direct identifiers with codes, such as replacing names with random IDs. It reduces risk, but the data can be re-linked using a key, so it is still personal data. Anonymisation aims to make re-identification impossible, which is very difficult in practice.

In 2006, Netflix released 'anonymised' movie ratings for a competition; researchers re-identified some users by matching ratings with public reviews. Techniques like k-anonymity (each record matches at least k people on quasi-identifiers) and differential privacy (adding carefully calibrated noise to results) give stronger protection.

🖐️ **TRY IT**

Generalise this record for k-anonymity: Age 19, Pin 629001, Female. (Hint: age range, district, keep gender?)

🔐 4. Privacy in everyday life and AI projects

As a user: review app permissions, deny those not needed, use strong unique passwords with two-factor authentication, think before uploading photos of others, and read what data a 'free' app collects. As a builder: practise privacy by design by collecting the minimum data, securing it, limiting access, documenting purposes and deleting data on schedule.

Be especially careful with AI tools: do not paste personal or confidential information about others into public chatbots. Privacy is not about having something to hide; it is about having the right to decide who knows what about you.

🤔 **PAUSE & THINK**

List two permissions a photo-editing app should NOT need. Why?

✅ Key takeaways

- Personal data identifies people directly or through combinations.
- DPDP Act 2023: lawful purpose, consent, minimisation, security, rights.
- Pseudonymised data is still personal; true anonymisation is hard.
- k-anonymity and differential privacy strengthen protection.
- Practise privacy by design as both a user and a builder.

❓ Quick check: answer in one or two lines.

1. Differentiate data principal and data fiduciary.

2. Why is pseudonymised data still personal data?

3. Name two rights of data principals under the DPDP Act.

🔬 CONCEPT DEEP DIVE: Data Privacy Matters

🔷 Personal data identifies a person directly (name, Aadhaar-linked ID) or indirectly (combinations like age + pin code + gender).

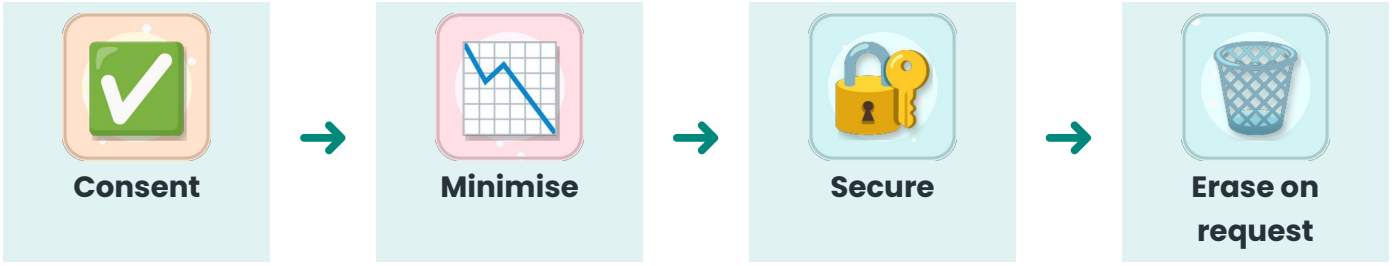
🔷 DPDP Act 2023 key ideas: lawful purpose, consent (free, specific, informed), data minimisation, accuracy, security, accountability.

🔷 Data principals' rights: access information, correction & erasure, grievance redressal and nominating someone to act for them.

🔷 Pseudonymisation replaces identifiers with codes (reversible with a key); anonymisation aims to make re-identification impossible.

🔷 Privacy techniques: k-anonymity (each record matches ≥ k people), differential privacy (calibrated noise), encryption, access control.

🔄 How it works



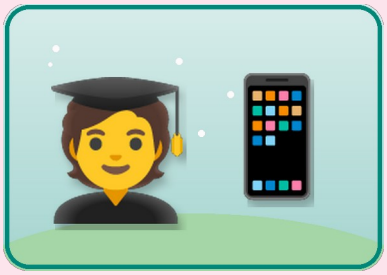
💻 Pseudonymising with a hash

```
import hashlib
def pseudo(name, salt='s3cr3t'):
    return hashlib.sha256((salt + name).encode()).hexdigest()[:10]
print(pseudo('Asha'))   # e.g. '7c2e...' (not the real name)
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Free Photo App

Characters: *Vikram (B.Com student), his sister Anu (a law student), a 'free' AI photo editor.*



SCENE 1

Vikram downloaded a free AI photo editor that turned selfies into cartoons. It asked for access to his contacts, location, microphone and all photos. Eager to try it, he tapped 'Allow' on everything.

Pause & predict: Which permissions does a photo editor actually need?

SCENE 2

His sister Anu, studying data protection law, frowned. She opened the privacy policy: 'We may share your data with partners for advertising and research.' 'Free isn't free,' she said. 'You're paying with your data, and your friends' contacts too.'

Pause & predict: Whose privacy did Vikram put at risk besides his own?

SCENE 3

Anu explained the DPDP Act: apps must ask for consent for specific purposes, collect only what is necessary and let users withdraw consent and request erasure. Together they revoked unnecessary permissions and emailed the company asking it to delete his data.

Pause & predict: Which DPDP rights did Vikram use?

SCENE 4

Vikram switched to an app that worked offline and asked only for photo access. He then helped his parents review the permissions on their phones too. 'Privacy isn't about hiding,' he told them. 'It's about choosing.'

⏸️ **Pause & predict:** What will you check before installing your next app?

🌟 The big idea

When an app is free, ask what data you are paying with. Privacy is the right to choose who knows what about you.

💬 Discuss with your classmates

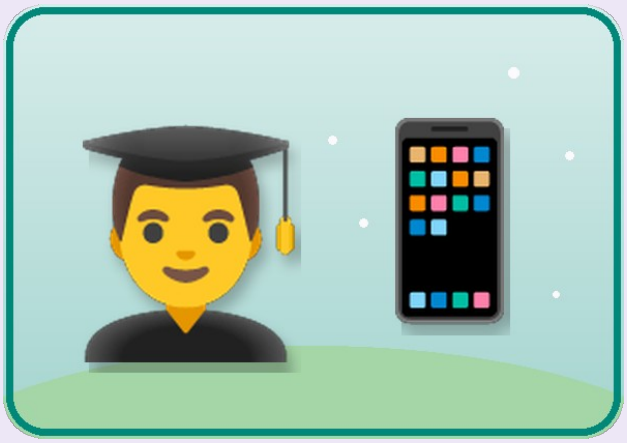
1. Review the permissions on one app on your phone. What surprised you?

2. Should apps be allowed to share data with 'partners'? Under what conditions?

🎞️ SCENARIO: Data Privacy Matters

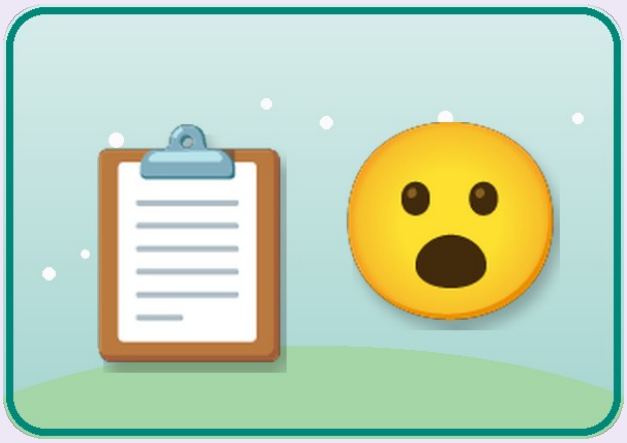
Read the scenario. Think about the choices made at each step.

1



Vikram installs a free 'AI photo editor' app.

2



It asks for contacts, location and microphone!

3

He reads the policy: data shared with 'partners'.

4

He denies permissions and picks a privacy-respecting app.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Privacy in daily digital life

App permissions

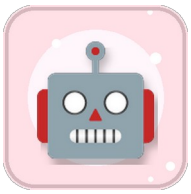
Cookie banners

Student records

Health apps

Face recognition

Shopping profiles



BOLT SAYS:

If an app is free, ask: what data am I paying with?

FUN FACT!

Under the DPDP Act 2023, penalties for data breaches can go up to ₹250 crore!

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The 'Anonymous' Netflix Prize Data

CASE FILE

In 2006, Netflix released 100 million 'anonymised' movie ratings for a competition. Researchers showed many users could be re-identified by matching a few ratings and dates with public reviews on another website. A planned second competition was cancelled after privacy complaints and a lawsuit.

🔍 Analyse the case

1. Why did removing names fail to protect users?

2. How could k-anonymity or differential privacy have helped?

3. What does this case teach about releasing 'anonymous' data?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Re-identification Challenge

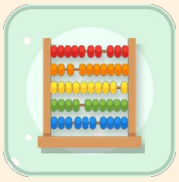
You need:

Data table

Clues sheet

Teams of 3

Steps:

1. Get a table with names removed (age, pin, gender).
2. Teams use public clues to re-identify 3 'people'.
3.  Apply k-anonymity: age ranges, districts.
4. Try again. Is it harder now?

THINK & TALK

Why is removing names alone not enough?

Record your observations & results:

💻 CODE LAB: Pseudonymise & Generalise

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  import hashlib
2  def pid(roll):
3      return hashlib.sha256(roll.encode()).hexdigest()[:8]
4  def age_band(a): return f'{a//5*5}-{a//5*5+4}'
5  rows = [('21CS001', 19), ('21CS002', 22)]
6  for roll, age in rows:
7      print(pid(roll), age_band(____))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What age bands are printed?**

EXPLAIN **T3. Is pid() output anonymous? Why or why not?**

EXTEND **T4. Add a secret salt to pid(). Why does it help?**

WORKSHEET: Data Privacy Matters (Part A)

Name: _____ Date: _____

1. Match the DPDP term to its meaning.

Data principal ●

● **Use data only for the stated reason**

Data fiduciary ●

● **The person the data is about**

Consent ●

● **Free, specific, informed agreement**

Purpose limitation ●

● **Organisation deciding how data is used**

2. Personal data or not? Tick ✓.

	Personal	Not
Mobile number		
City's average temperature		
Your live location		
Total students in India		

WORKSHEET: Data Privacy Matters (Part B)

Name: _____ Date: _____

1. Pseudonymised or anonymised? Tick ✓.

	Pseudo	Anon
Names replaced by codes; key stored separately		
Only district-level totals published		
Roll numbers swapped for random IDs with lookup table		
Aggregated averages with noise added		

2. Handling a data-erasure request. Write 1, 2, 3, 4.

Confirm erasure to the person ☐

Receive the request ☐

Delete data from all systems ☐

Verify the requester's identity ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Data Privacy Matters

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Data principal

Meaning: _____

Example: _____

Data fiduciary

Meaning: _____

Example: _____

Pseudonymise

Meaning: _____

Example: _____

? QUIZ TIME: Data Privacy Matters

Circle the right answer!

Q1. The person data is about:

a) Data principal

b) Data company

c) Data box

Q2. India's data protection law:

a) Motor Act

b) DPDP Act 2023

c) RTI Act

Q3. Collect only what's needed:

a) Maximise

b) Randomise

c) Minimise

Q4. Replacing names with codes:

a) Pseudonymise

b) Paint

c) Delete all

Q5. Removing names alone is...

a) Always safe

b) Often not enough

c) Fun

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Data Privacy Matters

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
👤	Age + pin code + gender can identify a person.		
🎭	Pseudonymised data is no longer personal data.		
✅	DPDP consent must be free and specific.		
🧮	Differential privacy adds calibrated noise.		
🎬	The Netflix Prize data could not be re-identified.		
🗑️	Data principals can request erasure.		

HOME CONNECT

Review app permissions on a family phone together. Turn off any that aren't needed.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Data Privacy Matters

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can combinations of harmless fields identify a person?**

COMPARE **Q2. Compare pseudonymisation and anonymisation.**

APPLY **Q3. A table has 3 rows sharing (20–25, Chennai, F). What k does this group give?**

EVALUATE **Q4. Should apps be allowed 'all-or-nothing' consent?**

CREATE **Q5. Draft a privacy notice (4 lines) for a college AI attendance app.**

📝 EXAM PREP: Data Privacy Matters

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Define personal data.

2. Differentiate data principal and data fiduciary.

3. What is k-anonymity?

4. Pseudonymisation vs anonymisation?

PART B • Long answer (5 marks)

5. Explain the main principles and rights under India's DPDP Act 2023, and how an AI app should comply.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Ratio 0.36 / 0.60 is...

a) 0.6 | b) 0.96 | c) 0.24

Q2. Ratio < 0.8 suggests...

a) Fairness | b) Possible bias | c) Nothing

Q3. Organisation deciding data use:

a) Principal | b) Proxy | c) Fiduciary

Q4. Each record matches ≥ k people:

a) k-anonymity | b) k-password | c) k-pop

Q5. Netflix Prize data showed...

a) Perfect privacy | b) Re-identification | c) New films

Q6. Hiring AI downgraded 'women's'. Cause:

a) Chance | b) Weather | c) Historical bias

WORD SEARCH: AI CARE

🔍 **Find these words. They go across → or down ↓.**

INTEGRITY

PRIVACY

CONSENT

BIAS

FAIRNESS

HASH

F	G	U	T	I	W	C	I	O	L
A	C	M	O	I	P	D	N	G	K
I	D	B	U	G	C	M	T	R	G
R	S	U	S	A	O	M	E	H	W
N	H	O	K	M	N	U	G	D	C
E	A	S	S	G	S	B	R	D	W
S	S	I	K	G	E	I	I	N	R
S	H	H	E	W	N	A	T	G	S
A	G	F	A	L	T	S	Y	T	W
W	K	M	P	R	I	V	A	C	Y

I finished AI CARE! Teacher's sign: _____

🔤 AI GLOSSARY (1/2)

🎯
Agent
Perceives & acts

🧠
AGI
General human-level AI

🔢
Array
Grid of numbers

🤝
Augmentation
AI assists humans

🤖
Automation
AI does the task

⚖️
Bias
Systematic unfairness

🪜
Chain-of-thought
Step-wise reasoning

🚫
Constraint
Limit on output

📉
Data drift
Data changes over time

🏢
Data fiduciary
Org deciding data use

👤
Data principal
Person the data is about

🐼
DataFrame
Labelled table

🌳
Decision tree
If-then splits

🏋️
Deliberate practice
Focused improvement

🎯
fit()
Train a model

🤵
Freelancing
Project-based work

😜
Hallucinated API
Made-up function

🔍
Heuristic
Smart rule of thumb

AI GLOSSARY (2/2)

Integrity
Honest academic work

JD
Job description

Narrow AI
Task-specific AI

Notebook
Code + notes + output

Pair programming
Two minds, one code

Pilot
Small real trial

Prediction
Estimate unknowns

Pseudonymise
Replace identity with code

Regression
Predict a number

Reproducible
Same steps, same result

Roadmap
Learning plan

Skill gap
Missing skills

Structured output
JSON / table format

T-shaped
Broad + deep skills

Task analysis
Break job into tasks

Transparency
Openness about AI

Unit test
Checks one function

Version control
Track code changes

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. PEAS for a robot vacuum: wheels are...

a) Actuators

b) Sensors

c) Environment

Q2. AlexNet error 26% → 15%: relative cut ≈

a) 11%

b) 42%

c) 15%

Q3. y = 3x - 2; MSE on (1,1),(2,5) is...

a) 1

b) 2

c) 0.5

Q4. BFS explores...

a) Level by level

b) Toward goal first

c) Randomly

Q5. Iris 150 rows, test_size 0.2 → test rows:

a) 20

b) 30

c) 120

❓ OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. sum([3, 4, 5]) returns...

a) 345

b) [12]

c) 12

Q7. 'Sure! {...}' fails json.loads because of...

a) Extra text

b) Numbers

c) Colour

Q8. Comparing prompt versions needs...

a) One lucky run

b) Fixed tests & rubric

c) No checks

Q9. Old mean 100, new 130. Drift above 20%?

a) No

b) Can't tell

c) Yes, 30%

Q10. AI contract review then lawyer check is...

a) Augmentation

b) Full automation

c) No AI

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Exposure 0.8 on 10 h, 0.2 on 8 h → assisted:

a) 18

b) 9.6

c) 1.0

Q12. Python in 4/4 JDs, SQL 3/4. Learn first:

a) Spark

b) Excel

c) Python

Q13. Look-alike domain + fee = ...

a) Fake offer

b) Genuine job

c) Scholarship

Q14. Public Kaggle notebooks show...

a) Birthdays

b) Applied skill

c) Taste

Q15. random.seed(42) ensures...

a) Faster code

b) Security

c) Same shuffle

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. clean('+91-98765-43210') returns...

a) 9876543210

b) +919876543210

c) Error

Q17. AI-invented package attack is called...

a) Phishing

b) Slopsquatting

c) Gaming

Q18. Selection 50% vs 30%: ratio is...

a) 0.8

b) 1.67

c) 0.6

Q19. Hashed roll numbers are...

a) Pseudonymous

b) Fully anonymous

c) Public

Q20. Age 22 falls in the 5-year band...

a) 15-19

b) 20-24

c) 25-29

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

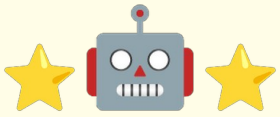
of AI Explorer

This is to say that

_____

of UG 1st Year finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. AI Demystified

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Supervised - Labelled spam; Unsupervised - Grouping customers; Reinforcement - Robot rewards; Self-supervised - Next word
Part A 2: Chess engine: Narrow; Learns any job like a human: AGI; Speech-to-text app: Narrow; Solves any new problem unaided: AGI
Part B 1: 1-Rule-based expert systems, 2-Classical machine learning, 3-Deep learning, 4-Generative AI
Part B 2: Performance - Cleanliness; Environment - Rooms; Actuators - Wheels & brushes; Sensors - Bump & dirt sensors
Challenge hints: Q1: Shows DL is one ML approach; ML is one AI approach; rules-based AI also exists. • Q2: Supervised: labelled spam; RL: game agent learning from rewards. • Q3: P: safety, time; E: roads; A: steering, brakes; S: cameras, LiDAR, GPS. • Q4: Uncertain; experts disagree; define AGI; note hype vs evidence. • Q5: 1956 Dartmouth, 1997 Deep Blue, 2012 AlexNet, 2016 AlphaGo, 2017 Transformer, 2022 ChatGPT.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 26; Output: 15 heat / 22 idle / 30 cool; Debug: string vs int, use float(temp); Extend: elif temp >= 24: return 'fan' (before the cool check).
Exam prep key points: 1: AI: machines doing intelligent tasks; ML: learning from data (subset). • 2: Judges whether a machine's conversation is indistinguishable from a human's. • 3: Spam filter, chess engine, face unlock, etc. • 4: Labels created from data itself, e.g., predict the masked word. • 5: Dartmouth 1956, expert systems, Deep Blue 1997, AlexNet 2012, AlphaGo 2016, Transformers 2017, GenAI; data + GPUs + algorithms.

2. The Intelligence Behind Machines

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Predict house price in ₹: Regress.; Email: spam or not: Classif.; Tomorrow's temperature: Regress.; Disease: yes or no: Classif.
Part A 2: 7, 13, 3
Part B 1: 4, 1, 0.5
Part B 2: BFS - Level by level; A* - Cost + heuristic; Tree - Split on best feature; kNN - Vote of closest
Challenge hints: Q1: Heuristic h(n) guides search toward the goal; fewer nodes expanded. • Q2: Tree: trains once, explainable rules; kNN: no training, slow prediction, less explainable. • Q3: Preds 1, 4; errors 0, 1; MSE = (0 + 1) / 2 = 0.5. • Q4: No: still used where rules are clear and auditable (tax, safety interlocks). • Q5: Clouds? → rain forecast > 50%? → long walk? → yes/no leaves.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: len(xs); preds [3,5,7,9], mse 0.0; b = 2 → errors 1 each → MSE 1.0; loop: for w in [1,2,3]: ... print(w, mse).
Exam prep key points: 1: f(n) = g(n) + h(n): path cost + heuristic estimate. • 2: Regression predicts numbers; classification predicts categories. • 3: Majority vote of the k nearest training points. • 4: Rule-based system encoding human expert knowledge. • 5: Split on best feature (Gini/entropy), recurse to leaves; explainable; prone to overfitting.

3. Python + AI Starter

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: NumPy - Numeric arrays; pandas - Tables; Matplotlib - Charts; scikit-learn - ML models
Part A 2: 1-Load data, 2-train_test_split, 3-model.fit(X_train, y_train), 4-Score on test data
Part B 1: 12, [2, 4, 6], 6
Part B 2: head - First 5 rows; describe - Summary stats; dropna - Remove missing; groupby - Aggregate by category
Challenge hints: Q1: To measure performance on unseen data and detect overfitting. • Q2: List: flexible types, slow maths; array: one type, fast vectorised maths. • Q3: [x**2 for x in range(1, 11) if x % 2 == 0] → [4, 16, 36, 64, 100]. • Q4: Often a sign of overfitting; check test accuracy. • Q5: read_csv → clean → groupby subject mean → plot bar chart.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 5; test samples: 30; Debug: X and y lengths differ (30 vs 120) — use Xtr; Extend: for k in range(1,16): fit, score, track max.
Exam prep key points: 1: NumPy arrays, pandas tables, Matplotlib plots, scikit-learn ML. • 2: Splits data into training and testing sets. • 3: [x**3 for x in range(1, 6)]. • 4: To make results reproducible. • 5: Load → split → choose model → fit → predict → score/metrics; mention preprocessing.

4. Prompt with Purpose

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-F 5-T 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Few-shot - 2 solved examples; CoT - Step by step; Persona - Act as an examiner; Self-critique - Review your answer
Part A 2: Tell me about AI: Vague; Explain CNNs in 150 words for B.Sc students with 1 example: Purposeful; Write something on climate: Vague; List 5 Python data types as a JSON array: Purposeful
Part B 1: 1-Fix a set of test inputs, 2-Run each prompt version several times, 3-Score outputs with a rubric, 4-Compare scores across versions
Part B 2: {"term": "AI", "year": 1956}: Valid; {term: AI, year: 1956}: Invalid; ["NumPy", "pandas"]: Valid; Sure! Here is the JSON: {...}: Invalid
Challenge hints: Q1: Examples show the exact pattern; model imitates structure and style. • Q2: CoT: shows steps, fewer errors, checkable; direct: faster, error-prone. • Q3: Specify keys: question, options[4], answer; level; count = 3; JSON only. • Q4: Yes for practice, if verified against textbook; not for graded work. • Q5: Accuracy, clarity, level-fit, examples, length compliance.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 2; prints OVERFITTING; line 7: extra text before JSON and single quotes; try: json.loads(bad) except json.JSONDecodeError: print('Invalid JSON').
Exam prep key points: 1: Goal, audience, context, constraints/format. • 2: Giving examples in the prompt to guide output. • 3: Machine-readable, easy to parse and validate. • 4: Asking the model to reason step by step. • 5: Fixed inputs, rubric, multiple runs, one variable at a time, compare scores, check facts.

5. AI in the Real World

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Predict - Electricity demand; Detect - Bridge cracks; Generate - Public notice; Optimise - Bus routes
Part A 2: 21, 11, 8
Part B 1: 30%, Yes, No
Part B 2: 1-Collect data, 2-Build model, 3-Integrate into workflow, 4-Measurable outcome
Challenge hints: Q1: Drift, poor integration, user distrust, wrong metric, edge cases. • Q2: Prediction: demand forecast; optimisation: delivery route planning. • Q3: |130 - 100| / 100 = 30% > 20% → drift. • Q4: No; experts verify; combine with official protocols. • Q5: Leaf photos → CNN → app & SMS → farmer sprays → yield saved.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: m1; m1 = 100, m2 = 130, change = 0.3; yes, prints Retrain!; Extend: print(st.stdev(train), st.stdev(live)).
Exam prep key points: 1: Predict, detect, generate, optimise. • 2: Change in data distribution over time, harming accuracy. • 3: Tests value and risks at small scale before full rollout. • 4: Crop advisories, cyclone alerts, health screening, etc. • 5: Data → model → integrate → action → outcome; drift, cost, trust, integration, language.

6. AI in Your Profession

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Copying invoice data into forms: Automate; Doctor's final diagnosis with AI scan: Augment; Resetting a password: Automate; Lawyer drafting strategy with AI research: Augment
Part A 2: Civil engineer - Structural simulation; Accountant - Ledger anomalies; Lawyer - Contract review; Agronomist - Satellite crops
Part B 1: 6, 12, 2.5
Part B 2: Circle: Comforting a patient, Negotiating a deal, Final legal judgement, Mentoring a junior
Challenge hints: Q1: Jobs mix tasks with different AI exposure; titles hide this. • Q2: Radiologist: image analysis augmented; nurse: physical care & empathy less exposed. • Q3: 0.6 × 10 + 0.2 × 30 = 12 hours. • Q4: No: domain knowledge needed to guide & judge AI; degrees evolve. • Q5: Modules: tools in field, data skills, case studies, ethics, project.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: values; assisted = 8 + 1.6 + 0.6 = 10.2, total 30 → 34%; with emails: 13.2 h; max(tasks, key=lambda t: tasks[t][0]) → 'reports'.
Exam prep key points: 1: Automation: AI does task; augmentation: AI assists human. • 2: Jobs contain tasks with different AI exposure. • 3: Domain expertise plus AI literacy and data skills. • 4: Discipline-specific answer (e.g., CAD generative design). • 5: Task list, automate/augment/unchanged, new tasks, skills plan, ethical issues.

7. Navigate the AI Job World

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Analyst - Dashboards & SQL; ML engineer - Deploy & monitor; Product manager - Decides features; Governance - Policies & audits
Part A 2: SQL: Must; Excel / spreadsheets: Must; Video editing: Nice; Deep learning research: Nice
Part B 1: Circle: Asks you to pay a fee, 'Accept in 1 hour or lose it', Spelling errors in the offer, No interview at all
Part B 2: Python, SQL, Spark
Challenge hints: Q1: Patterns reveal core vs niche skills; avoid overfitting to one post. • Q2: Analyst: SQL, dashboards, descriptive; scientist: ML models, predictive. • Q3: Python and SQL (highest frequency). • Q4: Often yes, with strong projects; learn the gaps quickly. • Q5: Mon JD scan, Wed skill block, Fri 3 tailored applications, Sun network.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: split; [('python', 3), ('sql', 3), ('excel', 2)]; use jd.lower().split(); [w for w, c in Counter(words).items() if c == 1].
Exam prep key points: 1: Data, ML engineering, product/design, governance/policy, research. • 2: Difference between required and current skills. • 3: Payment requests, look-alike domains, urgency, no interview. • 4: Internships, hackathons, placements, open-source, freelancing. • 5: Collect JDs, tally skills, must vs nice, gap analysis, projects, applications.

8. Build Your AI Skillset

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: 1-Set a specific goal, 2-Learn the basics, 3-Build a project, 4-Share & get feedback
Part A 2: Binge-watching tutorials: Passive; Building a model from scratch: Active; Debugging your own code: Active; Only reading slides: Passive
Part B 1: 27, 6.75, Yes
Part B 2: Child's own work
Challenge hints: Q1: Active recall, debugging, feedback, transfer to new problems. • Q2: Certificate: proves completion; project: proves application & depth. • Q3: 27 / 4 = 6.75 h → yes, on track. • Q4: T-shaped: broad base, one deep specialism. • Q5: W1 basics, W2 cleaning, W3 groupby/plots, W4 mini-project + blog.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: target; weeks_met = 4, avg = 6.17; final streak = 0 (last week 3); track best = max(best, streak) inside the loop → 2.
Exam prep key points: 1: Broad foundations plus depth in one area. • 2: Focused practice on weaknesses with feedback. • 3: Show real applied skills, beyond certificates. • 4: NPTEL, Kaggle Learn, university OCW, etc. • 5: 3 × 30-day sprints: learn → build → share; metrics; reflection; accountability.

9. AI Workbench

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-T 3-F 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Colab - Notebooks with GPU; GitHub - Share & version code; Kaggle - Datasets; Streamlit - Quick web app
Part A 2: 1-git init / clone, 2-git add file.py, 3-git commit -m 'msg', 4-git push
Part B 1: Circle: Fixed random seed, requirements.txt, README with steps, Git commit history
Part B 2: data.gov.in - Indian government data; Kaggle - Competitions; UCI - Classic academic; Model hubs - Pre-trained models
Challenge hints: Q1: Data splits & weight initialisation vary; seed makes runs repeatable. • Q2: Notebook: exploration, visuals; script: production, testing, reuse. • Q3: git init; git add .; git commit -m 'init'; git push. • Q4: Good for quick prototypes; risky without data understanding. • Q5: Title, problem, data, method, results, how to run (+ licence).
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 42; prints 8 2; yes, same seed → same shuffle; none needed (standard library) — note Python version in README.
Exam prep key points: 1: Interactive document combining code, output and notes. • 2: add → commit → push (after init/clone). • 3: Different versions can change results; reproducibility. • 4: Kaggle, UCI, data.gov.in, model hubs. • 5: Seeds, versions, data access, code sharing, README, documented hyperparameters, Git.

10. AI Coding Companion

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Run tests on AI-written code: Good; Paste API keys into a public AI chat: Risky; Ask AI to explain unfamiliar code: Good; Submit AI code without reading it: Risky
Part A 2: 1-Describe the task clearly, 2-Get AI draft code, 3-Review & test the code, 4-Commit when tests pass
Part B 1: 9876543210, 1234567890, Fail
Part B 2: Hallucinated package - Check registry; Leaked key - Env vars; Logic bug - Tests & review; Licence - Licence scanning
Challenge hints: Q1: Running ≠ correct; edge cases, logic and security bugs hide. • Q2: Write: speed, risk of errors; explain: learning, verify understanding. • Q3: assert add(-2, 3) == 1; assert add(0, 0) == 0. • Q4: Debate: real-world skill vs measuring fundamentals; maybe separate tasks. • Q5: Test all code, no secrets, verify packages, disclose AI use.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: '9876543210'; yes all pass; better to raise ValueError if len(digits) < 10; assert clean('') == '' or pytest.raises(ValueError).
Exam prep key points: 1: Generate, explain, debug, refactor, write tests. • 2: A function/package suggested by AI that doesn't exist. • 3: Verify functions behave correctly, incl. edge cases. • 4: Risk of leaks; data may be stored or exposed. • 5: Specify → draft → test → review → commit; prevents logic bugs, fake packages, leaks.

11. Responsible Digital Intelligence

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Fairness - Test across groups; Transparency - Tell users; Safety - Red-team; Accountability - Named owner
Part A 2: Using AI to brainstorm, then writing yourself (disclosed): OK; Submitting AI essay as your own: Misconduct; Asking AI to explain a concept: OK; Citing AI-invented references: Misconduct
Part B 1: 0.36, 0.6, Possible bias
Part B 2: Historical - Past hiring; Sampling - Cities only; Label - Annotators' prejudice; Feedback loop - Policing data
Challenge hints: Q1: Proxies (clubs, colleges, words) correlate with gender. • Q2: Transparency: openness that AI is used & how built; explainability: why a decision. • Q3: 0.36 / 0.60 = 0.6 < 0.8 → possible adverse impact. • Q4: Bans are hard to enforce; better clear policies, disclosure, AI literacy. • Q5: Tools used, for what, what I wrote, how verified, my responsibility.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: max; rate {A: 0.5, B: 0.3}, ratio 0.6; prints Review!; C rate 0.5 → min 0.3 / max 0.5 = 0.6 (still Review!).
Exam prep key points: 1: Fairness, transparency, accountability, safety, privacy. • 2: Feature that indirectly encodes a sensitive attribute. • 3: Selection-rate ratio below 0.8 suggests adverse impact. • 4: Document describing a model's use, data, limits & metrics. • 5: Data/sampling/label/proxy/feedback bias; audits, metrics, human review, docs, monitoring.

12. Data Privacy Matters

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Principal - Person; Fiduciary - Organisation; Consent - Free, specific, informed; Purpose limitation - Only stated reason
Part A 2: Mobile number: Personal; City's average temperature: Not; Your live location: Personal; Total students in India: Not
Part B 1: Names replaced by codes; key stored separately: Pseudo; Only district-level totals published: Anon; Roll numbers swapped for random IDs with lookup table: Pseudo; Aggregated averages with noise added: Anon
Part B 2: 1-Receive the request, 2-Verify the requester's identity, 3-Delete data from all systems, 4-Confirm erasure to the person
Challenge hints: Q1: Quasi-identifiers together become unique (age + pin + gender). • Q2: Pseudo: reversible with key, still personal data; anon: irreversible (ideally). • Q3: k = 3 for that group. • Q4: No under good practice; consent should be specific and granular. • Q5: Data collected, purpose, retention, rights & contact.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: age; bands 15-19 and 20-24; no — pseudonymous: roll numbers can be hashed again to match; salt stops dictionary/lookup attacks.
Exam prep key points: 1: Data identifying a person directly or indirectly. • 2: Principal: the person; fiduciary: the organisation deciding use. • 3: Each record matches at least k people on quasi-identifiers. • 4: Pseudo: reversible with key; anon: irreversible. • 5: Consent, purpose limitation, minimisation, security, rights (access, correction, erasure, grievance).

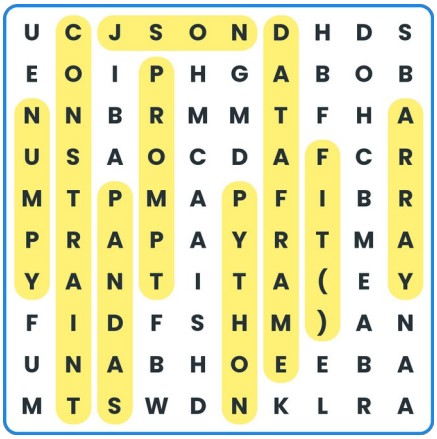
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

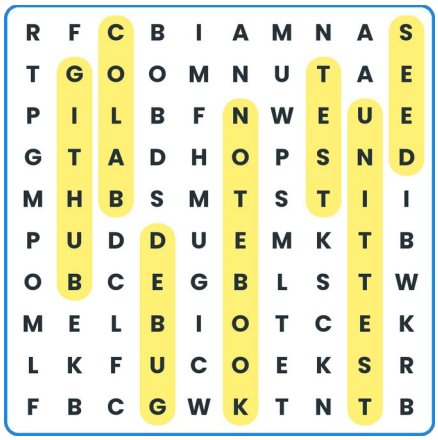
AI DISCOVER



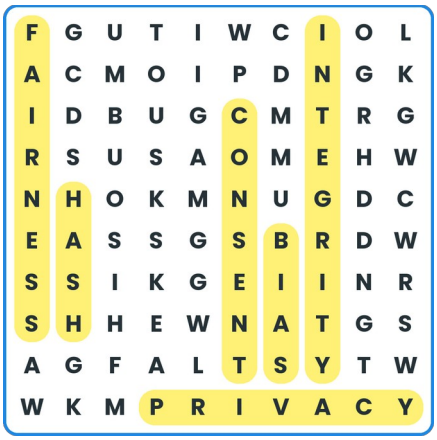
AI CONNECT

AI SOLVE

AI RISE



AI CREATE



AI CARE

MY NOTES

MY NOTES

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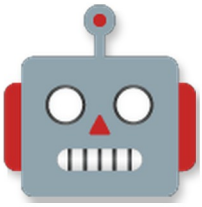

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